INCH-POUND

MIL-R-55333A(CR)
28 November 1990
SUPERSEDING
MIL-R-55333(EL)
17 December 1971

MILITARY SPECIFICATION

RADIO SET AN/GRC-144 ( ) (UNITS OF) AND INDICATOR, ANTENNA ALIGNMENT ID-1708( )/GRC

This specification is approved for use by the Communications-Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the following units of a general purpose, frequency modulated tactical radio set designated Radio Set AN/GRC-144( ). The radio set is capable of providing a full binary system for the transmission and reception of 48 reliable, high quality, voice channels in the frequency range of 4.4 to 5.0 Gigahertz (GHz) when operating with PCM multiplex equipment over line-of-sight paths (see 6.1 and 6.6):

Transmitter, Radio T-1054( )/GRC-144
Receiver, Radio R-1467( )/GRC-144

This specification also covers Antenna Alignment Indicator ID-1708( )/GRC which is used to adjust the antenna in both azimuth and elevation for maximum signal reception.

1.2 Appendices. The following appendices form a part of this specification:
Appendix A - Reliability Test Requirement for Radio Set AN/GRC-144( ).
Appendix B - Reconfiguration Kit Requirements for Asynchronous Digital Combining (ADC) Mode of Operation for Radio Set AN/GRC-144( ).

Beneficial comments (recommendations,additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Communications-Electronics Command, ATTN: AMSEL-ED-TD, Fort Monmouth, NJ 07703-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5820

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2. APPLICABLE DOCUMENTS

2.1. Government documents.

2.1.1. Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-P-11268 - Parts, Materials and Processes Used in Electronic
MIL-M-13231 - Marking of Electronic Equipment Items
MIL-F-14072 - Finishes for Ground Electrical Equipment
MIL-I-46058 - Insulating Compound, Electrical (For coating Printed Circuit Assemblies)

STANDARDS

MILITARY

MIL-STD-252 - Wired Equipment, classification of Visual and Mechanical Defects
MIL-STD-275 - Printed Wiring for Electronic Equipment
MIL-STD-454 - Standard General Requirements for Electronic Equipment
MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the control of Electromagnetic Interference
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of
MIL-STD-810 - Environmental Test Methods & Engineering Guidelines

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

GL-SM-B-323439 Gages for Receiver, Radio R-1467( )/GRC-144
GL-SM-B-323356 Gages for Synthesizer, Electrical Frequency
GL-SM-B-323440 Gages for Transmitter, Radio T-1054( )/GRC-144

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GL-SM-B-405738 Test Fixture Set, Electronic-Amplifier, AF-RF
GL-SM-B-405740 Test Fixture Set, Electronic-Demodulator
GL-SM-B-405742 Test Fixture Set, Ellectronic-Amplifier, IF, 70 MHz
GL-SM-B-405744 Test Fixture set, Electronic-Amplifier, Mixer
GL-SM-B-405750 Test Fixture Set, Electronic-Alarm Monitor
GL-SM-B-405752 Test Fixture Set, Electronic-Tunnel Diode Amplifier
GL-SM-B-405762 Test Fixture Set, Electronic-Frequency Multiplier, 2nd Stage (Transmitter)
GL-SM-B-405764 Test Fixture Set, Electronic-Frequency Multiplier, 3rd Stage
GL-SM-B-405766 Test Fixture Set, Electronic-Mixer Crystal, XMTR
GL-SM-B-405768 Test Fixture Set, Electronic-Modulator, Radio Transmitter
GL-SM-B-405770 Test Fixture Set, Electronic-Control, Electronic Frequency
GL-SM-B-405772 Test Fixture Set, Electronic-Regulator, Voltage, 15/28V
GL-SM-B-405774 Test Fixture Set, Electronic-Regulator, Voltage, 12V
GL-SM-B-405776 Test Fixture Set, Electronic-Regulator, Voltage, 5/6V
GL-SM-B-405778 Test Fixture Set, Electronic-Oscillator, RF, 100 MHz
GL-SM-B-405794 Test Fixture Set, Electronic-Regenerator, Digital Cable
GL-SM-B-405800 Test Fixture Set, Electronic-Amplifier, AF
GL-SM-B-405802 Test Fixture Set, Electronic-Alarm Monitor
GL-SM-B-405816 Test Fixture Set, Electronic-Oscillator Voltage Controlled (4.608 MHz)
GL-SM-B-405948 Test Fixture Set, Electronic-Amplifier Detector, Radio
GL-SM-B-405968 Test Fixture Set, Electronic-Processor, Digital, Radio
GL-SM-B-405987 Test Fixture Set, Electronic-Amplifier-Frequency Multiplier, Transmitter
GL-SM-B-405991 Test Fixture Set, Electronic-Amplifier-Frequency Multiplier, Receiver
GL-SM-B-406619 Test Fixture Set, Electronic-Regenerator, Digital, Radio
GL-SM-B-406633 Test Fixture Set, Electronic-Control-Comparator, Radio
GL-SM-B-407123 Test Fixture Set, Oscillator, RF, Voltage Controlled
GL-SM-B-407124 Test Fixture Set, Converter, Frequency, Electronic
GL-SM-B-407125 Test Fixture Set, Frequency Divider, Variable No. 1
GL-SM-B-407126 Test Fixture Set, Detector, AF, Phase Error
GL-SM-B-407127 Test Fixture Set, Frequency Divider, Variable No. 2

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GL-SM-B-407128	Test Fixture Set, Power Supply
GL-SM-B-407129	Test Fixture Set, Frequency Divider, Fixed
GL-SM-B-407130	Test Fixture Set, Oscillator, RF, Standard
GL-SM-B-407138	Test Fixture Set, Electronic-Frequency Multiplier Oscillator - 220 MHz
GL-SM-B-407151	Test Fixture Set, Electronic-Control Comparator, Cable
GL-SM-B-407211	Test Fixture Set, Electronic-Orderwire Daughter Board #1
GL-SM-B-407212	Test Fixture Set, Electronic-Orderwire Daughter Board #2
GL-SM-B-407213	Test Fixture Set, Electronic-Orderwire Daughter Board #3
GL-SM-B-407214	Test Fixture Set, Electronic-Orderwire Daughter Board #4
GL-SM-B-407215	Test Fixture Set, Electronic-Orderwire Daughter Board #5
GL-SM-B-407216	Test Fixture Set, Electronic-Processor, Digital Cable
SC-DL-531016	Cable Assembly, RF, CG-2437( )/TCC (-FT -IN)
SC-DL-531017	Cable Assembly, RF, CG-2438( )/TCC (-FT -IN)
SM-D-650500	Amplifier, AF-RF
SM-F-650502	Modulator, Radio Transmitter
SM-F-650503	Control, Electronic Frequency
SM-F-650504	Alarm Monitor
SM-C-650508	Power Supply, Transmitter Assy
SM-F-650909	Oscillator, RF, 100 MHz
SM-D-650511	Synthesizer Assembly
SM-D-650512	Orderwire Assembly
SM-F-650518	Demodulator
SM-D-650520	Amplifier Mixer
SM-D-650522	Mixer Stage Frequency (RCVR)
SM-C-650523	Power Supply, Receiver Assembly
SM-D-650527	Mixer Crystal (XMTR)
SM-D-650528	Frequency Multiplier, 2nd Stage
SM-D-650529	Freq. Mult. 3rd Stage
SM-D-650530	Regulator, Voltage 15/28V
SM-D-650531	Regulator, Voltage 12V
SM-D-650532	Regulator, Voltage 5/6V
SM-D-650539	Regenerator, Digital Cable
SM-D-650545	Alarm Monitor (DDM)
SM-D-650548	Oscillator, R.F. Voltage Controlled
SM-D-650549	Freq. Divider, Var. #1
SM-D-650550	Freq. Divider, Fixed
SM-D-650551	Converter, Freq. Electronic
SM-D-650552	Freq. Divider, Var. #2
SM-D-650553	Detector, A.F., Phase Error
SM-B-650580	Oscillator, Voltage Controlled (4.608 MHz - DDM)
SM-B-651379	Amplifier, Tunnel Diode
SM-B-651395	Specification-Coupler, Directional CU-1890( )/GRC

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SM-D-651609	Daughter Board #1
SM-D-651610	Daughter Board #2
SM-D-651611	Daughter Board #3
SM-D-651612	Daughter Board #4
SM-D-651613	Daughter Board #5
SM-C-651807	Power Supply
SM-C-651989	Oscillator, R.F., Standard
SM-F-652590	Amplifier, I.F., 70 MHz
SM-B-652910	Connector, Plug, Electrical
SM-B-652971	Connector, Plug, Electrical (Type MHV)
SM-B-652972	Connector, Plug, Electrical (Type MHV)
SM-A-653688	Transmitter Test Set and Power Cable
SM-A-653691	Orderwire Daughter Board Extenders
DL-SM-B-683614	Indicator, Antenna Alignment ID-1708( )/GRC
DL-SM-B-683616	Transmitter, Radio T-1054( )/GRC-144
DL-SM-B-683618	Receiver, Radio R-1467( )/GRC-144
SM-F-683619	Receiver, Radio R-1467( )/GRC-144
SM-D-683635	Attenuator Assembly
SM-C-683640	Filter Low Pass, 4.5 MHz
SM-F-683648	Frequency Multiplier-OSC, 220 MHz
SM-F-683663	Amplifier-Freq. Mult, (XMTR)
SM-D-683686	Modem, Digital Data Assy
SM-D-683689	Control-Comparator, Cable
SM-D-683690	Processor, Digital, Cable
SM-D-683691	Amplifier, A.F.
SM-D-683693	Amplifier-Detector, Radio
SM-D-683694	Regenerator, Digital, Radio
SM-D-683695	Control-Comparator, Radio
SM-D-683696	Processor, Digital, Radio
SM-B-683715	Dummy Load, Electrical DA-541( )/GRC
SM-B-683716	Adapter, Waveguide UG-1883( )/GRC
SM-B-683838	Connector, Plug, Electrical
SM-B-683846	Connector, Receptacle, Electrical
SM-F-684377	Amp-Freq. MUlt, (RCVR)
SM-D-684447	Filter, Bandpass, 70 MHz
SM-D-684499	Filter, Bandpass
SM-B-684756	Receiver, Cabinet Level Test Set
SM-A-684761	Power Panel-Transmitter Test
SM-A-684762	Capacitive Decoupler (1.0 pF)
SM-A-684766	System Test Set and Interconnect Cable

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1. First article. The contractor shall furnish eight (8) first article samples for approval, if required by the invitation for bids and the contract (see 4.3 and 6.2c (1)).

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3.2. Construction. The equipment shall be constructed in accordance with the requirements of this specification and the following drawings and Drawing Data Lists:

DL-SM-B-683614 Indicator, Antenna Alignment ID-1708( )/GRC
DL-SM-B-683616 Transmitter, Radio T-1054( )/GRC-144
DL-SM-B-683618 Receiver, Radio R-1467( )/GRC-144

3.3. Parts, materials, and processes: General. In addition to the requirements of this specification, the requirements of MIL-P-11268 including the selection requirements therein shall apply (see 4.4).

3.4. Finish, Protective. The equipment shall be finished in accordance with MIL-F-14072 and the equipment drawings. This includes finish of hardware, such as handles, brackets, chassis, screws, etc., and necessary touch-up after mounting. All colored final film finishes shall match color chips provided by the procuring agency (see 4.4 and 6.3).

3.5. Marking.

3.5.1. General. Marking shall conform to MIL-M-13231 and applicable construction drawings (see 4.4). Front panel marking shall be Group I as described in MIL-M-13231.

3.5.2. Visibility. Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.5.3. Serial Numbers. The following items shall have serial numbers:

Transmitter, Radio T-1054( )/GRC-144
Receiver, Radio R-1467( )/GRC-144
Indicator, Antenna Alignment ID-1708( )/GRC

3.5.4. Identification numbers. All modules, sub-modules, assemblies, sub-assemblies, units and parts thereof shall be numbered where practicable to afford a means for configuration control and through-production traceability. The contractor, at his option, may prefix or suffix these numbers with a lot, batch or control number or letter according to his own requirements. These identification nmumbers may be rubber-stamped.

3.6. Conformal coating. Printed wiring assemblies shall be conformally coated in accordance with MIL-STD-275 with a coating material that conforms to MIL-I-46058, type UR (Polyurethane).

3.7. Tropicalization.

3.7.1. Treatment of materials. Fungus-inert materials shall be used where practicable. These materials shall be in accordance with Requirement 4 of MIL-STD-454. When materials are required to be teated, they shall be cleaned in accordance with Requirement 9 of MIL-STD-454.

3.7.2. Fungus. The equipment (including accessories) shall show no evidence of viable fungus or corrosion when subjected to the test specified in 4.8.5.

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Corrosion is any visible degradation that can be attributed to flaky, pitted, blistered, or otherwise loosened finish or metal surface.

3.8. Service Conditions. The equipment shall meet the following service conditions: NOTE: A stabilization (warm-up) period of up to thirty (30) minutes is required before taking data (see 4.8.1.3, 4.8.2.1, 4.8.3.1).

3.8.1. Temperature. The equipment shall be subjected to the temperature procedures as shown in Method 501.1 and 502.1 of MIL-STD-810.

3.8.1.1. Low. During testing as specified in 4.8.1.1, the equipment shall meet the following requirements:

PARAGRAPH	MEASUREMENT
3.12.1	Radio Test Set Operation
3.12.2	Power Output
3.12.3	Deviation
3.12.4	Digital Data Modem (PCM Threshold Level) Operation
3.12.8	Transmitter Frequency Accuracy
3.12.11	Orderwire Peak Limiting
3.12.14 & 3.12.15	Transmitter Meter Readings and Alarm Operation
3.13.1	Frequency Accuracy
3.13.3	Receiver Noise Figure
3.13.4	Carrier Level Threshold
3.13.6	Output Capability and Stability
3.13.7	Receiver Bandwidth
3.14.2	Video Frequency Response
3.13.8 & 3.13.9	Receiver Meter Readings and Alarm Operation
3.15 & 3.15.1	Antenna Alignment Indicator Operational Check (Perform with Receiver under Test)

After testing, the equipment shall meet full specification performance for these measurements.

3.8.1.2. High. During testing as specified in 4.8.1.2, the equipment shall meet the following requirements:

Repeat all measurements for the Transmitter, Receiver and Antenna Alignment Indicator as delineated under 3.8.1.1 for Low Temperature. After testing, the equipment shall meet full specification performance for these measurements.

3.8.2. Humidity (Moisture-Resistance). During testing as specified in 4.8.2, the equipment shall meet the following requirements:

Repeat all measurements for the Transmitter, Receiver and Antenna Alignment Indicator as delineated under 3.8.1.1. After testing, the equipment shall meet full specification performance for these measurements.

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3.8.3. Altitude (Elevation). The equipment (Transmitter, Receiver and Antenna Alignment Indicator) shall meet full specification performance for the requirements listed under 3.8.1.1 prior to, during, and after testing as specified in 4.8.3. During step 3 of para 4.8.3, the primary meter readings of the equipment under test shall be restricted to only those readings which can be monitored through the test chamber window.

3.8.4. Vibration.

3.8.4.1. Bounce, vehicular. Prior to and after testing as specified in paragraph 4.8.4.1 the equipment shall meet the performance requirements of paragraph 3.17, and there shall be no physical damage.

3.9. Preconditioning. (See 4.7).

3.9.1. Bounce Preconditioning. All equipments on order (transmitter, receiver and spare modules) shall be capable of meeting the requirements herein, without subsequent processing, after subjection to the bounce preconditioning of 4.7.1.

3.9.2. Electrical preconditioning (burn-in). All equipments on order (transmitter, receiver and spare modules) shall be operated for forty eight (48) hours. The control alarm circuits, RF power output, and simulated PCM traffic error rates shall be continously monitored to detect failures. The equipment shall meet the contractor performed Group A Electrical Tests (Table III), without subsequent processing, after completion of the vibration and temperature periods (see 4.7.2).

3.10. Interchangeability. Like units, modules, chassis, assemblies, sub-assemblies and replaceable parts shall conform to Requirement 7 of MIL-STD-454 (see 4.9).

3.11. Reliability (see 4.5.5). Radio Sets that undergo reliability testing shall demonstrate a specified Mean-Time-Between-Failure (MTBF) of six-hundred (600) hours. The required reliability shall be demonstrated by a reliability test in accordance with the Reliability Test Requirement (Appendix A).

3.12. Transmitter, Radio T-1054( )/GRC-144. The radio transmitter (cabinet level) shall meet the following electrical requirements.

3.12.1. Radio Test Set operation. The Radio Test Set output shall be -33dBm minimum at 4.7 GHz with its controls set for HI Output level, and -63dBm minimum at LOW Output level (see 4.11.2.6).

3.12.2. Power output. The transmitter power output shall be greater than 0.25 watts (250mW) through the frequency range of 4.4 to 5.0 GHz when operated into a matched load. The power output shall be no less than 0.16 watts (160mW) during or 0.20 watts (200mW) after any environmental test (see 4.11.5).

3.12.3. Transmitter Deviation (48/96 Chan PCM): The deviation shall be adjustable such that with input levels of 1.0V to 4.0V peak-to-peak, the deviation (Traffic) meter shall indicate 100 $\pm$ 10 divisions and the rated peak deviations shall be as follows:

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48 Channels Binary (or 96 Channels Biternary): ± 720 kHz Peak

96 Channels Binary: ± 1750 kHz Peak

The peak deviation shall be verified using the First Carrier Disappearance Method for an FM signal.

FM Peak Deviation : F Carrier Disappearance X 2.40 (see 4.11.4).

3.12.4. Digital Data Modem (PCM Threshold Level) Operation. The transmitter and receiver operating in a 48 PCM channel made in a closed loop through the cable and radio combiner shall have a bit error rate of less than 1 in 10^6 (see 4.11.7 and Appendix B).

3.12.5. Orderwire Signal-to-noise (S/N) ratio. The Recovered Orderwire and the Emergency Ordewire signal-to-noise ratio shall be 29dB minimum F1A, and 32 dB minimum F1A, respectively, when measured at a carrier-to-noise ratio of 13 dB using a test receiver (see 4.11.6).

3.12.6. Orderwire Levels. The orderwire levels shall be checked and tested in accordance with procedures and limits specified in paragraph 4.11.8. Level measurements shall be made for Local Handset, Remote Phone, Thru Orderwire, and Transmitter Orderwire outputs for inputs of Recovered Oerwire, Local Handset, Receiver In, Remote Phone In, and Thru Orderwire (Receiver) (see 4.11.8).

3.12.7. Power Output with supply line variations. With the AC power source adjusted between 110 VAC, 47 Hz and 120 VAC, 63 HZ, the requirement of (3.12.2) for transmitter power output shall be met (see 4.11.9).

3.12.8. Frequency Accuracy. When tested in accordance with the requirements of paragraph 4.11.10, the carrier frequency accuracy shall be within ± 40 KHz of the desired frequency.

3.12.9. DDM operation with supply line variations. The AC power source shall be changed to 110 VAC, 47 Hz and 120 VAC, 63 Hz. The requirement of 3.12.4 for Digital Data Modem operaton shall be maintained (see 4.11.11).

3.12.10. Orderwire S/N ratio with supply line variations. The AC power source shall be changed to 110 VAC, 47 Hz and 120 VAC, 63 Hz. The requirement of 3.12.5 for Orderwire S/N ratio shall be maintained (see 4.11.12).

3.12.11. Orderwire peak limiting (see 4.11.13). The Orderwire shall compress the input of the local and remote handset orderwire signals such that limiting will start at 0 dBm, the level required to produce rated orderwire deivation of the transmitter, with the following characteristics:

INPUT	COMPRESSION
10 dB Below Initial Limiting	None
At Initial Limiting (0 dB)	0.4 dB Max.
15 dB Above Initial Limiting	7 dB Minimum

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3.12.12. Orderwire distortion. With a 0 dBm local handset input level, the distortion in any set of orderwire output terminals shall be less than 5% between 300 and 1800 Hz (see 4.11.14).

3.12.13. Transmitter video input impedance. The transmitter video input impedance shall be (50) ohms ± 10% (see 4.11.16).

3.12.14. Alarm systems. The following alarm systems shall operate properly when tested in accordance with 4.11.3.

3.12.14.1. Equipment status alarm. Shall provide local visible, and local and remote audible alarm indications for transmitter fault conditions.

3.12.14.2. Traffic status alarm. Shall provide Green or Red indications of PCM traffic conditions at input, intermediary, and output points in the transmitter.

3.12.15. Transmitter front panel meter functions. The meter functions shall be in the indicated color band with the Refl RF Power meter indicating less than 30. The "Off-Transit" position shall indicate zero (see 4.11.3).

3.12.16. Orderwire Frequency Response. The Orderwire Frequency Response for the Recovered and Emergency Orderwire levels shall be measured with inputs between 300 Hz to 1800 Hz and meet the limits specified in paragraph 4.11.15.

3.13. Receiver, Radio R-1467( )/GRC-144. The radio receiver (cabinet level) shall meet the following electrical requirements (see Appendix B).

3.13.1. Frequency Accuracy. The maximum error of the local oscillator frequency measured at the output of Local Oscillator Filter (FL-6) shall be ± 15 kHz (see 4.12.3).

3.13.2. Frequency Accuracy with supply line variations. When tested in accordance with the requirements of paragraph 4.12.4, the requirements in 3.13.1 shall be met.

3.13.3. Receiver noise figure. The receiver noise figure throughout the tunable frequency range shall be 5.5 dB plus 0.3 dB, minus any (see 4.12.5).

3.13.4. Carrier level threshold (48 CHNL). The receiver threshold shall be reached by signals no greater than -88.5 dBm (see 4.12.7).

3.13.5. Carrier level threshold with supply line variations. The AC power source shall be changed to 110 VAC, 47 Hz and 120 VAC. 63 Hz. The requirement of 3.13.4 for carrier level threshold shall be met (see 4.12.8).

3.13.6. Output capability and stability. The baseband output capability of the receiver operating into a 50 ohm termination shall be 1.0 volts peak to peak minimum. The output level shall not vary by more than ± 1 dB with input levels between -88.5 dBm to -38.5 dBm at room ambient and ± 2 dBm during or after any environmental test (see 4.12.10).

3.13.7. Receiver bandwidth. The 3 dB receiver IF bandwidth shall be 5.0 MHz ± 500 kHz for 48/96 channel PCM signals (see 4.12.6).

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3.13.8. Alarm systems. The following alarm systems shall operate properly when tested in accordance with 4.12.9.

3.13.8.1. Equipment status alarm. Shall provide a local visible, audible, and remote audible alarm indications for receiver fault conditions.

3.13.8.2. Traffic status alarm. Shall provide an alarm system covering the monitoring and display as Green or Red indications of PCM conditions at input, intermediary, and output points in the receiver.

3.13.9. Meter functions. The meter functions shall be in the indicated color band or as otherwise specified. The "Off-Transit" position shall indicate zero (see 4.12.9).

3.14. Transmitter/Receiver characteristics.

3.14.1. Transmitter and receiver system PCM phase response. The system phase response shall have an absolute phase shift of 37 ± 10 degrees at 300 Hz. The amplitude response at 300 Hz shall be 1 dB maximum below the 1 kHz level (see 4.13.3).

3.14.2. Video Frequency response (see 4.13.4). The frequency response using a test transmitter operating at rated deviation shall be:

FREQUENCY (kHz)	
48 Channel	Relative Response (dB)
50	0 Reference
460	1.0 dB Max.
650	1.5 ± 0.5 dB
920	3.0 ± 0.6 dB
1300	6.0 ± 0.9 dB
1840	12.0 ± 1.2 dB
2600	23.0 ± 2.4 dB
3300	42.0 dB Max

3.15. Indicator, Antenna Alignment ID-1708( )/GRC. The antenna alignment indicator operates in conjunction with Receiver, Radio R-1467( )/GRC-144 and is used to assist in orientation of the antenna in azimuth and elevation angles. The meter when interconnected through Receiver, Radio R-1467( )/GRC-144 indicates signal strength of the receiver when the 70 MHz IF amplifier AGC SEL switch is in DIV OFF position. ID-1708( )/GRC also contains a phone circuit for connection through twisted pair field telephone wire to a remote Telephone Set TA-312( )/PT which provides audio (orderwire circuit) communications during antenna set-up and alignment (see 4.14).

3.15.1 Operational check. The meter shall indicate input current ± 20% when the meter ON-OFF switch is in the ON position and shall indicate zero when switch is in OFF position. Meter reading varies with input level (see 4.14.2).

3.15.2. Continuity check. The continuity of the telephone pair within Cable W1 shall be checked between the input connector and the ends of the telephone connector leads (see 4.14.3).

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3.16. Electromagnetic interference characteristics. The equipment shall meet the electromagnetic interference requirements specified in MIL-STD-461 for Class I, except as specifically indicated below. Tests and test method shall be as specified in MIL-STD-462 (see 4.4). The applicable procedures are per Table I.

TABLE I. Electromagnetic interference requirements.

TEST METHOD	DESCRIPTION/ FREQUENCY RANGE(Hz)	OPERATIONAL MODE	
		XMTR	RCVR
CE01	Power leads (30 - 20 K)	X	X
CE02	Control and Signal Leads (30 - 20 K)	X	X
CE03	Power leads (20 K - 50 M)	X	X
CE04	Control and Signal Leads (20 K - 50 M)	X	X
CE06	Antenna Terminal (10 K - 12.4 G)	X	X
CS01	Power Leads (30 - 50 K)	X	X
CS02	Power Leads (50 K - 50 M)	X	X
CS03	Intermodulation (30 - 10 G)		X
CS04	Rejection of Undesired Signals (30 - 10 G)		X
CS05	Cross Modulation (30 - 10 G)		X
CS06	Spike, Power Leads	X	X

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TABLE I. Electromagnetic interference requirements - Continued.

TEST METHOD	DESCRIPTION/ FREQUENCY RANGE(Hz)	OPERATIONAL MODE	
		XMTR	RCVR
RE02	Electric Field NB, 14 K - 10 G BB, 14 K - 1 G	X	X
RE03	Spurious and Harmonics, Radiated Techniques (10 K-15 G)	X	
RS03	Electric Field (14 K - 10 G)*		X

"X" Denotes test is applicable and shall be performed as described in MIL-STD-462.

"*" The equipment shall not malfuntion when subjected to an electric field of 1 volt-per-meter IAW paragraph 6.19 of MIL-STD-461. This requirement need not to be met within Fo ± 25 MHz (where Fo is the receiver tuned frequency).

Susceptibility criteria shall be defined as follows:

a. Transmit mode - Any degradation of performance beyond the limits stated in 3.12 and 4.11 shall be considered a failure.

b. Receive mode - Any degradation of performance beyond the limits stated in 3.13 and 4.12 shall be considered a failure.

3.17. Operational test (Radio Set) The operational test specified in 4.15 shall be performed to determine that each radio set is operating satisfactory prior to packaging and at such other times as specified eleswhere herein (see Appendix B). For EMI testing, the procedure specified in 4.15 shall be used.

3.18. Technical literature, tools and running spare parts. Technical literature, tools and running spare parts shall be furnished if specified in the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order (see 40.1.6, Appendix B).

3.19. Workmanship. The equipment shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs of MIL-P-11268 (see 4.10).

General requirements for plastic material and plastic parts

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Wiring and cabling, including:

Slack Protection Clearance Splicing Connections, general Grounding, general Shielding on wire and cable

Soldering and brazing General requirements for securing of parts Cleaning of Units Process for welding Cleaning after welding Controls Self-tapping screws Cleaning prior to welding

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1. Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspection. Inspections shall be classified as follows:

a. First article inspection (does not include packaging (See 4.3.).

b. Inspection covered by subsidiary documents. (See 4.4.).

c. Quality conformance inspection of equipment before packaging (see 4.5).

4.3 First article inspection. This inspection shall consist of all the inspections specified in Table II and shall be performed in the order shown on the number of sample units specified therein, unless otherwise specified in the contract or order (see 40.1.7, Appendix B).

NOTES

a. The priority order of the inspections 1 through 6 below and the scope of each of these inspections, shall be performed on all First Article Sample units shown in the table before subjecting these units to any other inspection requirements specified in the table.

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b. The contractor shall expedite the implementation of inspections covered by 7, 8, 9 below to satisfy contract delivery requirements.

c. Reliability qualification (Demonstration) testing for the Transmitter T-1054( )/GRC-144/Receiver R-1467( )/GRC-144 shall be conducted on First Article Sample units (4 units shall be the sample size required for qualification testing) in accordance with the Reliability Test Requirement shown in Appendix A of this specification.

Table II. Order of first article inspection.

INSPECTION	REQ. PARA	INSP. PARA	UNIT 1	UNIT 2	UNIT 3	UNIT 4	UNIT 5 THRU 8
1. Inspection covered by subsidiary doc. except EMI	4.4	4.4	Inspections to be performed on all units				
2. Bounce Pre-conditioning	3.9.1	4.7.1	Inspections to be performed on all units				
3. Electrical Preconditioning	3.9.2	4.7.2	Inspections to be performed on all units				
4. Group A Inspection	(See Table III)		Inspections to be performed on all units				
5.Group B Inspection	(See Table IV)		Inspections to be performed on all units				
6. Group C Inspection (Electrical)	(See Table V)		All electrical inspections listed in sub Group I of the table to be performed on all units				
7. EMI	3.16	4.4	1				
8. Service Conditions							
Non-damaging							
Low Temp.	3.8.1.1	4.8.1.1		1			
High Temp.	3.8.1.2	4.8.1.2		2			
Altitude	3.8.3	4.8.3			1		
Potential Damaging							
Humidity	3.8.2	4.8.2		3			
Bounce, Vehicular	3.8.4.1	4.8.4.2			2"		
Fungus	3.7.2	4.8.5				1	
9. Reliability Test	3.11	4.5.5					1

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4.4. Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection before packaging.

ITEM	WHERE REQUIRED	
Parts, Materials and Processes	3.3	
Finish	3.4	
Marking	3.5	
Conformal coating	3.6	
Electromagnetic Compatibility	3.16	Workmanship 3.19

4.5. Quality conformance inspection of equipment before packaging. Each unit of each lot of equipment shall be preconditioned (see 3.9) prior to inspection. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government, at its discretion, may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements (see 6.7). Test equipment for Government verification inspection shall be made available by the contractor.

4.5.1. Group A inspection. Units presented for acceptance shall be inspected for conformance with the criteria specified in Table III. Lots shall be formed and subjected to sampling in accordance with Table III-1. Group A inspection shall be performed in any order which is satisfactory to the Government, except that the operational inspection (4.15) shall be last (see 6.5), and immediately prior to packaging (see 40.1.8, Appendix B).

Table III. Group A inspection.

VISUAL AND MECHANICAL	REQUIREMENT PARAGRAPH	INSPECTION PARAGRAPH
Transmitter, Radio T-1054( )/GRC-144		
Cabinet, Electrical Equip. Including Meter Panel & Radio Patch Panel	3.19	4.10
Plate Assy (Transmitter)	3.19	4.10
Mixer, Crystal	3.19	4.10
Attenuator Variable (Turret)	3.19	4.10
Amplifier, AF-RF	3.19	4.10
Oscillator, RF, 100 MHz	3.19	4.10
Modulator, Radio Transmitter	3.19	4.10

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Table III. Group A inspection - Continued.

VISUAL AND MECHANICAL	REQUIREMENT PARAGRAPH	INSPECTION PARAGRAPH
Frequency Multiplier, Group, Transmitter	3.19	4.10
Frequency Multiplier, 2nd Stage, Transmitter	3.19	4.10
Frequency Multiplier, 3rd Stage, Transmitter	3.19	4.10
Circulator, Coaxial (2270 to 2430 MHz)	3.19	4.10
Control, Electronic Frequency	3.19	4.10
Mixer Stage Frequency, Transmitter	3.19	4.10
Mixer, Crystal	3.19	4.10
Circulator, Coaxial (4400 to 5000 MHz)	3.19	4.10
Alarm-Monitor	3.19	4.10
Power Supply Chassis Assembly (Transmitter)	3.19	4.10
Regulator, Voltage 15/28 VDC	3.19	4.10
Regulator, Voltage 12 VDC	3.19	4.10
Regulator, Voltage 5/6 VDC	3.19	4.10
Orderwire Chassis Assy	3.19	4.10
Daughterboard No. 1	3.19	4.10
Daughterboard No. 2	3.19	4.10
Daughterboard No. 3	3.19	4.10
Daughterboard No. 4	3.19	4.10
Daughterboard No. 5	3.19	4.10
Filter, Low Pass (FL1 & FL2)	3.19	4.10
Synthesizer, Electronic Frequency Assembly	3.19	4.10
Oscillator RF, Voltage Controlled	3.19	4.10
Converter, Frequency Electronic	3.19	4.10
Frequency Divider, Variable No. 1	3.19	4.10
Frequency Divider, Variable No. 2	3.19	4.10

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Table III. Group A inspection - Continued.

VISUAL AND MECHANICAL	REQUIREMENT PARAGRAPH	INSPECTION PARAGRAPH
Frequency Divider, Fixed	3.19	4.10
Detector, AF, Phase Error	3.19	4.10
Power Supply	3.19	4.10
Oscillator, RF Standard	3.19	4.10
Modem, Digital Data Chassis Assembly	3.19	4.10
Regenerator, Digital, Cable	3.19	4.10
Control-Comparator, Cable	3.19	4.10
Alarm-Monitor	3.19	4.10
Processor, Digital, Radio	3.19	4.10
Oscillator, Voltage Controlled (4.608 MHz)	3.19	4.10
Regenerator, Digital, Radio	3.19	4.10
Control-Comparator, Radio	3.19	4.10
Processor, Digital Cable	3.19	4.10
Amplifier, AF	3.19	4.10
Amplifier-Detector,Radio	3.19	4.10
Crystal Unit (4.608 MHz)	3.19	4.10
Amplifier-Frequency Multiplier,Transmitter	3.19	4.10
Attenuator Assembly	3.19	4.10
Filter, Low Pass, 4.5 MHz	3.19	4.10
Filter, Band Pass, RF	3.19	4.10
Coupler, Directional, 5 Port	3.19	4.10
Filter, Low Pass, RF	3.19	4.10
Circulator, Coaxial, 1135-1215 MHz, Transmitter	3.19	4.10
Receiver, Radio R-1467 ( )/GRC-144		
Cabinet, Electrical Equip. Including Meter Panel	3.19	4.10
Plate Assembly Receiver	3.19	4.10
Amplifier-Mixer	3.19	4.10

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Table III. Group A inspection - Continued.

VISUAL AND MECHANICAL	REQUIREMENT PARAGRAPH	INSPECTION PARAGRAPH
Mixer, Crystal	3.19	4.10
Bandpass Filter (4.5-4.8 GHz)	3.19	4.10
Mixer Stage, Frequency Receiver	3.19	4.10
Mixer, Crystal	3.19	4.10
Filter, Bandpass, 220 MHz	3.19	4.10
Circulator, Coaxial 4330 to 5070 MHz	3.19	4.10
Synthesizer, Electronic Frequency Assembly	Modular Break-out - Same as for Transmitter	
Power Supply Chassis Assembly (Receiver)	3.19	4.10
Regulator, Voltage 15/ 28 VDC	3.19	4.10
Regulator, Voltage 12 VDC	3.19	4.10
Frequency Multiplier-Oscillator, 220 MHz	3.19	4.10
Oscillator, RF, 110 MHz	3.19	4.10
Amplifier-Frequency Multiplier	3.19	4.10
Filter, Bandpass, 70 MHz	3.19	4.10
Amplifier, AF-RF	3.19	4.10
Amplifier-Frequency Multiplier, Receiver (Octupler)	3.19	4.10
Circulator (4.5-4.9 GHz)	3.19	4.10
Amplifier, IF, 70 MHz	3.19	4.10
Demodulator	3.19	4.10
Alarm-Monitor	3.19	4.10
Filter, Low Pass, 1.0 MHz	3.19	4.10
Filter, Bandpass, Post Selector	3.19	4.10
Filter, Bandpass Local Oscillator	3.19	4.10
Filter, Bandpass, Pre-selector	3.19	4.10
Amplifier, Tunnel Diode	3.19	4.10

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Table III. Group A inspection - Continued.

AND MECHANICAL	REQUIREMENT PARAGRAPH	INSPECTION PARAGRAPH
Indicator, Antenna Alignment ID-1708( )/GRC		
Drawer No. 1 (Receiver Spares)	3.19	4.10
Drawer No. 2 (Transmitter Spares)	3.18	4.10
Drawer No. 3 (Digital Data Modem and Orderwire Spares)	3.18	4.10
Drawer No. 4 (Common Equipment Spares)	3.18	4.10
Transmitter, Radio T-1054( )/GRC-144		
Radio Test Set Operation	3.12.	4.11.2.6
Power Output	3.12.2	4.11.5
Transmitter Deviation	3.12.3	4.11.4
Digital Data Modem (PCM Threshold Level Operation)	3.12.4	4.11.7
Orderwire Signal-to-Noise Ratio & percentage of PCM Signal	3.12.5	4.11.6
Alarm operation and metering	3.12.14 3.12.15	4.11.3
Receiver, Radio R-1467 ( )/GRC-144		
Noise Figure	3.13.3	4.12.5
Receiver threshold	3.13.4	4.12.7
Alarm operation and metering	3.13.8 3.13.9	4.12.9
Indicator, Antenna Alignment ID-1708( )/GRC		
Operational test	3.15.1	4.14.2
Continuity test	3.15.2	4.14.3
Operational test (Radio System)	3.17	4.15

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TABLE III-1. Sampling plan for group A inspection.

Lot Size	Sample Size
2 to 8	*
9 to 15	13
16 to 25	13
26 to 50	13
51 to 90	13
91 to 150	13
151 to 280	20
281 to 500	29
501 to 1200	34
1201 to 3200	42
3201 to 10,000	50
10,001 to 35,000	60
35,001 to 150,000	74
150,001 to 500,000	90
500,001 and over	102

NOTES:
1. * - Indicates entire lot must be inspected.
2. Acceptance number in all cases is zero.

4.5.2. Group B Inspection. This inspection shall consist of the inspections listed in Table IV and shall be performed on lots that have passed Group A inspection. Each lot shall be subjected to sampling inspection in accordance with Table IV-1.

4.5.2.1. Order of inspection within Group B. Group B inspection shall be performed in any order which is satisfactory to the Government, except the mechanical interchangeability inspection, 4.9.1, shall be performed at the module/subassembly acceptance level.

TABLE IV. Group B inspection.

INSPECTION	REQUIREMENT PARAGRAPH	INSPECTION PARAGRAPH
Electrical		
Transmitter T-1054( )/GRC-144		
Orderwire Levels	3.12.6	4.11.8
Receiver R-1467( )/GRC-144		
I.F. Bandwidth	3.13.7	4.12.6
Interchangeability	3.10	4.9

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TABLE IV-1. Sampling plan for group B inspection.

Lot Size	Sample Size
2 to 8	3
9 to 15	3
16 to 25	3
26 to 50	5
51 to 90	6
91 to 150	7
151 to 280	9
281 to 500	11
501 to 1200	15
1201 to 3200	18
3201 to 10,000	22
10,001 and over	29

NOTE: Acceptance number in all cases is zero.

4.5.3. Group C Inspection. This inspection shall consist of the tests specified in Table V and shall be performed on sample units that have been subjected to and met Group A and B inspection. Sample units shall be selected in accordance with 4.5.3.1.

4.5.3.1. Sampling for inspection. Units shall be selected at random for each Group C inspection subgroup in accordance with the following.

4.5.3.2. Subgroup I. For this subgroup, two (2) units from the first month's production shall be selected for the first Group C inspection. For subsequent Group C inspections, two (2) from each successive 100, or two (2) units per month if less than 100 are produced monthly, shall be selected. For this subgroup, the maximum number of units to be selected and tested in a single month shall not exceed eight (8) units.

4.5.3.1.2. Subgroup II. For this subgroup, two (2) units from the first month's production shall be selected for the first Group C inspection. For subsequent Group C inspections, two (2) units shall be selected and tested during each two (2) month period throughout production (see 40.1.9, Appendix B).

4.5.3.1.3. Subgroup III. For this subgroup, excluding First Article testing, one (1) unit shall be selected at random, and tested at midpoint during the production schedule (see Appendix B).

NOTE: One Unit equals one (1) of each of the following items:

1 - Transmitter, Radio T-1054( )/GRC-144

1 - Receiver, Radio R-1467( )/GRC-144

1 - Indicator, Antenna Alignment ID-1708( )/GRC

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TABLE V. Group C inspection.

INSPECTION	REQUIRED PARAGRAPH	INSPECTION PARAGRAPH
Subgroup I (see 4.5.3.1)		
Transmitter T-1054( )/GRC		
Power Output w/Supply Line Variations	3.12.7	4.11.9
Frequency Accuracy	3.12.8	4.11.10
Digital Data Modem (PCM Threshold Level) Operation w/Supply Line Variations	3.12.9	4.11.11
Orderwire Signal-to-Noise Ratio w/Supply Line Variations	3.12.10	4.11.12
Orderwire Peak Limiter	3.12.11	4.11.13
Orderwire Distortion	3.12.12	4.11.14
Orderwire Frequency Response	3.12.16	4.11.15
Xmtr. Video Input Impedance	3.12.13	4.11.16
Receiver R-1467( )/GRC-144		
Frequency Accuracy	3.13.1	4.12.3
Frequency Accuracy w/ Supply Line Variations	3.13.2	4.12.4
Threshold Characteristics w/Supply Line Variations	3.13.5	4.12.8
Receiver Output Capability and Stability	3.13.6	4.12.10
Transmitter/Receiver Characteristics		
System PCM Phase Response	3.14.1	4.13.3
Video Frequency Response	3.14.2	4.13.4
Subgroup II (see 4.5.3.1.2)		
Temperature		
Low	3.8.1.1	4.8.1.1
High	3.8.1.2	4.8.1.2
Humidity (moisture-resistance)	3.8.2	4.8.2
Vibration		
Bounce-Vehicular	3.8.4.1	4.8.4.1
Electromagnetic Interference	3.16	4.4

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TABLE V. Group C inspection - Continued.

INSPECTION	REQUIRED PARAGRAPH	INSPECTION PARAGRAPH
Receive Mode: CE06, RE02, and RS03 Transmit Mode: RE02 only	3.16	4.4
Subgroup III (see 4.5.3.1.3)		
Altitude (elevation)	3.8.3	4.8.3
Fungus	3.7.2	4.8.5
Electromagnetic Interference	3.16	4.4

4.5.3.2. Noncompliance. The contractor shall immediately report, in writing, each Group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (i) the process and materials, as applicable, and, (ii) all units of product which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to Group C reinspection (all inspections, or the inspections which the sample failed, at the option of the responsible technical Government activity) and Group A and B inspection may be reinstituted; however, final acceptance and shipment will be withheld until the Group C inspection results have shown that the corrective action was effective and approved by the technical activity (see 6.4).

4.5.4. Reinspection of conforming Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on the contract or order provided all visible and known damage is repaired and the sample units are resubjected to and pass Group A inspection.

4.5.5. Group D inspection. This inspection shall consist of the tests specified in Table VI and shall be performed on units from lots which have been subjected to and met Group A and Group B inspection and the following electrical inspections listed under Subgroup I of Group C inspection (see Table V):

For Receiver R-1467( )/GRC-144: Frequency Accuracy Receiver Output Capability and Stability

For Transmitter/Receiver Characteristics: System PCM Phase Response Video Frequency Response

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TABLE VI. Group D inspection.

INSPECTION	REQUIRED PARAGRAPH	INSPECTION PARAGRAPH
Reliability Test	3.11	Appendix A

4.5.5.1. Sampling for reliability testing of equipment. Four (4) samples shall be selected at random in accordance with the procedures referenced in Appendix A.

4.5.5.2. Noncompliance. The contractor shall immediately investigate the cause of all failures in any lot in which a reject decision is reached and further report in writing within 48 hours to the procuring activity the results of the investigation and the proposed corrective action applicable to all production units which were manufactured under the same conditions which the Government considers subject to the same failures. The procuring activity shall have the final determination as to the nature of the failures; i.e., whether the failures are random in nature (no corrective action required) or subject to pattern failures. Acceptance by the Government of all in-house items shall be withheld until the contractor's proposed corrective actions have been accepted by the procuring activity and implemented as directed by the Government on all contract items in-house. A retest shall be imposed if it is deemed necessary by the Government to demonstrate the effectiveness of the corrective actions wherein the reliability of the equipment shall be demonstrated in accordance with the Reliability Test Requirement in Appendix A of this specification to determine compliance with 3.11.

4.5.6. Reinspection of conforming Group D sample units. Unless otherwise specified, sample units which have been subjected to and passed Group D inspection may be accepted on the contract or order provided all visible and known damage is repaired and the sample units are resubjected to and pass Group A inspection.

4.6. Standard test conditions. Unless otherwise specified, all measurements and tests shall be conducted as follows:

4.6.1. Standard ambient conditions. The standard ambient conditions are:

Temperature	23° ± 10° C (73° ± 18° F) Relative
Humidity	50 percent ± 30 percent
Atmospheric Pressure	725 + 50 -115 mm. of mercury (28.5 + 2.0 -4.5 in. of mercury)

4.6.2. Standard line voltage and frequency. Unless otherwise specified, all measurements and tests shall be conducted with a line voltage of 115 ± 5 VAC, single phase, 60 Hz.

4.7. Preconditioning.

4.7.1. Bounce Preconditioning. The unit, with shock mounts (if any) removed or blocked, shall be placed in its normal operating position on the table of the Package Tester, Type 1000-SC, as made by the L.A.B. Corporation,

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Skaneatales, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is possible to insert a 1/32 - inch-thick strip of material under one corner or edge of the unit to distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the unit shall not be repaired, aligned, cleaned, or otherwise changed prior to subjection to inspections, (see 3.9.1).

4.7.2. Electrical preconditioning (burn-in). The equipment (Receiver, Transmitter, and Spare Modules) shall be operated for forty-eight (48) hours in accordance with the following temperature and vibration conditions (see 3.9.2 and Appendix B).

4.7.2.1. Temperature Test Condition: 40° C ± 5° C.

4.7.2.2. Vibration test condition: 2.2G ± 10% peak acceleration value at any non-resonant frequency between 20 and 40 Hz measured at the mounting points on the equipment. The duration of vibration shall be a minimum of ten (10) minutes during each hour of equipment operating time.

4.7.2.3. Test condition. The testing shall be conducted on a continuous twenty-four (24) hour basis.

4.7.2.4. Test measurements. The following tests and recording of data shall be performed on all units as applicable: (See Appendix A).

a.	Receiver Meter Readings	20.7.5.1
b.	Transmitter Meter Readings	20.7.5.2
c.	Orderwire Performance Monitor Check	20.7.5.3
d.	Transmitter Alarm Status	20.7.5.4
e.	Receiver Alarm Status	20.7.5.5
f.	Receiver Indicator Lamp Status	20.7.5.6
g.	Transmitter Indicator Lamp Status	20.7.5.7
h.	Transmitter Power Output	20.7.5.8
i.	Receiver Video Output Level	20.7.5.9
j.	Orderwire Operation	20.7.5.10
k.	Antenna Alignment Indicator Operation	20.7.5.11

4.7.2.5. Monitoring. Monitoring of test measurements for Transmitter Power Output, Receiver Traffic, and Transmitter and Receiver central alarm conditions shall be on a continuous basis throughout the entire preconditioning phase so that any relevant malfunction of the equipment will be detected at the time of occurrence. All other test measurements as delineated under 4.7.2.4 shall be conducted twice during each twenty-four (24) hour period of test time (once at the start and once at the end of 24 hours) of the equipment in order to detect relevant failures. 4.7.2.6. Failures. Failures occurring during the forty-eight (48) hour electrical pre-conditioning period shall be repaired and the test continued. If a failure occurs during the final ten (10) hours of test, the equipment shall be repaired and the test shall be continued until a ten (10) hour failure-free period has been achieved. Equipment with component failures encountered during Group A, B, or C testing (following electrical pre-conditioning) which require component replacement need not be subjected to re-cycling through electrical pre-conditioning provided the repair is made at the component level. If higher levels of assemblies are used to correct or repair component

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failures, the equipment shall be reycled through electrical pre-conditioning or the replacement assembly shall have previously been subjected to electrical pre-conditioning. Data and documentation of pre-conditioned replacement modules and assemblies shall be maintained by the contractor.

4.7.2.6.1. Failure definitions. Failures include the inability to meet the test parameter and limits for the measurements delineated in 4.7.2.4, and alarm circuit operation. Operation of any alarm circuit indicating reduction in traffic carrying capability is a failure. A reduction in transmitter traffic carrying capability is indicated by operation of all transmitter alarm indicators except "RADIO TEST SET," "WAVE GUIDE SW", "RING", and "REFL RF POWER". A reduction in receiver traffic carrying capability is indicated by operation of all receiver alarm indicators except "SPARE".

4.7.2.7. Spare Modules. Spare modules, either individual or multiple modules installed as part of a sub-assembly shall receive forty-eight (48) hours of electrical pre-conditioning on a 100% basis and shall be tested to the same criteria as established for the "Active" modules which are installed in the equipment (cabinet level) for test. Acceptance and failure criteria is the same for all modules (see 40.1.13, Appendix B).

4.8. Service condition tests. (See 3.8).

4.8.1. Temperature test.

4.8.1.1. Low temperature. The equipment shall be tested in accordance with Method 502.1, Procedure I of MIL-STD-810. The storage temperature shall be -65° F, and the operating temperature shall be -25° F. The storage time shall be maintained for 24 hours and Step 3 of Procedure shall be omitted. Prior to testing, the equipment shall meet full specification performance for the measurements specified in 3.8.1.1.

4.8.1.2. High temperature. The equipment shall be tested in accordance with Method 501.1, Procedure I of MIL-STD-810. The storage temperature shall be +160° F, and the operating temperature shall be +125° F. Prior to testing, the equipment shall meet full specification performance for the measurements specified in 3.8.1.2.

4.8.1.3. Stabilization (Warm-Up) Time. Prior to making any measurements where data is to be recorded, a thirty (30) minute maximum warm-up periods allowed upon initial turn-on of the equipment under test to allow for frequency stabilization of the 8 MHz output of the Standard Oscillator within the Frequency Synthesizer.

4.8.2. Humidity (moisture-resistance) test. The equipment shall be tested in accordance with Method 507.1, Procedure II of MIL-STD-810. Step 2 and Step 3 may be omitted provided that humidity testing commences immediately following the temperature test and the test hardware has been located (after temperature testing) in an environment (factory floor, test chamber, etc.) with temperature and humidity conditions not exceeding those stipulated in Step 3. Prior to and at the completion of humidity testing, the equipment shall meet full specification performance for the measurements specified in 3.8.2. Twice, during each humidity cycle, the measurements specified in 3.8.2. shall be made as follows: once during the 24th to 28th hour period and once during the 44th to 48th hour period. The equipment shall remain in the humidity chamber for

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measurements and shall not be powered except while measurements are being taken (see 3.8.2).

4.8.2.1. Stabilization (Warm-Up) Time. Same as 4.8.1.3. herein.

4.8.2.2. Equipment failures during/after humidity. If any equipment fails to meet the specified performance during cycling, it shall be realigned or readjusted once and the test continued. No repair or replacement of parts shall be permitted. If the equipment still fails to meet the specified performance during cycling, or fails subsequently, it does not pass the test. In addition, if the equipment fails to meet specified requirements after final conditioning and adjustment, it does not pass the test. At the conclusion of these tests, the equipment shall be examined and there shall be no visible evidence of corrosion, rust or deterioration of finishes or changes in tolerance limits of any internal or external parts which could in any manner prevent the test item from meeting operational service.

4.8.3. Altitude (Elevation). The equipment shall be tested as follows:

Step 1. Prepare the test item in accordance with General Requirements, 3.2 of MIL-STD-810, and maintain standard ambient temperature during the entire test.

Step 2. Reduce the changer internal pressure to 87.5 mm of Hg (3.44 in. of Hg or 50,000 ft. above sea level). The rate of pressure change may be the maximum attainable by the chamber. Maintain the chamber pressure for a period of not less than 1 hour.

Step 3. Increase the chamber pressure to 523 mm of Hg (20.6 in. of Hg or 10,000 ft above sea level) and then operate the test item and compare the results with the data obtained in Step 1.

Step 4. With the test item not operating, return the chamber to standard ambient conditions.

Step 5. Operate and inspect the test item and obtain results in accordance with General Requirements, 3.2 of MIL-STD-810. In addition, the equipment shall be checked visually to determine if any damaging effects of low pressure (cracks in transformer cases, other sealed items, etc.) have occurred during the test.

4.8.3.1. Stabilization (Warm-Up) Time. Same as 4.8.1.3. herein.

4.8.4. Vibration.

4.8.4.1. Bounce, Vehicular. The equipment shall be tested in accordance with Method 514.2, Procedure IX, Part 2, of MIL-STD-810. A package tester of suitable capacity for testing military equipment, as made by L.A.B. Corporation, Skaneateles, New York, or equal, shall be used as the test apparatus (see 3.8.4.2).

4.8.5. Fungus (see Appendix B). The equipment (including accessories) shall be tested in accordance with the following:

4.8.5.1. Preparation of Equipment. The equipment, unless heremtically

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sealed, shall be opened and all surfaces, parts and materials, on and under the chassis and the interior of the cabinets, shall be sprayed with the spore suspension. Demountable chassis, modules and electronic circuit board assemblies shall be removed and opened as applicable for thorough coverage of the fungi spore. The antenna alignment indicator case shall be open and the chassis removed for the test. The equipment shall remain in an open condition during the test. Operation is not required.

4.8.5.2. Apparatus. The apparatus required to conduct this test consists of chambers or cabinets together with auxiliary instrumentation capable of maintaining the specified condition of temperature and humidity. Provisions shall be made to prevent condensation from dripping on the test item. There shall be free circulation of air around the test item and the surface area of fixtures supporting the test item shall be kept to a minimum.

4.8 5.3. Procedure.

4.8.5.3.1. Preparation of mineral-salts solution. The solution shall contain the following:

Potassium dihydrogen orthophosphate (KH_2PO_4)	0.7 g
Potassium monohydrogen orthophosphate (K_2HPO_4)	0.7 g
Magnesium sulfate ($MgSO_4 \cdot 7H_2O$)	0.7 g
Ammonium nitrate (NH_4NO_3)	1.0 g
Sodium chloride (NaCl)	0.005 g
Ferrous sulfate ($FeSO_4 \cdot 7H_2O$)	0.002 g
Zinc sulfate ($ZnSO_4 \cdot 7H_2O$)	0.002 g
Manganous sulfate ($MnSO_4 \cdot 7H_2O$)	0.001 g
Distilled water	1000 ml

Sterilize the mineral salts solution by autoclaving at 121° C (250° F) for 20 minutes. Adjust the pH of the solution by the addition of 0.01 normal solution of NaOH so that after sterilization the pH is between 6.0 and 6.5. Prepare sufficient salts solution for the required tests.

4.8.5.3.1.1. Purity of reagents. Reagent grade chemicals shall be used in all tests. Unless otherwise specified, it is intended that all reagents shall conform to the specifications of the Committee on Analytical Reagents of the American Chemical Society, where such specifications are available.

4.8.5.3.1.2. Purity of water. Unless otherwise specified, references to water shall be understood to mean distilled water or water of equal purity.

4.8.5.3.2. Preparation of mixed spore suspension. The following test fungi

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shall be used:

Fungi	ATCC No. 1	NLABS No. 2
Aspergillus niger	9642	386
Aspergillus flavus	9643	380
Aspergillus versicolor	11730	432
Penicillium funiculosum	9644	391
Chaetomium globosum	6205	459

1. American Type Culture Collection, 12301 Parklawn Drive, Rockville, Maryland 20852

2. Pioneering Research Division, U.S. Army Natick Laboratories, Natick, Massachusetts 01760.

Maintain cultures of these fungi separately on an appropriate medium such as potato dextrose agar. However, the culture of Chaetomium globosum shall be cultured on strips of filter paper on the surface of mineral - salts agar. (Mineral salts agar is identical to mineral salts solution described in 3.1.1., but contains in addition 15.0 g of agar per liter.) The stock cultures may be kept for not more than 4 months at 6 ± 4° C (43° F). Use subcultures incubated at 29° C (84° F) for 7 to 20 days in preparing the spore suspension.

Prepare a spore suspension of each of the five fungi by pouring into one subculture of each fungus a sterile 10-ml portion of water or of a sterile solution containing 0.05 g per liter of a nontoxic wetting agent such as sodium dioctyl sulfosuccinate. Use a sterile platinum or nichrome inoculating wire to scrape gently the surface growth from the culture of the test organism. Pour the spore charge into a sterile 125-ml glass-stoppered Erlenmeyer flask containing 45 ml. of sterile water and 10 to 15 solid glass beads, 5 mm. in diameter. Shake the flask vigorously to liberate the spores from the fruiting bodies and to break the spore clumps.

Filter the shaken suspension through a thin layer of sterile glass wool in a glass funnel into a sterile flask in order to remove mycelial fragments.

Centrifuge the filtered spore suspension aseptically, and discard the supernatant liquid. Resuspend the residue in 50 ml. of sterile water and centrifuge.

Wash the spores obtained from each of the fungi in this manner three times. Dilute the final washed residue with sterile mineral-salts solution in such a

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manner that the resultant spore suspension shall contain 1,000,000 ± 200,000 spores per ml. as determined with a counting chamber.

Repeat this operation for each organism used in the test and blend equal volumes of the resultant spore suspension to obtain the final mixed spore suspension.

The spore suspension may be prepared fresh each day or may be held at 6° ± 4° C (43° F) for not more than 4 days.

4.8.5.3.3. Viability of inoculum control. With each daily group of tests place each of three pieces of sterilized filter paper, 1 in. square, on hardened mineral-salts agar in separate Petri dishes. Inoculate these with the spore suspension by spraying the suspension from a sterilized atomizer1 so that the entire surface is moistened with the spore suspension. Incubate these at 29° C (84° F) at a relative humidity not less than 85 per cent and examine them after 14 days incubation. There shall be copious growth on all three of the filter paper control specimens. Absence of such growths requires repetition of the test.

4.8.5.3.4. Control items. In addition to the viability of inoculum control, a number of known susceptible substrates shall be inoculated along with the test item to insure that proper conditions are present in the incubation chamber to promote fungus growth. The control items shall include three pieces each of preservative free vegetable tanned leather and protein-glue bonded cork.

[1]DeVilbiss No. 154 atomizer or equivalent has been found satisfactory for this purpose. 4.8.5.3.5. Inoculation of test and control items.

a. Mount the test and control items on suitable fixtures or suspended from hangers.

b. Precondition the chamber and its contents at 29° C (84° F) and 95 percent R.H. for at least 4 hours.

c. Inoculate the test and control items with the mixed fungus spore suspension (4.8.5.3.3) by spraying it on the test and control items in the form of a fine mist from a previously sterilized atomizer or nebulizer until they are thoroughly wet with the spray. Incubation is t be started immediately following the inoculation.

4.8.5.3.6. Incubation.

a. Maintain the test chamber at 29° C (84° F) and 95 per cent R. H. (minimum) during the life of the test. Keep the test chamber closed during the incubation period except during inspection or for addition of other test items.

b. After 14 days, inspect the control items. They should show an abundant growth of fungus. If the control items do not show an abundant growth, the entire test shall be repeated.

c. If the control items show satisfactory fungus growth, continue the test for a period of 28 days from the time of inoculation or as specified in

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the equipment specification.

4.8.5.4. Criteria for passing test. At the end of the incubation period, the test item shall be removed from the test chamber and inspected, and a record made of any damage or deterioration resulting from the test. There shall be no visible evidence of deterioration, corrosion or any change in tolerance limits of any internal or external parts which could in any manner prevent the test item from meeting operational service (see 3.7.2.).

4.9. Inspection for interchangeability (See Appendix B). Radio sets which have been subjected to and passed all Group A inspections and Group B inspections shall be inspected and tested for compliance with the requirements of 3.10. The specified number of sample units shall be selected from the lot, and the modules therein shall be identified with the serial number of the unit from which they were removed. Following testing, the modules shall be replaced into the units from which they were removed.

4.9.1. Inspection for mechanical (dimensional) interchangeability. The gage points identified on gage lists GL-SM-E-323439 (Gages for Receiver, Radio R-1467( )/GRC-144), GL-SM-B-323440 (Gages for Transmitter, Radio T-1054( )/GRC-144), and GL-SM-B-323356 (Gages for Synthesizer, Electrical Frequency), shall be gaged or measured for compliance with the mechanical interchangeability requirements. Modules, chassis, and sub-assemblies which fail to meet the specified gage limits shall be considered not acceptable. Radio sets which fail to meet the dimensional interchangeability requirement or which contain modules, chassis or sub-assemblies which fail to meet the specified gage limits shall be considered to have failed the interchangeability requirement.

4.9.2. Inspection for electrical interchangeability. The following modules and sub-assemblies shall be tested to determine compliance with the performance requirements specified in the test procedure drawing referenced in the production drawings of the module or sub-assembly if a Government owned test fixture does not exist. Modules and sub-assemblies which fail to meet test fixture limits or limits on the test procedure drawings, as applicable, shall be considered not acceptable. Radio sets which contain modules or sub-assemblies which fail to meet these limits shall be considered to have failed the interchangeability requirement.

TRANSMITTER

Transmitter Modules	Test Fixture
Oscillator, RF, 100 MHz, SM-F-650509	GL-SM-B-405778
Attenuator Assembly, SM-D-683635	----------
Amplifier, AF-RF*, SM-D-650500	GL-SM-B-405738
Filter, Low Pass, 4.5 MHz, SM-C-683640	----------
Modulator, SM-F-650502	GL-SM-B-405768
Control, Elec. Freq., SM-F-650503	GL-SM-B-405770
Alarm Monitor*, SM-F-650504	GL-SM-B-405750
Amplifier, Freq. Mult. (XMTR), SM-F-683663	GL-SM-B-405987
Freq. Mult., 2nd Stage, SM-D-650528	GL-SM-B-405762
Freq. Mult., 3rd Stage, SM-D-650529	GL-SM-B-405764
Mixer, Crystal (XMTR), SM-D-650527	GL-SM-B-405766
Modem, Digital, Data Assy., SM-D-683686	----------
Orderwire Assy., SM-D-683686	----------

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Synthesizer Assy.*, SM-D-650511	----------
Power Supply, Transmitter Assy., SM-C-650508	----------

DDM Modules

Oscillator, Voltage Controlled, (4.608 MHz) SM-B-650580	GL-SM-B-405816
Regenerator, Dig. Cable, SM-D-683689	GL-SM-B-405794
Control-Comparator, Cable, SM-D-683689	GL-SM-B-407151
Processor, Dig. Cable, SM-D-683690	GL-SM-B-407216
Amplifier, AF, SM-D-683691	GL-SM-B-405800
Alarm-Monitor, SM-D-650545	GL-SM-B-405802
Amplifier, Detector Radio, SM-D-683693	GL-SM-B-405948
Regenerator, Dig. Radio, SM-D-683694	GL-SM-B-406619
Control-Comparator, Radio, SM-D-683695	GL-SM-B-406633
Processor, Dig. Radio, SM-D-683696	GL-SM-B-405968

O.W. DAUGHTER BOARDS

Daughter Board #1, SM-D-651609	GL-SM-B-407211
Daughter Board #2, SM-D-651610	GL-SM-B-407212
Daughter Board #3, SM-D-651611	GL-SM-B-407213
Daughter Board #4, SM-D-651612	GL-SM-B-407214
Daughter Board #5, SM-D-651613	GL-SM-B-407215

RECEIVER

Receiver Modules

	Test Fixture
Filter, Bandpass, SM-D-684499	----------
Amplifier AF-RF*, SM-D-650500	GL-SM-B-405738
Amp. Freq. Mult, (RCVR), SM-F-684377	GL-SM-B-504991
Filter, Bandpass, 70 MHz, SM-D-684447	----------
Amplifier, IF, 70 MHz, SM-F-652590	GL-SM-B-405742
Demodulator, SM-F-650518	GL-SM-B-405740
Amplifier, Tunnel Diode, SM-B-651379	GL-SM-B-405752
Amplifier Mixer, SM-D-650520	GL-SM-B-405744
Mixer Stage, Freq. (RCVR), SM-D-650522	----------
Frequency Multiplier-Osc., 220 MHz, SM-F-683648	GL-SM-B-407138
Alarm Monitor*, SM-F-650504	GL-SM-B-405750
Synthesizer Assy*, SM-D-650511	----------
P.S. Receiver Assy., SM-C-650523	----------
Filter, Low Pass, 1.0 MHz, SM-C-684448	----------

Synthesizer Modules

Oscillator, R.F., Voltage Controlled, SM-D-650548	GL-SM-B-407123
Converter, Freq. Elec., SM-D-650551	GL-SM-B-407124
Detector, AF, Phase Error, SM-D-650553	GL-SM-B-407126
Freq. Divider, Var. #1, SM-D-650549	GL-SM-B-407125
Freq. Divider, Var. #2, SM-D-650552	GL-SM-B-407127
Freq. Divider, Fixed, SM-D-650550	GL-SM-B-407129
Osc. R.F., Standard, SM-C-651989	GL-SM-B-407130
Power Supply, SM-C-651807	GL-SM-B-407128

P.S. VOLTAGE REGULATORS

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Regulator 15/28V*, SM-D-650530	GL-SM-B-405772
Regulator 12V*, SM-D-650531	GL-SM-B-405774
Regulator 5/6V, SM-D-650532	GL-SM-B-405776

NOTES:

*Represents modules which are commonly used in the transmitter and receiver. All common modules shall be subjected to the inspection requirements for electrical interchangeability.

---------(dash line) means that the item shall be tested for compliance with the performance requirements specified in the test procedure drawing referenced in the indicated production drawing for the item.

4.10. Visual and mechanical inspection. Equipment shall be examined for the defects delineated in MIL-STD-252, the defects listed under Requirement 5 (Soldering) of MIL-STD-454, standards of workmanship listed under Requirement 9 of MIL-STD-454. Visual and mechanical inspection shall be performed on all cabinets, chassis, plate assemblies modules, sub-modules, units, and parts with covers removed (see 3.19).

4.11. Transmitter, Radio T-1054( )/GRC-144 tests (see Appendix B). The following electrical tests shall be performed on each Transmitter, Radio T-1054( )/GRC-144 as specified below.

4.11.1. Test equipment required.

4.11.1.1. Standard test equipment. Standard test equipment indicated in the test figures are listed below. Unless otherwise specified standard test equipment shall be provided by the contractor in the quantities necessary to accomplish the testing herein:

4.11.1.1. Standard test equipment - continued.

Oscilloscope, TEKTRONIX 535 w/Dual Trace Head
Spectrum Analyzer Plug-In Unit, NELSON-ROSS PSA-511B
Spectrum Analyzer, H.P. 8551B/851B
Termination (50-ohm), MICROLAB TA-5MN
Type "N"" Short, MICROLAB TS-5MN
Attenuator, 3 dB, PRD 130A
Attenuator, 30 dB, PRD 130C (10 dB) & 130D (20 dB)
RMS Voltmeter, H.P. 3400A
AC VTVM, BALLANTINE 310B
Test Oscillator, H.P. 651B or WAVEFORMS 401E
Frequency Counter, H.P. 5245L
Frequency Converter, H.P. 5253B
Frequency Converter, H.P. 5255A
Calorimetric Power Meter, H.P. 434A
Transmission Measuring Set DAVEN 12B
Multimeter, SIMPSON 260
BNC "TEE", UG-274( )/U
Termination (50-ohm), BNC, MICROLAB TA-5MB
Adaptor, Waveguide-Coax, H.P. G281A
Adaptor, POMONA, Type 1296

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Variable Attenuator, w/Waveguide Adaptors, PRD-G101
Ear Phone w/Dual Banana Plug
Radio Frequency Adapter, UG-29( )/U
Distortion Analyzer, H.P. 331A

4.11.1.2. Special test equipment. Special test equipment indicated in the test figures are listed below. Unless otherwise specified, special test equipment shall be provided by the contractor in the quantities necessary to accomplish the testing herein:

Transmitter Test Set and Power Cable per SM-A-653688-2. This test set is used to allow necessary control of alarm and other signals during the transmitter "cabinet-level" tests.

Transmitter Test Power Panel per SM-A-684761-1
Capacitive Decoupler (1.0pF) per SM-A-684762
TD-204( )/U Multiplexer Set (Cable Combiner) (GLE)
TD-353( )/U Multiplexer (GLE)
H-91( )/U Headset-Microphone (GLE)
Radio Receiver, R-1467( )/GRC-144
Orderwire Daughter Board Extender #3 per SM-A-653691-1
Orderwire Daughter Board Extender #4 per SM-A-653691-2
Orderwire Daughter Board Extender #5 per SM-A-653691-3
Termination (600-ohm), BNC Male to BNC Female
Termination (50-ohm, 1/2 Watt, 1%)

4.11.1.3. Interconnecting cables. Interconnecting cables indicated in the test figures are listed below. Unless otherwise specified, interconnecting cables shall be provided by the contractor in the quantities necessary to accomplish the testing herein:

Power cable (Transmitter and Receiver AC Input) 6-feet long, 3-conductor cable, #16 wire.

CG-2437( )/TCC Cable Assembly per SC-DL-531016 (10ft 0 in Length)
CG-2438( )/TCC Cable Assembly per SC-DL-531017 (10ft 0 in Length) 1/4 mile CX-11230( )/G Cable Assembly, Coaxial (GLE) Cable, 2 wire, shielded with ADC/PJ-1 and ADC/PJ-3 connectors RG-214( )/U cables w/Type "N" connectors RG-214( )/U cable w/Type "N" connectors, approx. 6-feet long, calibrated for insertion loss at 4.4, 4.5, 4.6, 4.7, 4.8, 4.9, and 5.0 GHz. RG-58( )/U cable w/BNC connectors RG-58( )/U cables w/BNC and dual banana connectors Cables, dual conductor, shielded, with ADC/PJ-1 Plug and 2-BNC Male connectors RG-62( )/U cables with BNC connectors Cable, coax, with dual banana plug to GCU-161/U connector (Pins A & B) Cable, coax, with BNC and ADC/PJ-3 plugs

4.11.2. Preliminary set-up and measurements.

4.11.2.1. Warm-up time. Allow a warm-up period not to exceed thirty (30) minutes prior to conducting tests where data is recorded.

4.11.2.2. Test Receiver (R-1467( )/GRC-144) check-out. The test receiver shall have been checked and calibrated for "Threshold Level" alignment prior to start of tests (see 4.11.2.6.4 and 4.11.2.6.5).

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4.11.2.3. Multiplex alarms and adjustments. The TD-204( )/U and TD-353( )/U alarms will sound on a change of operating conditions. The alarms may be silenced by pressing the "BUZZER OFF" switch on the front panel of either the TD-204( )/U or the TD-353( )/U. Operation of these alarms is not part of the transmitter test and should be disregarded.

4.11.2.3.1. Normal switch positions.

TD-204( )/U

CABLE POWER switch (front panel) to "OFF" (CAUTION - High Voltage is present when in ON position).

TRAFFIC SEL switch to "48-AR" position (located internal right-hand facility panel).

TONE OFF switch to "OFF" (located internal right-hand side 6A2 panel).

MILE switch to "1 MILE" (located internal right-hand side 6A4 and 6A5 panels).

TD-353( )/U

EVEN CHAN switch (selects even-numbered channels). ODD CHAN switch (selects odd-numbered channels).

SERVE SEL switch (selects signal applied to TEST ALIGN meter).

ADDRESS switch to "MASTER" position.

(All located internal right-hand side facility panel)

4.11.2.4. Module removal. When removing modules from the transmitter for Extender Board insertion, or any other purpose, the transmitter main power switch shall be set to "OFF" during removal and reinsertion of the module or extender board.

4.11.2.5. Test mode. The equipment, receiver and transmitter, shall be in 48 channel configuration unless otherwise indicated. Connect the transmitter and test equipment as shown in Figure 1.

4.11.2.5.1. The transmitter will be in 48 channel configuration by switching the attenuator assembly to 48 channels and by actuating the "TRAFFIC SELECT" switch in the Digital Data Modem to 48 Chan position.

4.11.2.5.2. The receiver will be in 48 channel configuration with the 1.0 MHz Low Pass Filter installed.

4.11.2.5.3. On the Transmitter Test Set SM-A-653688-2, set the W.G. switch to "ANT" position and all other switch to "NORM". Insure that the Remote Handset terminals are terminated in 600-ohms.

4.11.2.5.4. Connect P2 to J13 inside the Radio Patch Panel on top of the transmitter. Also on top of the transmitter, but to the rear of the patch panel, connect P3 to J20 (Fault Locator Power Supply interconnect). Turn all potentiometers, located under the transmitter meter panel, to full CCW position.

4.11.2.5.5. Turn the transmitter main power switch to ON and set the AC voltage on the transmitter power panel (SM-A-684761-1) to 115 VAC. All lights on the transmitter power supply shall be on, the cable terminal light on the right top front shall be lit, and the fan shall blow air out of the top of the transmitter. RECORD: Verify that power supply lights are lit and fan is

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operating normally and the cable terminal light is lit. Also RECORD: AC power in watts, amps, and volts as indicated on the transmitter power test panel. Allow the equipment to warm-up for 30 minutes and during this period, perform the following.

4.11.2.5.6. Set the front panel meter switch to the TRAFFIC position. Adjust R7 (under meter panel) fully CCW.

4.11.2.5.7. Remove the Alarm-Monitor module (1A5) from the transmitter. Connect the cable from the TRAFFIC LEVEL OUT jack on test set SM-A-653688-2 to the alarm monitor jack XA5B on the transmitter plate assembly.

4.11.2.5.8. Adjust test set SM-A-653688-2 Traffic Level to 50 μa. Adjust R7 for a reading of 90 on the transmitter panel meter with the test set meter remaining at 50 μa. Then lock R7. RECORD: R7 adjusted for 90 on transmitter meter.

4.11.2.5.9. Disconnect the test cable and re-install the Alarm-Monitor module (1A5).

4.11.2.5.10. Set the meter switch on the transmitter to "TEST LEAD" positive (+) or negative (-) as dictated by the voltages measured. Using the test lead provided with the transmitter, check that all power supply test points (TP-1 through TP-12) read in the orange range of the meter scale and that all voltage test points on the modules of the transmitter read in the Black range of the meter scale. RECORD: Verify power supply test points read in orange range and module voltage test points read in Black range.

4.11.2.5.11. Set the synthesizer and the R.F. Bandpass Filter (FL-3) for a transmitter frequency of 4800 MHz. Set the Synthesizer, the Preselector Bandpass Filter (FL4), Postselector Bandpass Filter (FL5), and the L.O. Bandpass Filter (FL6) for a test receiver frequency of 4700 MHz.

4.11.2.5.12. Disconnect Cable W9 from J1 on the ATTEN ASSY (A3). (When measuring the Modulator or R.F. frequency, disconnect the input to the Attenuator Assembly (W9). This disconnects the modulation signals to the Modulator. Any frequency measurement with modulation is incorrect and misleading).

4.11.2.5.13. After the 30 minute warm-up period, connect the Frequency Counter H.P. 5245L and Frequency Converter H.P. 5255A through a Capacitive Decoupler SM-A-684762 to the 150 MHz output of the XMTR MODULATOR (A8). (Use a BNC "TEE" to keep the modulator in the circuit). The modulator frequency shall be 150 MHz ± 15 kHz, with S1 switch on the modulator in normal position, and 150 MHz ± 350 kHz with S1 depressed.

4.11.2.5.14. Depress S1 and adjust C15 on the modulator for a free running modulator frequency of 150.500 MHz, release S1, then measure the modulator frequency and verify as follows:

RECORD: S1 - Normal _____MHz
Limit: 150 MHz ± 15 kHz
S1 - Depressed _____MHz
Limit: 150 MHz ± 350 kHz
Modulator Frequency _____MHz

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Limit: 150.030 MHz max.

4.11.2.5.15. Depress S1 and adjust C15 for 149.500 MHz, release S1, then measure the modulator frequency and verify as follows:

RECORD: Modulator Frequency_____MHz
Limit: 149.970 MHz min.

4.11.2.5.16. Set the meter function switch on the transmitter front panel to FREQ CONTROL-CTR FREQ. Depress S1 and adjust C15 for Zero on the meter. Release S1, then measure the modulator frequency and verify as follows:

RECORD: Modulator Frequency_____MHz
Limit 150 MHz ± 10 kHz

4.11.2.6. Radio test set check-out and threshold measurement. The test receiver shall be checked and calibrated for threshold before continuation of this test (see 4.11.2.6.4. and 4.11.2.6.5.).

4.11.2.6.1. Connect the equipment as shown in Figure 2. Set the LEVEL ADJ control on OSC, RF, 100 MHz module for "SET OSC LEVEL ADJ" mark on the meter of the Radio Test Set with its switch in the OSC LEVEL position. Next, set this switch to the CR1 and CR2 positions. The meter shall indicate in the upper right-hand 50% of the meter scale.

RECORD: CR1 Meter Indication_____
CR2 Meter Indication_____

4.11.2.6.2. Disconnect the cables with the Variable Attenuators from the Receiver (X) and the Transmitter (Y) and using a signal generator set at 4700 MHz and a micro-wave power meter, calibrate the insertion loss of the cables and attenuators (attenuators at zero dB).

RECORD: Insertion loss of cables and attenuator (X-Y)_____dB

Replace the cables to the transmitter and receiver.

4.11.2.6.3. On the test receiver (R-1467/GRC-144), set the AGC SEL switch on the 70 MHz IF Amplifier to "TEST" position, then set the meter switch to CARRIER (IF) position. Next set the OUTPUT LEVEL switch on the Radio Test Set of the transmitter to "LOW".

4.11.2.6.4. Adjust the Variable Attenuator No. 1 in the RF line for a reading of 150 on the receiver meter. Variable Attenuator No. 2 shall be at "zero".

RECORD: The attenuator setting. This is the setting for threshold.

Threshold setting_____dB

4.11.2.6.5. Add the insertion loss of the cables and attenuators as measured in 4.11.2.6.2. to the Threshold Setting. This value subtracted from 91.0 (Receiver Threshold), is the Radio Test Set "Threshold Output Level".

Receiver Threshold: - 91.0 dBm
RECORD: Threshold setting plus X-Y

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Insertion loss: _____dBm
RECORD: Radio Test Set "Low Level" (Threshold)
Output Level _____dBm
Limit: -63 dBm minimum

NOTE: Variable Attenuator No. 1 shall remain at Threshold Setting for remainder of all test, unless otherwise indicated (see 4.11.2.2.).

4.11.2.6.6. Set the OUTPUT LEVEL switch on the Radio Test Set of the transmitter to "HIGH" and then re-adjust Variable Attenuator No. 2 again for a meter reading of 150 on the receiver meter.

RECORD: No. 2 Attenuator reading for "HIGH LEVEL" setting_____dB
Limit: 30 ± 2 dB

Return Attenuator No. 2 to zero dB position.

4.11.2.6.7. On the test receiver, re-set the AGC SEL switch on the IF Amplifier to "DIV OFF" position.

4.11.2.6.8. Insure that the Cable Length Jumper Plug (P2) on Cable Digital Regenerator Board (A2) and Radio Digital Processor Board (A10) in the Digital Data Modem (PCM Combiner) chassis are set for 1/4 MILE cable length. Also set the MILE switches on plug-in panels A4 and A5 to the 1 MILE position on the internal right-hand side of the TD-204( )/U Multiplexer.

NOTE: If the 1/4 MILE CX-11230( )/G Cable Assembly is not used, set both the TD-204( )/U and the DDM to 1/2 MILE positions.

4.11.3. Transmitter meter readings and alarm operation (see 3.12.14 and 3.12.15). With the equipment set-up and adjusted as in the previous paragraphs, re-connect Cable W9 to J1 on the ATTEN ASSY (A3). Insure that the OSC, RF, 100 MHz is turned ON, then set the LEVEL ADJ control for a meter reading of 100 on the transmitter with the switch in the "TRAFFIC" position. Then set the TRAFFIC LEVEL ADJ control on the AMPL, AF-RF module for a meter reading of 100 on the receiver with the meter switch in the "TRAFFIC" position. 4.11.3.1. All alarm indicators and the external traffic indicators (on test set) shall be GREEN, except the Radio Test Set indicator on the transmitter shall light RED, and the synthesizer oven shall be WHITE.

RECORD: Radio Test Set indicator is RED_____

Synthesizer Oven indicator is WHITE_____(Indicator may be depressed to test light)

All other indicators and external Traffic indicators are GREEN_____

4.11.3.2. Rotate the meter function switch on the transmitter front panel through all switch positions as indicated below.

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RECORD: Indication of the meter for each switch position.

METER SWITCH	METER READING	GROUP "A" TEST LIMITS	ALL OTHER (Except Group "A") LIMITS
OFF (TRANISIT)		ZERO	ZERO
CW TEST[1/]		90-110 (ORANGE)	90-110
MODULATOR CONTROL		90-110	80-120 (BLACK)
MODULATOR OUTPUT		90-110	80-120
FREQ CONTROL-CENTER FREQ		10-0-10 (BLUE)	10-0-10
FREQ CONTROL OUTPUT		90-110	80-120 (BLACK)
TRAFFIC	100	__CHECK	90-110
SYNTHESIZER		60-140 (YELLOW)	30 - F.S. [2/] (YELLOW)
AMPL MULT		90-140 (YELLOW)	30 - F.S.
2ND MULT		90-140	30 - F.S.
3RD MULT		90-140	30 - F.S.
RF POWER		50 MINIMUM	42 - F.S.
REFL RF POWER		0 - 20	0 -30

NOTES:
1/ - See instruction in 4.11.3.2.1.
2/ - F.S. = Full Scale Meter Indication.

4.11.3.2.1. Slide the TD-204( )/U Cable Combiner partly from its case and actuate the TONE-OFF switch, located on the right side, to the "TONE" position in order to obtain a meter reading in the "OW TEST" position on the transmitter. Adjust the "THRU OW RECEIVE LEVEL ADJ" control (green) on the Digital Data Modem chassis (A12) if required. This adjustment sets the incoming orderwire level FROM CABLE (see NOTE 1/ above).

4.11.3.3. Return the TONE-OFF switch in the TD-204( )/U to its "OFF" position. Replace the TD-204( )/U in its case and actuate the RESET switch on test set SM-A-653688-2 to the "ON" position.

4.11.3.4. During each of the following alarm tests when the indicator turns RED, the "CNTRL ALARM" indicator will also turn RED and the audible alarm will sound. Observe the indicated alarm and disregard any others. Reset the audible alarm, as required after each test.

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4.11.3.4.1. Turn "OFF" the TD-204( )/U and check that the indicators shown below turn RED and an alternating 1.1 and 1.6 kHz tone is present.

RECORD: Test Set SM-A-653688-2 "TRAFFIC IN" indicator is RED:_____
Test Set SM-A-6536888-2 "TRAFFIC OUT" indicator is RED:_____
CNTRL ALARM indicator on transmitter is RED:_____
1.1 and 1.6 kHz tone present:_____

4.11.3.4.2. Turn "ON" the TD-204( )/U and check as shown below.

RECORD: All indicators return to GREEN_____

4.11.3.4.3. Set the synthesizer to 5090.0 MHz. The SYNTH LOCK indicator shall turn RED. Reset the synthesizer to 4800.0 MHz. The SYNTH LOCK indicator shall turn GREEN.

4.11.3.4.4. Disconnect Cable W6 from the synthesizer output (A14-J8). The RF POWER indicator shall turn RED. Replace Cable W6 and the RF POWER indicator shall turn GREEN.

4.11.3.4.5. Disconnect the 50-ohm RF termination from the transmitter RF OUTPUT (CP5) (see Figure 2). The REFL RF POWER indicator shall turn RED. Replace the 50-ohm termination and the indicator shall turn GREEN.

4.11.3.4.6. Actuate the W.G. switch on test set SM-A-653688-2 to the DUM LOAD position. The WAVE GUIDE SW indicator on the transmitter shall turn RED. Re-actuate the W.G. switch on the test set to the ANT. position. The WAVE GUIDE SW indicator on the transmitter shall turn GREEN.

4.11.3.4.7. Disconnect the cable to TRAFFIC-INPUT (RCVR) J6 (on top of transmitter), then depress the momentary action RING switch and observe that the RING indicator turns AMBER, the transmitter speaker emits a tone signal and the TD-204( )/U buzzer is actuated. Next, turn on the Radio Test Set, then depress the TALK-OFF-SIG switch on the TD-204( )/U to the "SIG" position. The RING indicator on the transmitter shall turn AMBER and the speaker shall sound. Check and verify indications and reconnect the TRAFFIC INPUT (RCVR) cable to J6.

4.11.3.4.8. Turn the transmitter POWER switch to "OFF". Remove the Cable Digital Processor board (A4) from the DDM and then turn the power "ON". The CABLE TO RADIO MODEM indicator shall be RED.

4.11.3.4.9. Turn the transmitter power "OFF" again. Replace the A4 card and remove the Radio Digital Processor board (A10) from the DDM and then turn the power "ON". The CABLE TO RADIO MODEM indicator shall be GREEN and the RADIO TO CABLE MODEM indicator shall be RED.

4.11.3.4.10. Turn the Transmitter power "OFF", replace the A10 card and return the power switch to "ON". The RADIO TO CABLE MODEM indicator shall be GREEN.

RECORD: Verify and check the operation of alarm tests for 4.11.3.4.3.

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through 4.11.2.4.10. monitored at 4.8 GHz as follows:

ALARM INDICATOR	FUNCTION	INDICATOR COLOR	TEST CONDITION
Radio Test Set	Status of Radio Test Set	GREEN	Radio Test Set Off
		RED	Radio Test Set On
Wave Guide Switch	Status of Waveguide Switch	GREEN	Transmitter Output to Antenna
		RED	Transmitter Output to Load
Ring	Status of Signalling	GREEN	No Signalling
		AMBER	Signalling (Remote or Local)
Ring (Switch Depressed)	Status of Signalling	AMBER	Speaker emits tone signal and TD-204 ( )/U buzzes
Central Alarm	Status of all Equipment (XMTR, RCVR, DDM, & External Traffic	GREEN	All equipment normal
		RED	Failure of one or more equipments
Traffic	Status of Transmitter FM Deviation	GREEN	Deviation normal
		RED	Deviation below that required or no traffic
RF Power	Status of Transmitter Power Output	GREEN	Output power normal
		RED	Output power failure
Refl RF Power	Status of Transmitter Reflected RF Power	GREEN	Refl power normal
		RED	Refl power excessive
Synthesizer Lock	Status of Synthesizer Phase Lock	GREEN	Normal Condition
		RED	Out of Lock

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ALARM INDICATOR	FUNCTION	INDICATOR COLOR	TEST CONDITION
Radio-to-Cable Modem	Status of Modem in One Direction	GREEN	Normal Operation
		RED	Failure (functional)
Cable-to-Radio Modem	Status of Modem in Other Direction	GREEN	Normal Operation
		RED	Failure (functional)

4.11.3.4.11. Rotate the "MODULE TEST" switch on Daughter Board No. 1 of the Orderwire unit through positions 1 to 12 and observe the indications listed below. After completion of this test, return the Module Test switch to position No. 1.

RECORD: Indication for each switch position.

Module Test Switch Position	Module Test Indicator	Central Alarm Indicator	Speaker Output
1. Depress Module Test Lamp	OFF	GREEN	NONE
	ON	GREEN	NONE
2.	OFF	GREEN	4 kHz
3.	OFF	GREEN	4 kHz
4.	OFF*	GREEN	4 kHz
5.	OFF*	GREEN	4 kHz
6.	OFF	GREEN	4 kHz
7.	OFF	GREEN	4 kHz
8.	OFF	GREEN	4 kHz
9.	OFF	GREEN	4 kHz
10.	OFF	GREEN	1.6 kHz

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Module Test Switch Position	Module Test Indicator	Central Alarm Indicator	Speaker Output
11.	OFF	GREEN	1.1 kHz
12.	OFF	RED	Alternate 1.1 and 1.6 kHz
13. Depress RESET switch on Transmitter Meter Panel	ON	RED	NONE

NOTES: 1 - (*) If no PCM signal is present, Indicator Lamp will be "ON".

2 - Turn speaker volume full CW for the above tests.

4.11.3.4.12. Return the RESET switch on test set SM-A-653688-2 to "NORMAL" position.

4.11.4. Transmitter Deviation (48/96 CHAN PCM) (See 3.12.3) Set up the equipment as shown in Figure 1, and connect the Test Oscillator H.P. 651B and the AC VTVM, Ballantine 310B to the "PCM INPUT" (J1) of the ATTEN ASSY - 48/96 CHAN module (A3). Set CH selector switch to 48 CH position. Adjust the frequency to 300 kHz and the level to 0.35 VRMS.

4.11.4.1. Plug the Spectrum Analyzer Head, NELSON-ROSS PSA-511B into the TEKTRONIX 535 Oscilloscope and set the scope controls as follows:

HORIZONTAL DISPLAY - A
TRIGGERING MODE - AUTOMATIC
TRIGGER SLOPE - EXT. -
STABILITY - PRESET (FULL CCW)
TRIGGERING LEVEL - ZERO (CENTERED)
SWEEP TIME - APPROX. 0.2 M SEC/CM

4.11.4.2. Connect the SWEEP-IN jack on the Analyzer Head to the SAWTOOTH-OUT connector on the scope thru the Banana-to-Banana Cable.

4.11.4.3. Preset the Spectrum Analyzer, H.P. 8551B controls as follows:

DISPLAY - "60 dB LOG"
BAND SELECTOR - +5
VERNIER TUNING - CENTERED
SCAN WIDTH/CM - 50
RESOLUTION - AUTOMATIC
IF ATTENUATOR - "ZERO dB"
kHz/MHz SWITCH - MHz
IF GAIN - FULLY CCW
VIDEO FILTER - OFF
CENTER FREQUENCY GHz - 4.7

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4.11.4.4. Connect the Spectrum Analyzer thru 30 dB of attenuators to the RF OUTPUT (CP5) of the transmitter.

4.11.4.5. Re-tune the transmitter to 4700 MHz with Synthesizer and B.P. Filter and adjust the TRAFFIC LEVEL control fully CW.

4.11.4.6. Tune the Spectrum Analyzer to the 4700 MHz signal and adjust the IF GAIN for a signal display within the screen graticule of the CR tube. Reduce the spectrum width, while keeping the signal centered on the screen, until the carrier and the side-bands are visible.

4.11.4.7. Adjust the TRAFFIC LEVEL control on the AMPL, AF-RF MODULE (A4) fully CCW and then CW for the First Carrier disappearance.

RECORD: Traffic Level adjusted for First Carrier disappearance _____

4.11.4.8. Check and verify the meter indication on the transmitter with the meter function switch in the "TRAFFIC" POSITION.

RECORD: Meter indication in TRAFFIC position.

Limit: 80 to 100 _____

4.11.4.9. Connect the AC VTVM thru a BNC "TEE" connector to the MODULATOR module (A8) INPUT jack (J1) and measure the input level.

RECORD: Modulator Input Level _____mVRMS.

Limit: 23 to 28 mVRMS

4.11.4.10. Switch the ATTEN ASSY (A3) to the 96 CH position. Adjust the Test Oscillator to 729 kHz and a level of 0.35 VRMS.

4.11.4.11. Repeat procedure of 4.11.4.7 and 4.11.4.9 and check that the transmitter meter in the "TRAFFIC" position indicates between 80 and 100.

RECORD: Traffic Level adjusted for First Carrier disappearance _____.

Modulator Input Level _____mVRMS.
Limit 56 to 68 mVRMS

Meter indicates between 80 and 100 _____

4.11.4.12. Insure that the ATTEN ASSY (A3) is switched back to the 48 CH position.

4.11.5. Transmitter Power Output (See 3.12.2). With the equipment set-up as shown in Figure 1, measure the power output of the transmitter with the Calorimetric Power Meter, H.P. 434A. Prior to actual measurement, check the "zero" setting of the Calorimetric Power Meter and check the calibration by connecting the Calibration Output to the Input of the Power Meter.

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4.11.5.1. The RF output shall meet the limits shown below at the output of the transmitter. Add the power loss of the cable between the transmitter and the calorimetric power meter to the actual power measured. Use the "DB" scale or calculate the correct power.

RECORD: Output Power Measured:

At 4400 MHz:_____ "DBW"_____mW
Cable Loss:_____dB
Output Power:_____"DBW"_____mw

At 4700 MHz:_____"DBW"_____mw
Cable Loss:_____dB
Output Power:_____DBW"_____mW

At 5000 MHz:_____"DBW"_____mW
Cable Loss:_____dB
Output Power:_____"DBW"_____mW

LIMITS: -6 "DBW" or 250 mW MINIMUM
(Standard Test Conditions - See 4.6)
-7 "DBW" or 200 mW MINIMUM
(After Environmental Testing)
-8 "DBW" or 160 mW MINIMUM
(During Environmental Testing)

4.11.5.2. De-tune the RF Bandpass Filter (FL3) to reduce the power level at the transmitter RF Output to 75 mW (calorimetric indication will be 75 mW minus cable loss).

RECORD: RF level adjusted to 75 mW:_____(Correct for Cable Loss)

4.11.5.3. Adjust R6 located under the meter panel, for an indication of 30 on the transmitter meter with the meter switch in the "RF POWER" position. Then lock R6.

RECORD: R6 adjusted for 30 in RF Power position_____.

4.11.5.4. Readjust the RF Band pass Filter (FL3) for an output level at the transmitter RF Output, of 250 mW (Calorimetric indication will be 250 mW minus cable loss).

RECORD: RF level adjusted to 250 mW_____ Front Panel meter reading_____

LIMIT: 50 minimum

4.11.5.5. Replace the cable at the transmitter RF Output with a 3 dB attenuator and a Type "N" Short. Set the transmitter meter switch to the REFL RF POWER position and adjust R5 under the meter panel for a meter indication of 18. Then lock R5.

RECORD: R5 adjusted for 18 in REFL RF PWR position _____

4.11.5.6. Replace the 3 dB attenuator and short with the calibrated RG-214(

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)/U cable from the calorimetric power meter and adjust the Bandpass Filter (FL3) to 4700 MHz. Set the transmitter meter switch to the "AMPL-MULT" position and adjust R4, under the meter panel, for a meter indication of 100.

RECORD: R4 adjusted for 100 in AMPL-MULT position _____

4.11.5.7. Set the meter switch to the "2ND MULT" position and adjust R3 for a meter indication of 100.

RECORD: R3 adjusted for 100 in 2ND MULT position _____

4.11.5.8. Set the meter switch to the "3RD MULT" position and adjust R2 for a meter indication of 100.

RECORD: R2 adjusted for 100 in 3RD MULT position _____

4.11.5.9. Adjust the synthesizer and RF Bandpass Filter (FL3) for a transmitter frequency of 4400 MHz.

4.11.5.10. Verify and check the transmitter meter indication with the meter switch in the following positions:

1 - AMPL-MULT
2 - 2ND MULT
3 - 3RD MULT
4 - RF POWER
5 - REFL RF POWER

4.11.5.11. Verify and check the RF Power at the output of the transmitter (calorimetric indication plus cable insertion loss).

4.11.5.12. Repeat procedures 4.11.5.9. thru 4.11.5.11 at 4500 MHz, 4600 MHz, 4700 MHz, 4800 MHz, 4900 MHz and 5000 MHz. If any of the meter indications for "AMPL-MULT", "2ND MULT" and "3RD MULT" are outside the limits on the following table, readjust the appropriate potentiometer under the meter panel and repeat any previous measurements. Then set and lock R4, R3, and R2.

4.11.5.13. Record required meter indications for procedures 4.11.5.10 thru 4.11.5.12 as follows:

NOTE: Paragraphs 4.11.5.2. through 4.11.5.13 are meter calibration adjustments which shall be performed once under Standard Test Conditions (See 4.6) in conjunction with Preliminary Set-up and Measurements (See 4.11.2).

METER SWITCH POSITION		METER INDICATION AT MHz						
		4400	4500	4700	4800	4900	5000	LIMIT
AMPL-MULT	INITIAL							90-140
	FINAL							
2ND MULT	INITIAL							90-140
	FINAL							

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METER SWITCH POSITION		METER INDICATION AT MHz						
		4400	4500	4700	4800	4900	5000	LIMIT
3RD MULT	INITIAL							90-140
	FINAL							
RF OUTPUT POWER	µA METER							------
	PWR METER	-dBW	-dBW	-dBW	-dBW	-dBW	-dBW	------
	INS. LOSS	dB	dB	dB	dB	dB	dB	------
	OUTPUT POWER	-dBW	-dBW	-dBW	-dBW	-dBW	-dBW	-6.0 MIN
		mW	mW	mW	mW	mW	mW	250 MIN
REFL RF POWER								0-20

4.11.6. Orderwire Signal-to-Noise Ratio and Percentage of PCM Signal (See 3.12.5). Use the test set-up of Figure 4, with the transmitter, receiver, multiplexer sets and coax termination interconnected as in Figure 2 and Figure 3 and with the transmitter adjusted to a frequency of 4800 MHz.

4.11.6.1. Adjust the Radio Test Set OUTPUT LEVEL control to "LOW" and the #1 Attenuator to obtain 3 dB less than adjusted value of 4.11.2.6.5 and set the transmitter Traffic Level to 100.

4.11.6.2. Connect the TEKTRONIX 535 scope to the receiver-transmitter traffic cable thru a BNC "TEE" and set the receiver Traffic Level for 1 volt peak-to-peak on the scope. NOTE: Set the Radio Test set switch to "HIGH" output when making the level adjustment, and then return switch to "LOW" position.

4.11.6.3. Connect the Transmission Measuring Set, DAVEN 12B to -4 dBm.

4.11.6.4. Connect the DAVEN 12B "RECEIVE" cable to the telephone jack (THRU O.W. J1) on orderwire Daughterboard Extender No. 3. The switch on Extener No. 3 shall be set to "NORMAL" position.

4.11.6.5. Measure the signal level with the Daven test set in "F1A" position.

RECORD: Signal Level _____dBm

LIMIT: -4 ± 2 dBm

4.11.6.6. Disconnect the "SEND" cable, and measure the noise with the Daven test set.

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RECORD: Noise Level _____dBm

4.11.6.7. Calculate the Recovered O.W. Signal-to-Noise ratio. In case of problem with signal-to-noise and any other test near "Threshold", recheck the receiver threshold calibration.

RECORD: Signal-to-Noise Ratio _____dB

LIMIT: 29 dB Minimum

4.11.6.8. Turn OFF the power to the TD-204( )/U Cable Combiner. Reconnect the "SEND" cable from the DAVEN test set to Extender No. 4 THRU O.W. jack (J1). Measure the signal level with the DAVEN test set still connected to Daughterboard Extender No.3 (THRU O.W.) jack (J1).

RECORD: Signal Level _____dBm

LIMIT: -4 ± 2 dBm

4.11.6.9. Disconnect the "SEND" cable and measure the noise.

RECORD: Noise Level _____dBm

4.11.6.10. Calculate the Emergency Orderwire signal-to-noise ratio.

RECORD: Signal-to-Noise ratio_____dB

LIMIT: 32 dB Minimum

4.11.6.11. Turn on the power to the TD-204( )/U Multiplexer. Adjust the Radio Test Set OUTPUT LEVEL control to "HIGH" and the #1 Variable Attenuator to "T" position (this is the Threshold setting of 4.11.2.6.5).

4.11.6.12. Insure that the Traffic Level on the transmitter reads 100 on the meter. Check that the receiver Traffic Level control is set for a 1 volt peak-to-peak signal on the scope.

4.11.6.13. Connect the "SEND" cable from the DAVEN test set to Orderwire Extension Card No. 4 THRU OW jack (J1). The level shall again be -4 dBm. Disconnect Cable W9 from PCM INPUT (J1) of the transmitter ATTEN ASSY (A3).

4.11.6.14. Measure the Orderwire Level with the scope still connected to the receiver Traffic Out jack.

RECORD: Orderwire Level-Percentage of PCM Signal _____mV P-P

LIMIT: 150 to 200 mV P-P

4.11.6.15. Reconnect Cable W9 to the ATTEN ASSY module.

4.11.7. Digital Data Modem (PCM Threshold Level) Operation (See 3.12.4). The equipment shall be connected, set-up and initially adjusted as in preceding procedure under 4.11.6.

4.11.7.1. Insure that the radio Test Set Output Level switch is set to

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"HIGH". The No. 1 Variable Attenuator shall be at Threshold setting (see 4.11.2.6.5).

4.11.7.2. Adjust the Transmitter TRAFFIC LEVEL control for an indication of 100 on the meter in the transmitter and adjust the receiver TRAFFIC LEVEL control for a level of 1 volt peak-to-peak as measured with the oscilloscope.

4.11.7.3. On the right-hand inside service facility panel of the TD-353( )/U Multiplexer, set the 4th switch from the top to either "PHONE ODD" or "PHONE EVEN" positions. Then select one of the "EVEN " OR "ODD" channels on the channel switches above or below depending on what has been selected in the PHONE position.

4.11.7.4. Set the Ballantine 310B switch to "AMPL" and the range switch to 10 mV. Then monitor the earphone connected thru the Ballantine 310B meter (See Figure 3) to the TALK MONITOR audio connector on the front panel of the TD-353( )/U.

RECORD: No clicks are audible._____

4.11.7.5. Set the Output Level switch on the Radio Test to "LOW". Clicks shall be audible on the phones. Then reduce the attenuator between the receiver and transmitter until an average of 1 to 2 clicks per second are heard. This indicates that PCM Threshold has been reached.

RECORD: Attenuator settings at PCM Threshold _____dB.

LIMIT: Within + ANY, -3 dB of Receiver Threshold

Threshold Attenuator Setting _____dB (from 4.11.2.6.5)

4.11.7.6. Connect the Transmission Measuring Set, DAVEN 12B "SEND" cable to the telephone jack (THRU OW) (J1) on Orderwire Daughterboard Extender No. 4 and adjust the Output Level on the DAVEN 12B to -4 dBm.

4.11.7.7. Check on the earphone. No degradation of PCM Threshold shall take place.

RECORD: No degradation of PCM Threshold, 1 to 2 clicks per second still audible: _____.

4.11.7.8. Remove the Daughter board Extenders Cards and replace the Daughter boards in the chassis. Disconnect the Ballantine 310B meter.

4.11.7.9. Check that the equipment is operating normally and that the TD-353( )/U is framing (FRAME Indicator Lamp Not Lit). Disconnect Cable W9 from the PCM Input (J1) of the ATTEN ASSY (A3). The FRAME light on the TD-353( )/U will be lit.

4.11.7.10. Actuate the CABLE LOOP TEST switch on the DDM chassis to the "TEST" position. The TD-353( )/U shall frame.

RECORD: CABLE LOOP TEST switch operating and frame light on TD-353( )/U is Not lit. _____.

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4.11.7.11. Throw the CABLE LOOP TEST switch back to "NORMAL" position and replace Cable W9 to the PCM INPUT of the ATTEN ASSY (A3).

4.11.8. Orderwire Levels (See 3.12.6). The equipment shall be connected, set-up and initially adjusted as in preceding procedure under 4.11.6.

4.11.8.1. The multiplex equipment and the Radio Test Set shall be turned OFF. The Radio Test Set shall be turned off at the OSC, RF, 100 MHz module (A2).

4.11.8.2. On Orderwire Board Extender No. 3, connect the test oscillator (H.P. 651B) to the"PCM recivered O.W" telephone jack (J2) and actuate switch (S1) to "SHORT" position

4.11.8.3. Adjust the test oscillator to 1000 Hz, 600-ohms and an output level of -4dBm (0.49 VRMS).

4.11.8.4. Check and measure the levels at the LOCAL HANDSET "SIGNAL OUTPUT" terminals and "REMOTE HANDSET" terminals on transmitter test set SM-A-653688-2 and the "THRU O.W." jack (J1) on Orderwire Board Extender No. 3. (Terminate the meter with 600-ohms when measuring the O.W Level at jacks (J1) on Extender Boards No. 3 and No. 5).

RECORD: Input: PCM RECOVERED O.W. 1000 Hz, -4 dBm (0.49 VRMS), 600-ohms

a. Local Handset Level: _____mVRMS

LIMIT: -18 ± 3 dBm (34 - 68 mVRMS) at 150-ohms

b. Remote Phone Level: _____VRMS

LIMIT: -14 dBm (0.112 - 0.225 VRMS) at 600 ohms

c. Thru O.W. Extender No. 3 (J1) at 600-ohm level: _____VRMS
LIMIT: -4 ± 3 dBm (0.35 - 0.70 VRMS)

4.11.8.5. Connect the cable from the test oscillator (H.P. 651B) to LOCAL HANDSET "OSC. INPUT" terminals on transmitter test set SM-A-653688-2 and set the operate switch on the test set to "OPER" position. The test oscillator output shall be 1000 Hz, 50-ohms and 0 dBm (0.22 VRMS) output level.

4.11.8.6. Check and measure the levels at the LOCAL HANDSET "SIGNAL OUTPUT" terminals and the "REMOTE HANDSET" terminals on test set SM-A-653688-2, the "THRU O.W." jack (J1) (600-ohms) on Orderwire Board Extender No. 3, and the"XMTR O.W." jack (J1) (600-ohms) on Orderwire Board Extender No. 5. Return the Operate Switch on the test set to "NORM" position.

RECORD: Input: Local HANDSET OSC. INPUT, 1000 Hz, 0 dBm (0.22 VRMS), 50-ohms.

a. Local Handset Level: _____mVRMS

LIMIT -18 ± 3 dBm (34 - 68 mVRMS) at 150-ohms

b. Remote Phone Level: _____VRMS

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LIMIT: -14 dBm (0.112 - 0.225 VRMS) at 600 ohms

c. Thru O.W. (Extender No. 3-J1) at 600-ohm level: _____VRMS

Limit: -4 ± 3 dBm (0.35 - 0.70 VRMS)

d. Xmtr, O.W. (Extender No. 5-J1) at 600-ohm level: _____VRMS

LIMIT 0 ± 3 dBm (0.55 - 1.10 VRMS

4.11.8.7. Remove the 50-ohm termination from the TRAFFIC INPUT (RVCR) connector (J6) on the transmitter cabinet top and connect the test oscillator, set to 50-ohms in its place. Adjust the test oscillator to 1000 Hz at an output level of -11 dBm (62 mVRMS) and actuate switch S1 on Orderwire Board Extender No. 3 to "NORMAL" position.

4.11.8.8. Check and measure the levels at the LOCAL HANDSET "SIGNAL OUTPUT" terminals and the "REMOTE HANDSET" terminals on test set SM-A-653688-2, and the "THRU O.W." jack (J1) (6-ohms) on Orderwire Board Extender No. 3.

RECORD: RECEIVER IN, 1000 Hz, -11 dBm (62 mVRMS), 50 ohms

a. Local Handset Level: _____mVRMS

LIMIT: -18 ± 3 DbM (34 - 68 mVRMS) at 150-ohms

b. Remote Phone Level: _____VRMS

LIMIT: -14 dBm (0.112 - 0.225 VRMS) at 600 ohms

c. Thru O.W. (Extender No. 3 - J1) at 600-ohm level: _____VRMS

LIMIT: -4 ± 3 dBM (0.35 - 0.70 VRMS)

4.11.8.9. Disconnect the test oscillator from the TRAFFIC INPUT Connector (J6) on top of the transmitter and replace with the 50-ohm termination. Return switch S1 on Orderwire Board extender No. 3 to the "SHORT" position.

4.11.8.10. Connect the test oscillator, adjusted to 1000 Hz, 0 dBm (0.775 VRMS) at 600-ohms to the "REMOTE HANDSET" terminals on test set SM-A-653688-2. Insure that 600-ohm termination on these terminals has been removed.

4.11.8.11. Check and measure the levels at the LOCAL HANDSET "SIGNAL OUTPUT" terminals on SM-A-653688-2, the "THRU O.W." jack (J1-600-ohms) on Orderwire Board Extender No. 3, and the "XMTR, OW." jack (J1-600-ohms) on Orderwire Board Extender No.5. After completion of measurements, disconnect the test oscillator from the "REMOTE HANDSET" terminals on test set SM-A-653688-2 and replace the 600-ohm termination.

RECORD: Input: REMOTE PHONE, 1000 Hz, 0 dBm (0.775 VRMS), 600-ohms

a. Local Handset Level: _____mVRMS

LIMIT: -18 ± 3 dBm (34 - 68 mVRMS) at 150-ohms

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b. Thru O.W. (Extender No. 3-J1) at 600-ohm level: _____VRMS

LIMIT: -4 ± 3 dBm (0.35 - 0.70 VRMS)

c. Xmtr O.W. (Extender No. 5-J1) at 600-ohms level: _____VRMS

LIMIT: 0 ± 3 dBm (0.55 - 1.10 VRMS)

4.11.8.12. Connect the test oscillator, adjusted to 1000 Hz, -4 dBm (0.49 VRMS) at 600-ohms to the "THRU O.W. RECEIVER" jack (J1) on Orderwire Board Extender No. 4.

4.11.8.13. Check and measure the levels at the LOCAL HANDSET "SIGNAL OUTPUT" terminals and the "REMOTE HANDSET" terminals on test set SM-A-653688-2 and "XMTR O.W." jack (J1) (600-ohms) on Orderwire Board No. 5 Extender.

RECORD: Input: THRU O.W. (RECEIVER), 1000 Hz, -4 dBm (0.49 VRMS), 600-ohms

a. Local Handset Level: _____mVRMS

LIMIT: -18 ± 3 dBm (34 - 68 mVRMS) at 150-ohms b. Remote Phone Level: _____VRMS

LIMIT: -14 dBm (0.112 - 0.225 VRMS) at 600-ohms

c. Xmtr O.W. (Extender No. 5-J1) at 600-ohm level: _____VRMS

LIMIT: 0 ± 3 dBm (0.55 - 1.10 VRMS)

4.11.8.14. Upon completion of Orderwire Level tests, return all equipment to Normal test conditions.

4.11.9. Transmitter Power Output with Supply Line Variations (See 3.12.7).

4.11.9.1. This test shall be conducted in conjunction with transmitter test procedures for Power Output as delineated under 4.11.5 with the equipment set-up as shown in Figure 1, and with Cable W9 disconnected from the "PCM INPUT" (J1) of the ATTEN ASSY (A3). Prior to actual measurement, check the "ZERO" setting of the Calorimetric Power Meter, H.P. 434A and check the calibration by connecting the calibration Output to the Input of the Power Meter.

4.11.9.2. Adjusting the Transmitter Test Power Panel SM-A-684761-1 to provide an input of 110 VAC at 47 Hz to the transmitter.

4.11.9.3. Check and measure the transmitter output power at 4400 MHz. Add the power loss of the cable between the transmitter and the calorimetric power meter to the actual power measured. Use the "DB" scale or calculate the correct power. Check and observe the transmitter front panel meter reading with the meter function switch in the "RF POWER" position.

RECORD: Line Input: 110 VAC - 47 Hz

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Output at 4400 MHz measured: _____ "DPW" _____mW

Cable Loss at 4400 MHz: _____dB

Power at Transmitter Output: _____ "DBW" _____mW

Limit: 250 MW Minimum (-6.0 "DBW")

Front Panel Meter Reading_____

Limit: 50 Minimum

4.11.9.4. Repeat procedure of 4.11.9.3. Check and measure the transmitter output power at 4700 MHz and 5000 MHz.

a. RECORD: Line Input: 110 VAC-47 Hz

Output at 4700 MHz Measured: _____"DBW" _____mW

Cable Loss at 4700 MHz: _____dB Power at Transmitter Output: _____"DBW"_____

Limit 250 mW Minimum (-6.0 DBW)

Front Panel Meter Reading _____

Limit: 50 Minimum

b. RECORD: Line Input: 110 VAC-47 Hz

Output at 5000 MHz: _____ "DBW" _____mW

Cable Loss at 5000 MHz: _____dB

Power at transmitter Output: _____"DBW" _____mW

Limit: 250 mW Minimum (-6.0 DBW)

Front Panel Meter Reading

Limit: 50 Minimum

4.11.9.5. Adjust the Transmitter Test Power Panel SM-A-684761-1 to provide an input of 120 VAC at 63 Hz to the transmitter. Prior to actual measurement, check the "ZERO" setting of the Calorimetric Power Meter, H.P. 434A and re-check the calibration by connecting the Calibration Output to the Input of the Power Meter.

4.11.9.6. Check and measure the transmitter output power at 5000 MHz. Add the power loss of the cable between the transmitter and the calorimetric power meter to the actual power measured. Use the "DB" scale or calculate the correct power. Check and observe the transmitter front panel meter reading with the meter function switch in the "RF POWER" position.

RECORD: Line Input: 120 VAC-63 Hz

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Output at 5000 MHz Measured:_____ "DBW" _____mW

Cable Loss at 5000 MHz _____dB

Power at Transmitter Output: _____"DBW" _____mW

Limit: 250 mW Minimum (-6.0"DBW")

Front Panel Meter Reading _____

Limit: 50 Minimum

4.11.9.7. Repeat procedure of 4.11.9.5. Check and measure the transmitter output power at 4700 MHz and 4400 MHz.

a. RECORD: Line Input: 120 VAC-63 Hz

Output at 4700 MHz Measured : _____"DBW" _____mW

Cable Loss at 4700 MHz: _____dB

Power at Transmitter Output: _____ "DBW" _____mW

Limit: 250 mW Minimum (-6.0 "DBW")

Front Panel Meter Reading _____

Limit: 50 Minimum

b. RECORD: Line Input: 120 VAC-63 Hz

Output at 4400 MHz Measured: _____"DBW" _____mW

Cable Loss at 4400 MHz: _____ dB

Power at Transmitter Output: _____ "DBW" _____mW

Limit: 250 mW Minimum (-6.0 "DBW")

Front Panel Meter Reading _____

Limit: 50 Minimum

4.11.10. Transmitter Frequency Accuracy (See 3.12.8). With the equipment set-up as shown in Figure 2, measure the transmitter frequency accuracy.

4.11.10.1. Connect the Frequency Converter, H.P. 5255A of the Frequency Counter, H.P. 5245L thru a 30 dB Attenuator to the RF Output (CP5) of the transmitter. Insure that Cable W9 is disconnected from the "PCM INPUT" (J1) of the ATTEN ASSY (A3).

4.11.10.2. Adjust the transmitter Test Power Panel SM-A-684761-1 to provide an input of 120 VAC at 63 Hz to the transmitter.

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4.11.10.3. Set the transmitter to 4400.0 MHz, then check and measure the frequency and the frequency error.

RECORD: Frequency Adjusted to 4,400,000 kHz

Frequency Measured: _____kHz

Frequency Error: _____kHz

Limit: ± 40 kHz Maximum

4.11.10.4. Set the transmitter to 5000.0 MHz, then check and measure the frequency and the frequency error.

RECORD: Frequency Adjusted to 5,000,000 kHz

Frequency Measured: _____ kHz

Frequency Error: _____kHz

Limit: ± 40 kHz Maximum

4.11.10.5. Adjust the transmitter power panel SM-A-684761-1 to provide an input of 110 VAC at 47 Hz to the transmitter.

4.11.10.6. Check and measure the frequency and the frequency error at 5000 MHz.

RECORD: Frequency Adjusted to 5,000,000 kHz

Frequency Measured: _____ kHz

Frequency Error: _____ kHz

Limit: ± 40 kHz Maximum

4.11.10.7. Set the transmitter to 4400 MHz, then check and measure the frequency and the frequency error.

RECORD: Frequency Adjusted to 4,400,00 kHz

Frequency Measured: _____kHz

Frequency Error: _____kHz

Limit: ± 40 kHz Maximum

4.11.10.8. Adjust the transmitter power panel SM-A-684761-1 to provide an input of 115 FAC at 60 Hz to the transmitter.

4.11.10.9. Check and measure the frequency and the frequency error at 4400 MHz.

RECORD: Frequency Adjusted to 4.400,00 kHz

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Frequency Measured: _____kHz

Frequency Error: _____ kHz

Limit: ± 40 kHz Maximum

4.11.10.10. Set the transmitter to 5000 MHz, then check and measure the frequency and the frequency error.

RECORD: Frequency Adjusted to 5,000,000 kHz

Frequency Measured: _____kHz

Frequency Error: _____kHz

Limit: ± 40 kHz Maximum

4.11.10.11. Reconnect Cable W9 to the input of the ATTEN ASSY Module (A3).

4.11.11. Digital Data Modem (PCM Threshold Level) operation with Supply Line Variation (see 3.12.9). This test shall be conducted in conjunction with transmitter test procedures for DDM (PCM Threshold Level) operation as delineated under 4.11.7, using the test set-up of Figure 4, with the transmitter, receiver, multiplexer sets and coax termination interconnected as in Figure 2 and Figure 3 and with the transmitter adjusted to a frequency of 4800 MHz.

4.11.11.1. Adjust the Radio Test Set OUTPUT LEVEL control to "HIGH". The #1 Variable Attenuator shall be at Threshold setting (see 4.11.2.6.5).

4.11.11.2. Adjust the transmitter TRAFFIC LEVEL control for an indication of 100 on the meter in the transmitter and adjust the receiver TRAFFIC LEVEL control for a level of 1 volt peak-to-peak as measured with the oscilloscope.

4.11.11.3. Adjust the Transmitter Test Power Panel SM-A-684761-1 to provide an input of 110 FAC at 47 Hz to the transmitter and proceed as indicated in 4.11.7. Then monitor the ear phone connected through the Ballantine 310B AC VTVM (see Figure 3) to the TALK MONITOR audio connector of the TD-353( )/U.

RECORD: Line Input: 110 VAC-47 Hz

Audible clicks per minute: _____

Limit: No more than 1 click/minute

4.11.11.4. Adjust the transmitter power panel SM-A-684761-1 to provide an input of 120 FAC at 63 Hz to the transmitter and proceed as indicated in 4.11.7. Once again, monitor the earphone.

RECORD: Line Input: 120 FAC - 63 Hz

Audible clicks per minute: _____

Limit: No more than 1 click/minute

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4.11.12. Orderwire signal-to-noise ratio with supply line variation (see 3.12.10). This test shall be conducted in conjunction with transmitter test procedures for Orderwire Signal-to-Noise Ratio as delineated under (4.11.6) using the test set-up of Figure 4, with the transmitter, receiver, multiplexer sets and coax termination interconnected as in Figure 2 and Figure 3 and with the transmitter adjusted to a frequency of 4800 MHz.

4.11.12.1. Adjust the Radio Test Set OUTPUT LEVEL control to "HIGH". The No. 1 Variable Attenuator shall be at Threshold setting (see 4.11.2.6.5).

4.11.12.2. Adjust the transmitter TRAFFIC LEVEL control for an indication of 100 on the meter in the transmitter and adjust the receiver TRAFFIC LEVEL control for a level of 1 volt peak-to-peak as measured with the oscilloscope.

4.11.12.3. Adjust the Transmitter Test Power Panel SM-A-684761-1 to provide an input of 110 VAC at 47 Hz to the transmitter and repeat the procedure and tests in 4.11.6 and record as follows:

RECORD: Line Input: 110 VAC-47 Hz
Signal Level (F1A): _____dBm
Limit: -4 ± 2 dBm
Noise Level: _____dBm
Recovered O.W. Signal-to-Noise Ratio: _____dB.

Limit: 29 dB Minimum

Signal Level: _____dBm
Limit: -4 ± 2 dBm
Noise Level: _____dBm
Emergency O.W. Signal-to-Noise Ratio: _____dB.

Limit: 32 dB Minimum

4.11.12.4. Adjust the transmitter power panel SM-A-684761-1 to provide an input of 120 VAC at 63 Hz to the transmitter and repeat the procedures and tests in 4.11.6 and record as follows:

RECORD: Line Input: 120 VAC-63 Hz
Signal Level (F1A): _____db.
Limit: -4 ± 2 dBm
Noise Level: _____dBm
Recovered O.W. Signal-to-Noise Ratio: _____dB.

Limit 29 dB Minimum

Signal Level: _____dBm
Limit: -4 ± 2 dBm
Noise Level: _____dBm
Emergency O.W. Signal-to-Noise Ratio: _____dB

Limit: 32 db Minimum

4.11.13. Orderwire peak Limiter (See 3.12.11). Use the test set-up of figure

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4, with the transmitter, receiver, multiplexer sets and coax termination interconnected as in figure 2 and figure 3 and with the transmitter adjusted to a frequency of 4800 MHz.

4.11.13.1. Adjust the Transmitter Test Power Panel SM-A-684761-1 to provide an input of 115 VAC at 60 Hz to the transmitter.

4.11.13.2. Connect the Test Oscillator, WAVEFORMS 401E, to the Local Handset "OSC. INPUT" terminals on Transmitter Test Set SM-A-653688-2. Throw the NORM/OPER switch on this test set to the "OPER" position. Connect the RMS Voltmeter H.P. 3400A to the TRANSMIT THRU O.W. telephone jack (J1) on Orderwire Board Extender No.3, terminated in 600-ohms.

4.11.13.3. Using the 50-ohm output, adjust the test oscillator to 1000 Hz and check and measure the Local Handset input levels at J1 on Orderwire Board Extender No. 3 for each of the test oscillator signal levels indicated below.

RECORD: Local Handset Input

Input Level: 0.07 VRMS: (-10 dBm)
Output Level (REFERENCE) _____VRMS
_____dB

Input Level: 0.22 VRMS (0 dBm)
Output Level: _____VRMS
_____dB

Limit: REFERENCE PLUS + 9.6 dB Minimum

Input Level: 1.30 VRMS (+ 15 dBm)
Output Level: _____VRMS
_____dB

Limit: REFERENCE PLUS + 18 dB Maximum

4.11.13.4. Disconnect the test oscillator from the Local Handset terminals on Test Set SM-A-653688-2 and throw the NORM/OPER switch to the "NORM" position.

4.11.13.5. Remove the 600-ohm termination from the REMOTE HANDSET terminals on the test set and connect the Waveform test oscillator in its place.

4.11.13.6. Connect the RMS Voltmeter, H.P. 3400A, to the TRANSMIT THRU O.W. telephone jack (J1) on Orderwire Board No. 3 Extender.

4.11.13.7. Using the test oscillator adjusted to 1000 Hz, check and measure the Remote Handset input levels at J1 on Orderwire Board Extender No. 3 for each of the test oscillator signal levels indicated below.

RECORD: Remote Handset Input

Input Level: 0.25 VRMS (-10 dBm)
Output Level: (REFERENCE) _____VRMS
_____dB

Input Level: 0.78 VRMS (0 dBm)
Output Level: _____VRMS
_____dB

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Limit: REFERENCES PLUS + 9.6 dB Minimum

Input Level: 4.35 VRMS (+ 15 dBm)
Output Level: _____VRMS
_____dB
Limit: REFERENCE PLUS + 18 dB Maximum

4.11.14. Orderwire Distortion (See 3.12.12). Use the test set-up of figure 4, with the transmitter, receiver, multiplexer sets and coax termination interconnected as in figure 2 and figure 3 and with the transmitter adjusted to a frequency of 4800 MHz.

4.11.14.1. Adjust the Transmitter Test Power Panel SM-A- 684761-1 to provide an input of 115 VAC at 60 Hz to the transmitter.

4.11.14.2. Connect the Test Oscillator, H.P. 651B, to the Local Handset "OSC.INPUT" terminals on Transmitter Test Set SM-A-653688-2 and throw the NORM/OPER switch to the "OPER" position.

4.11.14.3. Adjust the test oscillator to 300 Hz with an output signal level of 0 dBm (0.22 VRMS) at the 50-ohm output.

4.11.14.4. Connect the cable from the AF-INPUT of Distortion Analyzer, H.P. 331A, to the output (J2 OUT) of the AMPL, AF-RF module (A4). Next, connect a cable from AMPL, AF-RF module (J1-IN) to the Remote Handset terminals (with the 600-ohm termination removed) on transmitter test set SM-A-6536888-2. The cables from the ATTEN ASSY (A3) and the Low Pass, 4.5 MHz Filter (A6) to the AMPL-AF-RF module shall be disconnected.

4.11.14.5. Adjust the TRAFFIC LEVEL control on AMPL, AF-RF (A4) fully CCW.

4.11.14.6. Set up the H.P 331A Distortion Analyzer as follows:

FUNCTION SELECTOR: Switch to SET LEVEL
METER RANGE: Switch to 100% (SET LEVEL)
INPUT SENSITIVITY: Set to MAX

4.11.14.7. Adjust Traffic Level control on the A4 module for Full Scale Deflection of 1.0 on the H.P 331A meter.

4.11.14.8. Adjust the Distortion Analyzer as follows:

FUNCTION SELECTOR: Switch to DISTORTION
FREQUENCY RANGE: Switch to X10
FREQUENCY COURSE: Adjust control until meter pointer dips sharply.

FREQUENCY FINE: Adjust control for maximum dip of meter pointer. If at low end of meter scale, decrease setting of Meter Range switch. FREQUENCY FINE & BALANCE: Readjust until no further reduction of meter reading can be obtained.

4.11.14.9. Check and measure distortion at frequency of 300 Hz as indicated by the Distortion Analyzer meter reading times the range switch setting.

RECORD: Output frequency at 300 Hz:

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a. Distortion at Remote Handset Terminals _____%

b. Distortion at THRU O.W. Jack (#3 Extender) _____%

c. Distortion at XMTR O.W. Jack (#5 Extender) _____%

Limit: 5% Maximum

4.11.14.10. Disconnect the cable between the AMPL, AF-RF module (J1-IN) and the Remote Handset terminals on transmitter test set SM-A-653688-2. Replace the 600-ohm termination to the test set. Next, connect a cable from the AMPL, AF-RF module (J1-IN) to the THRU O.W. telephone jack (J1) on orderwire Board No. 3 Extender.

4.11.14.11. Repeat procedures of 4.11.14.5 through 4.11.14.8.

4.11.14.12. Move the cable from Orderwire Board No. 3 Extender and plug into the XMTR O.W. jack (J1) on Orderwire Board No. 5 Extender.

4.11.14.13. Recheck procedures for 4.11.14.5 through 4.11.17.8 except that the Distortion Analyzer FREQUENCY RANGE switch shall be set at X100 position.

4.11.14.14. Adjust the H.P.. 651B Test Oscillator to 1000 Hz with an output signal level of 0 dBm (0.22 VRMS) and reconnect the input (J1) of the AMPF, AF-RF module to the Remote Handset terminals (with the 600-ohm termination removed) on transmitter test set SM-A-653688-2.

4.11.14.15. Recheck procedures of 4.11.14.5 through 4.11.14.8 and 4.11.14.10 through 4.11.14.13, then check and measure distortion at frequency of 1000 Hz as indicated by the Distortion Analyzer meter reading times the range switch setting.

RECORD: Output frequency at 1000 Hz:

a. Distortion at Remote Handset Terminals _____%

b. Distortion at THRU O.W. Jack (#3 Extender) _____%

c. Distortion at XMTR O.W. Jack (#5 Extender) _____%

Limit: 5% Maximum

4.11.14.16. Adjust the H.P. 651B test Oscillator to 1800 Hz with an output signal load of 0 dBm (0.22 VRMS), and reconnect the input (J1) of the AMPL, AF-RF module to the Remote Handset terminals (with the 600-ohm termination removed) on transmitter test set SM-A-653688-2.

4.11.14.17. Recheck procedures of 4.11.14.5 through 4.11.14.8 and 4.11.14.10 through 4.11.14.13, then check and measure distortion at frequency of 1800 Hz as indicated by the Distortion Analyzer meter reading time the range switch setting.

RECORD: Output frequency at 1800 Hz:

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a. Distortion at Remote Handset Terminals _____%

b. Distortion at THRU O.W. Jack (#3 Extender) _____%

c. Distortion at XMTR O.W. Jack (#5 Extender) _____%

Limit: 5% Maximum

4.11.14.18. Disconnect the orderwire board extenders and replace with the respective Daughterboards in the Orderwire Assembly.

4.11.14.19. Reconnect the AMPL, AF-RF module (A4) to the ATTEN ASSY (A3) and the L.P 4.5 MHz FILTER (A6). Throw the NORM/OPER switch on Transmitter Test Set SM-A-653688-2 to the "NORM" position.

4.11.15. Orderwire frequency response (See 3.12.16). Use the test set-up and transmitter, receiver and multiplexer interconnections as in 4.11.14 for Orderwire Distortion. The RF output (CP5) shall be terminated.

4.11.15.1. Remove Daughterboards No. 3 and No. 4 and insert the Daughterboard Extenders and then replace the Daughterboards.

4.11.15.2. Insure that the OUTPUT LEVEL switch on the Radio Test Set is set at "HIGH" position. The attenuators shall be at the threshold settings and the traffic level (Transmitter and Receiver) shall be adjusted to meter indications of 100.

4.11.15.3. Connect the H.P. 651B (600-ohm output) to the THRU O.W. jack (J1) on Daughterboard No. 4 Extender. The H.P. 651B output frequency shall be 1000 Hz with an output level of -4 dBm (0.49 VRMS).

4.11.15.4. Connect the RMS Voltmeter, H.P. 3400A, terminated in 600-ohms to the THRU O.W. jack (J1) on Daughterboard No. 3 Extender.

4.11.15.5. Check and measure the meter indication. This shall be the "Reference" reading for "RECOVERED" Orderwire. Then repeat this measurement at all the frequencies indicated below. The H.P. 651B output level shall remain at -4 dBm (0.49 VRMS) at all frequencies.

4.11.15.6. Turn off the TD-204( )/U, reset the H.P. 651B to 1000 Hz with an output level of -4 dBm (0.49 VRMS).

4.11.15.7. Check and measure the meter indication. This shall be the "Reference" reading for "EMERGENCY" Orderwire. Then repeat this measurement at all the frequencies below. The H.P. 651B output level shall remain at -4 dBm (0.49 VRMS) at all frequencies.

RECORD: Frequency Response for Recovered and Emergency Orderwire as follows:

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Orderwire Frequency Response

INPUT LEVEL: -4 dBm (0.49 VRMS)

RECOVERED				EMERGENCY		
Frequency (Hz)	Output (dBm)	Ref. to 1 kHz (dB)	Limit (dB)	Output (dBm)	Ref. to 1 kHz (db)	Limit (dB)
300			-1.0,+2.0			-1.0, +2.0
500			-1.0,+1.8			-1.0, +1.8
700			-1.0,+1.8			-1.0, +1.8
900			± 1.0			± 1.0
1000		0	(0) Ref.		0	(0) Ref.
1200			± 1.0			± 1.0
1400			± 1.0			± 1.0
1600			± 1.0			± 1.0
1800			± 1.0			± 1.0

4.11.15.8. Remove the Daughterboard Extenders and replace the Daughterboards.

4.11.16. Transmitter video input impedance (See 3.12.13). Use the test set-up of figure 5, with the transmitter, receiver, multiplexer sets and coax termination interconnected as in figure 2 and with the transmitter adjusted to a frequency of 4800 MHz.

4.11.16.1. Adjust the Transmitter Test Power Panel SM-A-684761-1 to provide an input of 115 FAC at 60 Hz to the transmitter.

4.11.16.2. Carefully remove cable and Plug-P2 from J13 inside the Radio Patch Panel (top of transmitter) and reconnect P2 to Jack-J15.

4.11.16.3. Connect the H.P. 651B Test Oscillator through the 50-ohm, 1/2 Watt, 1% Resistor to J11, the "TO RADIO" connector on top rear of Radio Patch Panel through a BNC "TEE" to J1-PCM INPUT on ATTEN ASSY (A3).

4.11.16.4. Adjust the H.P. 651B to 50 kHz at an approximate output level of 0.5 VRMS. 4.11.16.5. Using the H.P. 3400A RMS Voltmeter, check and measure the voltage at V1 (see figure 5) and adjust the test oscillator for an output level of 0.050 VRMS at this point. Verify and make note that the input voltage at V1 is 0.50 VRMS.

4.11.16.6. Next, using the H.P. 3400A, check and measure the voltage at V2 (see figure 5).

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4.11.16.7. Record and calculate the transmitter input impedance as follows:

INPUT VOLTAGE (V1): 0.50 VRMS _____Verify VOLTAGE (V2): _____VRMS Limit: 0.237 to 0.262 VRMS

IMPEDANCE:

Z IN = ____V2_____ x 50-OHMS V1 - V2

4.11.16.8. Carefully disconnect Plug-P2 from Jack-J15 (inside patch panel) and reconnect to Jack-J13.

4.12. Receiver, Radio R-1467( )/GRC-144 tests (See Appendix B. The following electrical tests shall be performed on each Receiver, Radio R-1467( )/GRC-144 as specified below.

4.12.1. Test equipment required.

4.12.1.1. Standard test equipment. Standard test equipment indicated in the test figures are listed below. Unless otherwise specified, standard test equipment shall be provided by the contractor in the quantities necessary to accomplish the testing herein.

Noise Figure Meter, H.P. 342A
Noise Generator, H.P. G347A
Wave Guide Termination, H.P. G910A
Sweep Generator (Base Unit), H.P. 8690B
Sweep RF (Plug-In) Unit, H.P. 8693A
Variable Attenuator, w/Waveguide Adaptors, PRD-G101
Oscilloscope, TEKTRONIX, RM-35A (535A)
Plug-In (Scope), High Gain, TEKTRONIX H
Signal Oscillator, GR-1215C7R
Attenuator, Variable, Coax, KAY 442C
Frequency Counter, H.P. 5246L
AC-VTVM, BALLANTINE 310B (Used with Test Transmitter)
Frequency Converter, H.P. 5255A
Isolator, SPERRY D44C2-2
Signal Generator, H.P. 618 CR
Frequency Selective Voltmeter, WANDEL-GOLTERMAN TFPM-43
Attenuator, Coax, 10 DB, NARDA 757-10 OR PED 130C
Test Oscillator, H.P. 651B (USED WITH TEST TRANSMITTER)
Adapter, Waveguide-Coax, H.P. G281A
Power Meter, BOONTON 41A
Detector Mount, BOONTON 41A-4A
Directional Coupler, 5 GHz, 10 DB, NARDA 3044-10
Detector, Crystal, H.P. 423A
Attenuator, W.G. 10 DB, WAVELINE 30066
Attenuator, Fixed, Coax, 10 DB, MICROLAB AH-10B
Waveguide Bend, 90o -E Plane, NARDA 353E
Waveguide Attenuator, Variable, H.P. G382A
RMS Voltmeter, H.P. 3400A

4.12.1.2. Special test equipment. Special test equipment indicated in the test figures are listed below. Unless otherwise specified, special test

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equipment shall be provided by the contractor in the quantities necessary to accomplish the testing herein.

Receiver Cabinet Level Test Set per SM-B-684756. This test set is used to allow necessary control of alarm and other signals during the receiver "Cabinet-Level" tests.

Test, Transmitter, T-1054( )/GRC-144.
Shielded Test Room - 10' x 10' x 8', ACE R10H - 10G.
Variable Voltage & Frequency Generator - 110 VAC to 120 VAC at 47 Hz to 63 Hz, 1 Phase, 5 Amperes.
Adaptor, Coax-TNC (F) to BNC (F), GRFF #1367
Adaptor, Coax:
Type TNC-F to Type N-F
Type TNC-M to Type N-F
Type MICON-F to BNC-F
Type MICON-M to BNC-F
Type NF-F
Type N-M-M
Type BNC-F-F
Type BNC-M-M
Type BNC-TEE

4.12.1.3. Interconnecting cables. Interconnecting cables indicated in the test figures are listed below. Unless otherwise specified, interconnecting cables shall be furnished by the contractor in the quantities necessary to accomplish the testing herein.

Power cables (Receiver and Transmitter AC Input) 6-feet long, 3-conductor cable, #16 wire.
Cable, Coax, RG-214( )/U, w/Type N(M) connectors, Both Ends, 8-feet long, (W1).
Cable, Coax, RG-214( )/U, w/Type N(M) connectors, Both Ends, 6-feet long, (W2).
Cable, Coax, RG-214( )/U, w/Type N(M) to N(F) connectors, 5-feet long, (W3).
Cable, Coax, RG-58( )/U, w/Type BNC(M) connectors, Both Ends, 6-feet long, (W4).
Cable, Coax, RG-58( )/U w/Type BNC(M) connectors, Both Ends, 3-feet long, (W5).
Cable, Coax, RG-58( )/U, w/Type BNC(M) connectors, Both Ends, 16-feet long, (W6).
Cable, Coax, RG-58( )/U, w/Type BNC(M) connectors, Both Ends, 6-feet long, (W7).
Cable, Coax, RG-58( )/U, w/Type BNC(M) connectors, Both Ends, 4-feet long, (W8).
Cable, Coax, RG-58( )/U, w/Type BNC(M) to Tip Plug and Clip, 3-feet long, (W9, W10, W11).
Cable, Coax, RG-59( )/U, w/Type BNC(M) connectors, 6-feet long, (W12).
Cable, Coax, RG-214( )/U, w/Type N(M) connectors, Both Ends, 1-feet long, (W13).
Cable, Coax, RG-58( )/U, w/Type BNC(M) to Dual Banana Plug, 4-feet long, (W14).
Cable, Coax, RG-58( )/U, w/Type BNC(M) to Tip Plug and Clip,

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6-feet long.
Cable, Coax, RG-58( )/U, w/Type BNC(M) to Tip Plug and Clip, 12-feet long.

4.12.2. Preliminary set-up and measurements.

NOTE: Adjustments and tests in 4.12.2.4.8 through 4.12.2.4.10, 4.12.2.4.12, 4.12.2.4.13 and 4.12.2.4.16 through 4.12.2.4.18 are test point checks and calibration adjustments which shall be performed only once under Standard Test Conditions (see 4.6) at the beginning of Government inspection and test.

4.12.2.1. Warm-up time. Allow a warm-up period not to exceed thirty (30) minutes prior to conducting tests where data is recorded.

4.12.2.2. Test Transmitter (T-1054( )/GRC-144) check-out. The test transmitter shall have been checked and calibrated prior to start of tests where the transmitter is used as the signal source. The transmitter synthesizer frequency controls and the RF, B.P. Filter shall be tuned for a 4700.0 MHz carrier frequency at rated deviation (traffic meter reading of 90) at a modulating frequency of 50 kHz.

4.12.2.3. Module removal. When removing modules from the receiver for replacement or change-over during tests, the receiver main power switch shall be set to "OFF" during removal and reinsertion of the module.

4.12.2.4. Test mode. The receiver and transmitter shall be in 48 channel configuration unless otherwise indicated.

4.12.2.4.1. The transmitter will be in 48 channel configuration by switching the ATTEN ASSY (A3) to the "48 CH" position and by actuating the TRAFFIC-SELECT switch on the Digital Data Modem chassis to the "48 CHAN" position.

4.12.2.4.2. The receiver will be in 48 channel configuration with the 1.0 MHz L.P. Filter (A3) installed.

4.12.2.4.3. Connect the receiver and test equipment as shown in figure 7.

4.12.2.4.4. Unless otherwise specified in this procedure, the receiver shall be set up for Normal operation. Normal operation is defined as follows:

1. All module interconnect cables in place and connected in accordance with SM-F-683619. Check each coax connector to insure tight connections.

2. AGC SEL switch on 70 MHz IF Amplifier (A5) set to "DIV OFF" position.

3. TRAFFIC LEVEL ADJ control on AMPL, AF-RF module (A11) set for minimum gain (Max. CCW rotation).

4.12.2.4.5. Place the controls of the receiver under test as follows:

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CONTROL	LOCATION	POSITION
Power Switch	Meter Panel	OFF
Meter Function Switch	Meter Panel	OFF (TRANSIT)
AGC SEL switch	I.F. Ampl.	TEST
TRAFFIC LEVEL ADJ	Ampl, AF-RF	FULL CCW
Filter, B.P. Preselector	(FL4) Inside Cab.	4700 MHz
Filter, B.P. Post Selector	(FL5) Inside Cab.	4700 MHz
Filter, B.P. Local Oscillator	(FL6) Inside Cab.	4700 MHz
Synthesizer	Inside Cabinet	4700 MHz

Before applying power to the receiver cabinet, connect a ground lead (flat-braided strap wire 3/4" to 1" wide) between cabinet ground stud (E2) and an electrical grounding point.

4.12.2.4.6. Apply power (115 VAC @ 60 Hz) to the receiver under test via the receiver cabinet AC Power Input connector (J5) on top of receiver cabinet.

4.12.2.4.7. Allow the equipment to warm-up for 30 minutes and during this period, perform and verify the following. No receiver input signal is present at this time.

RECORD: Cabinet Blower exhausts air through filter at top of cabinet: _____

TDA Oven Indicator lights WHITE: _____

Alarm Indicator Status:

CARR (IF) lights RED _____

SPARE lights GREEN _____

SYNTH LOCK lights GREEN _____

TRAF lights RED _____

Power Supply Assembly inside cabinet:

DS1 through DS4 shall glow GREEN _____

Extinguished lamp on power supply assembly indicates associated voltage regulator is inoperative, or indicator lamp is defective. TDA Oven indicator will extinguish when oven attains operating temperature (Maximum-30 minutes to cycling) and will then cycle ON/OFF-Verify: _____. When TDA indicator has cycled to OFF, press indicator to test lamp - Verify: _____. Synthesizer 28V and 5V power indicators shall be on-GREEN-Verify: _____. TEST (Black push-button switch) when depressed, lights fault indicator lamps on Frequency Synthesizer modules to RED: _____. Fault indicator lamp shall normally be Off: _____.

Synthesizer Oven indicator on meter panel shall be OFF when AC power is first applied and ON after warm-up. WHITE: _____. When OFF, press indicator to test lamp. Verify: _____.

While the TDA oven indicator is lit, temporarily remove the protective fuse (TDA .1 AMP) on the meter panel. Note that the TDA oven indicator extinguishes. Restore fuse. Verify: _____. TDA neon fuse indicator lights when fuse has blown.

Remove the POWER fuse (5 Amp) on the meter panel and note that all power is removed from the equipment as denoted by blower stoppage and all indicators

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(except cycling TDA oven light) extinguish. Verify:_____. Restore the Power fuse. POWER neon fuse indicator lights when fuse has blown.

4.12.2.4.8. Set the receiver meter selector switch to "OFF (TRANSIT)". If necessary, adjust the meter mechanical "zero" for a zero indication on the meter.

RECORD: Panel meter mechanical zero adjusted: _____

4.12.2.4.9. Rotate the receiver meter selector switch to the (+) TEST LEAD position. Use a test lead (similar to one normally supplied with each receiver) connected to the meter panel TEST LEAD jack (Black) and check and verify each of the following module test point measurements for a reading of 100 ± 20 divisions on the receiver cabinet meter.

RECORD: Module Test Point Check: Limit: All readings shall be within BLACK Band on meter scale or 100 ± 20 divisions on meter.

MODULE	TEST POINT (TP)	METER READING (DIV)
REG VOLT 15/28V	TP1 + 28V SIG	
REG VOLT 15/28V	TP2 + 15V SIG	
REG VOLT 12V	TP3 - 12V SIG	
REG VOLT 15/28V	TP4 + 28V AUX	
AMPLIFIER MIXER (A7)	TP2 + 15 VDC	
AMPLIFIER, I.F. (A5)	TP3 + 15 VDC	
DEMODULATOR (A4)	TP7 + 15 VDC	
ALARM-MON (A12)	TP1 + 28 VDC	
ALARM-MON (A12)	TP2 - 12 VDC	
AMPL, AF-RF (A11)	TP1 + 15 VDC	
FREQ MULT OSC, 220 MHz (A13)	TP1 + 28 VDC	
AMPL FREQ MULT, RCVR (A10)	TP1 + 28 VDC	
TUNNEL DIODE AMPLIFIER TDA (A2)	BIAS #1 - (LEFT)	
	BIAS #2 -(RIGHT)	

4.12.2.4.10. Set the receiver meter selector switch to the (-) TEST LEAD position and use the test lead as in 4.12.2.4.9 to check Power Supply TP3 and Alarm Monitor TP2 test points for a reading of 100 + 20 divisions on the receiver cabinet meter.

RECORD: Minus (-) measurements in table above.

4.12.2.4.11. Check and measure AC power line voltage at which the receiver is being tested.

RECORD: Receiver Power Line Voltage _____VAC
Limit: 115 VAC ± 5 VAC

4.12.2.4.12. Remove power from the receiver under test by placing power switch on the meter panel to the OFF position. Then remove the cover assembly on the rear of the meter panel to permit access to the "Depot" adjustments behind the meter panel assembly.

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4.12.2.4.13. Remove Alarm Monitor module (A12) from the receiver. Then adjust the Receiver Test Set SM-B-684756-1 controls as follows:

CONTROL	POSITION
DC PWR (S3)	OFF
TEST SELECTOR (S2)	TFC ADJ
TFC METER METER SELECTOR (S1)	TFC
CONTROL (R3)	MAX. CCW

4.12.2.4.14. Connect Receiver Test Set SM-B-684756-1 to the receiver under test as follows:

TEST SET CONNECTOR	RECEIVER CIRCUIT POINT
P2	Equipment Interconnect J2 (Top of Rcvr Cabinet)
P3	Antenna Alignment J3 (Top of Rcvr Cabinet)
P4/P5	Alarm Monitor Connector XA12B (On Rcvr Plate Assembly)

4.12.2.4.15. Set the meter function switch on receiver meter panel to "TRAFFIC" position. Next, turn "ON" the DC PWR switch (S3) on Rcvr Test Set SM-B-684756-1.

4.12.2.4.16. While observing the test set panel meter, rotate the TFC CONTROL (R3) on the test set for a 50 - Micro-amp reading on the test set meter.

4.12.2.4.17. Adjust R7 (under receiver meter panel) and the TFC CONTROL (R3) on the receiver test set for a simultaneous 100 division indication on the receiver cabinet panel meter and a 50 micro-amp indication on the meter of test set SM-B-684756-1. Tighten the lock-nut on R7, making sure that this action does not change the meter setting.

RECORD: TRAF ADJUST (R7) set for 100 division on Rcvr and 50 micro-amps on test set - Verify: _____

4.12.2.4.18. Turn "OFF" the DC PWR switch (S3) on the receiver test set. Turn OFF the main power switch on the receiver cabinet. Disconnect the test cable with connector P4/P5 from the receiver ALARM MONITOR connector XA12B on the receiver plate assembly and re-install the ALARM-MONITOR module (A12) which was removed in 4.12.2.4.13.

4.12.2.4.19. Restore power to the receiver under test.

4.12.2.4.20. For those tests which require the use of a signal generator as a signal source, the signal frequency should be adjusted to the Test Frequency ± 100 kHz (use Frequency Counter) prior to taking data. NOTE: When making low level measurements on the receiver, seal the "UNCAL OUTPUT" connector on the signal generator with the cap provided.

4.12.2.4.20.1. The following method should be used to set up the Signal Generator output level before making receiver adjustments or taking test data:

1. Connect the H.P. 618 CR Signal Generator to the receiver using test connections as shown in figure 7 (DO NOT connect cable to receiver RF INPUT).

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2. Set signal generator frequency to 4700 MHz.

3. Set Variable Attenuators No. 1 and No. 2 to 0-dB

4. Set signal generator Output Attenuator control for a -20dBm ± 0.2 dB level at the receiver RF Input connection.

4.12.2.4.20.2. The alignment level (for Threshold Test only) indicated in this procedure is for 48 channel operation where the acceptable Threshold level is established at -88.5 dBm.

4.12.2.4.20.3. Prior to testing the receiver, verify the I.F. Amplifier Gain as follows:

1. Set the receiver Frequency Synthesizer controls, the Bandpass Pre-selector (FL4), the Bandpass Post Selector (FL5), and the Filter, Bandpass L.O. (FL6) for operation at 4700 MHz.

2. Set the 70 MHz I.F. Amplifier (A5) AGC SEL switch to "TEST" position and the receiver meter function switch to "CARRIER (IF)".

3. Connect the H.P. 618 CR, adjusted for an output frequency of 4700 MHz ± 100 kHz at an output level to provide a receiver input level of -91 dBm, to the receiver RF INPUT port (FL4).

4. If necessary, adjust the I.F. Amplifier GAIN control (Green) for a receiver panel meter reading of 150 ± 10 divisions.

5. Reset I.F. Amplifier AGC SEL switch to "DIV OFF" position.

4.12.2.4.21. After completion of Preliminary Set-up and Measurements and the 30 minute warm-up period, proceed with the actual test phase and record data as specified.

4.12.3. Frequency Accuracy (3.13.1). Set the synthesizer unit of the receiver for a frequency of 4400 MHz and rotate the receiver meter function switch to the "SYNTH" position. Note that the meter indication is within the YELLOW Band marking.

4.12.3.1. Connect Cable W9 to the Input of the H.P. 5246L Frequency Counter. Use the H.P. 5255A Frequency Converter to measure frequencies between 4 GHz and 5 GHz.

4.12.3.1.1. Set the synthesizer frequency controls and the Filter, B.P., L.O. (FL6) for a frequency of 4570 MHz.

4.12.3.1.2. Check and measure the L.O. frequency on the counter. Measure to nearest 100 Hz. Record data as indicated below.

(Nominal Frequency is 4,500,000 kHz)

4.12.3.1.3. Set the synthesizer frequency controls and the Filter (FL6) for a frequency of 4830 MHz.

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4.12.3.1.4. Check and measure the L.O. frequency on the counter. Measure to nearest 100 Hz. Record data as indicated below. (Nominal Frequency is 4,900,000 kHz).

RECORD: Local Oscillator Frequency:

1	2	3	4
SYNTHESIZER & L.O. FILTER FREQ. (MHz)	NOMINAL FREQ. MEASURED (kHz)	TO BE MEASURED FREQ. (kHz)	ERROR FREQ. (kHz)
4570.0	4,500,000	_____kHz=	_____kHz
4830.0	4,900,000	_____kHz=	_____kHz

NOTE: Error Freq. in Col. 4 is equal to the difference between the nominal freq. in Col. 2 and the measured freq. in Col.3.

LIMIT: Max error frequency is 15 kHz

4.12.3.1.5. Disconnect the frequency counter from Cable W9.

NOTE: Paragraphs 4.12.3.1.6 through 4.12.3.2.4 are meter calibration adjustments which shall be done only once under Standard Test Conditions (see 4.6) in conjunction with Preliminary Set-up and Measurement under 4.12.2.

4.12.3.1.6. Reset the synthesizer frequency controls and the filter (FL6) to 4700 MHz.

4.12.3.1.7. Set the signal generator frequency to 4770 $\pm$ 0.1 MHz. Adjust its output attenuator for an output level of -6 dBm as measured by the power meter. 4.12.3.1.8. Connect the signal generator output cable to L.O. IN (J2) connector of the Amplifier Mixer (A7) module.

4.12.3.1.9. Adjust R5 (under receiver meter panel) for a 30 division indication on the "LOCAL OSC POWER" position of the meter function switch of the receiver. Tighten the lock-nut on R5, making sure that this action does not change the meter setting.

RECORD: L.O. Power meter control (R5) set, adjusted, and locked for 30 - division setting on RCVR meter. Verify:_____

4.12.3.1.10. Disconnect the signal generator cable from the Amplifier Mixer connector (J2) and reconnect the Rcvr. L.O. Input Cable (W9) in its place.

4.12.3.2. The adjustments (under receiver meter panel) for FREQ MIXER CURRENT 1 (R3), FREQ MIXER CURRENT 2 (R4), and MULT-OSC (R6) shall be set for a meter indication of 100 for each switch position. Record data as indicated below.

4.12.3.2.1. Tune the receiver synthesizer controls and the L.O. Filter (FL6) for 4400 MHz. Check and measure the receiver meter readings for each of the switch positions indicated below.

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4.12.3.2.2. Repeat procedure of 4.12.3.2.1 for receiver frequencies of 4550, 4850, and 5000 MHz.

4.12.3.2.3. Tighten the lock-nuts on R3, R4 and R6, making sure that this action does not change the meter setting. Record data as indicated below.

RECORD: Meter Indications and Adjustments.

RCVR. METER SWITCH POSITION	RCVR METER READING AT FREQ. (MHz)					METER ADJUSTMENT CHECKED & LOCKED	FINAL MEASUREMENT AT FREQ (MHz) 4700
	ADJ 0 4700	4400	4550	4850	5000		
SYNTHESIZER							
FREQ. MIXER CURRENT 1						R3_____	
FREQ. MIXER CURRENT 2						R4_____	
MULT-OSC						R6_____	
AMPL-MIXER CURRENT 1							
AMPL-MIXER CURRENT 2							
L.O. OSC. POWER							

LIMIT: 1. Meter indications shall remain within their respective Color Bands when adjusted to 100 divisions at the 4700 MHz frequency.

2. L.O. Power shall indicate 50 divisions minimum.

4.12.3.2.4. Reset the receiver filters, Pre-Selector (FL4), Post Selector (FL5) and Local Oscillator (FL6), and the Synthesizer to 4700 MHz. Record final measurements as indicated above.

4.12.4. Frequency Accuracy with Supply Line variations (See 3.13.2). This test shall be conducted in conjunction with receiver test procedures for Frequency Accuracy as delineated under 4.12.3 with the equipment set up as shown in figure 7. The identical measurements, recording of data, and limits in 4.12.3 shall apply for tests to be conducted with power sources of 110 VAC at 47 Hz and 120 VAC at 63 Hz. No readjustment of controls R3, R4, R5, R6, and R7 (located under meter panel) shall be made since this alignment has previously been accomplished.

4.12.5. Noise Figure measurement (See 3.12.3). Set up the equipment and connect to the receiver under test as shown in figure 8. Connect the H.P. G347A Noise Generator to the receiver RF Input (FL4) through the Waveline

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30066 10 dB Attenuator and the 90o waveguide bend, NARDA Type 353B.

4.12.5.1. Set the receiver synthesizer controls, the Bandpass Preselector (FL4), the Bandpass Post Selector (FL5), and the L.O. Filter (FL6) to 4400 MHz. 4.12.5.2. Using test interconnect Cable (W12), connect the "I.F. OUT" cable (W1) of the Amplifier-Mixer module (A7) to the "INPUT" connector of the H.P. 342A Noise Figure Meter. (Set Noise Figure Meter for "AUTO" measurement of the 70 MHz signal). The Noise Figure Meter shall also be interconnected to the H.P. G-347A Noise Generator with the special power cable provided with the meter.

4.12.5.3. Check and read the Noise Figure (N.F.) on the noise meter in dB and record as indicated below.

4.12.5.4. Check the value in dB of the calibrated pad connected in the receiver input circuit. Record as indicated below.

4.12.5.5. Calculate the actual Receiver Noise Figure. This may be accomplished by subtracting the data in 4.12.5.4 from the data noted in 4.12.5.3. Record as indicated below.

4.12.5.6. Repeat procedures of 4.12.5.1 through 4.12.5.5 for the remaining test frequencies of 4700 MHz and 5000 MHz. Record as indicated below.

RECORD: Receiver Noise Figure Measurements:

Limit: Noise Figure shall be 5.5 dB, + 0.3 dB, - Any.

RF FREQUENCY MHz	4400	4700	5000
NOISE FIGURE METER READING (dB)			
CALIBRATED PAD LOSS (dB)			
RCVR NOISE FIGURE (dB)			

4.12.6. Receiver Bandwidth (See 3.13.7). Set up the equipment and connect to the receiver under test as shown in figure 7 and figure 9. Set the receiver synthesizer controls and filters FL4, FL5, and FL6 for an operating frequency of 4700 MHz.

4.12.6.1. Set the I.F. Amplifier AGC SEL switch to the "TEST" position.

4.12.6.2. Adjust the receiver input level to -88 dBm.

4.12.6.3. Adjust the RM-35A Oscilloscope for sensitivity settings as follows:

Vertical Sensitivity - 0.02V/CM, DC Coupled Horizontal Sensitivity - 0.01V/CM, EXT. X1 (10:1 Variable, Mid Range)

These are suggested scope sensitivity settings. Alternate adjustments may be selected, however, in any case, verify linearity of the detected signal by

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adding 3 dB of attenuation in the W/G Attenuator and noting an approximate 3 dB decrease in detected waveform.

4.12.6.4. Adjust the Center Frequency and Sweep Width controls on the H.P.8690B Sweep Generator and the Vertical Sensitivity control on the Oscilloscope for a selectivity pattern on the scope of approximately 7 CM in amplitude centered on the scope graticule.

4.12.6.5. Insert 3 dB of attenuation in variable W/G Attenuator and set the new vertical deflection to the 5 CM line using the Vertical Position control on the scope. Note and record 3 dB Reference Line as indicated below. Remove the 3 dB of attenuation.

4.12.6.6. Set the CR 1215-C7R Marker Generator to 70 MHz and adjust its output as required to provide a suitable marker on the scope. The marker level should be high enough to easily observe its presence on the scope, but not of a level which raises the scope baseline when the marker frequency is varied through 70 MHz.

4.12.6.7. Increase the marker generator frequency slowly until the marker resets on the 3 dB Reference Line found in 4.12.6.5 and note as "F1 UPPER" the marker frequency displayed on the counter. Record as indicated below.

4.12.6.8. Repeat 4.12.6.7 in a decreasing frequency direction to find "F2 LOWER".

4.12.6.9. Check and measure the 3 dB Bandwidth. (The difference in the two marker frequencies of 4.12.6.7 and 4.12.6.8). Record as indicated below.

RECORD: Bandwidth (I.F.) at 4700 MHz (48 CHAN) Limit: 4500 to 5500 KHz - 3 dB BW

REFERENCE LINE _____	CM
F1 UPPER 3 dB FREQ. _____	kHz
F2 LOWER 3 dB FREQ. _____	kHz
3 dB BANDWIDTH _____	kHz

4.12.7. Threshold Characteristics-48 Channel (See 3.13.4). Set up the equipment and connect to the receiver under test as shown in figure 7. Ensure that the receiver is set for normal operation in the 48 channel mode. Adjust the WANDELL-GOLTERMANN TFPM-43 Selective Voltmeter to 150 kHz. The transmitter and its radio test set shall be operating and tuned to 4800 MHz.

4.12.7.1. Set the receiver synthesizer controls and filters FL4, FL5, and FL6 for an operating frequency of 4700 MHz.

4.12.7.2. Adjust the 4700 MHz receiver signal input level to -60 dBm with the H.P. G-32A Variable Attenuator No. 2 set for 0 dB. Set the TRAFFIC LEVEL ADJ control on the AMPL, AF-RF module (A11) to full CW position. Insure that the AGC SEL switch on 70 MHz I.F. Amplifier (A5) is set to "DIV OFF" position 5.

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4.12.7.3. Change the receiver signal input level to the levels indicated in the threshold data chart below and check and measure the Noise Output Level as indicated on the Selective VTVM for each level. Record as indicated below.

4.12.7.4. Plot a curve of Output Noise (dB) versus Receiver Input Level (dBm), and from this plot, determine the receiver threshold. Threshold is the Input Level in dBm at the intersection of the straight lines drawn through the two linear regions of the plotted points.

4.12.7.5. Plot the receiver threshold on standard graph paper of 10 x 10 divisions per inch or 10 x 10 divisions per CM, size 7 x 10. See figure 6 for a typical threshold plot.

RECORD: Threshold data (48 channel) - At 115 VAC-60 Hz

#2 W.G. ATTEN. dB	RCVR. INPUT -dBm	RCVR. NOISE OUTPUT dB
0	60	
10	70	
15	75	
20	80	
23	83	
24	84	
25	85	
26	86	
27	87	
28	88	
29	89	
30	90	
31	91	
32	92	
33	93	

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#2 W.G. ATTEN. dB	RCVR. INPUT -dBm	RCVR. NOISE OUTPUT dB
34	94	
35	95	
36	96	
38	98	
40	100	
THRESHOLD (dBm)		
LIMIT: (dBm)		-88.5 MAX

4.12.8. Threshold Characteristics - 48 Channel with Supply Line Variations (See 3.13.5). This test shall be conducted in conjunction with receiver test procedures for Threshold Characteristics as delineated under 4.12.7 with the equipment set up as shown in figure 7. the identical measurements and recording of data as in 4.12.7 shall be made with supply line variations for 110 VAC at 47 Hz and 120 VAC at 63 Hz. Record as indicated in Threshold Data Chart below.

4.12.8.1. Plot curves of Noise (dB) versus Receiver Input Level (DBM) and from these plots determine the Receiver Threshold at both 110 VAC, 47 Hz and 120 VAC, 63 Hz.

4.12.8.2. Plot and determine the threshold by the same method used in 4.12.7.4 to 4.12.7.5 and use graph outline as specified. Refer to figure 6 for a typical threshold plot.

RECORD: Threshold Data (48 Channel) - at 110 VAC and 120 VAC

#2 W.G. ATTEN. dB	RCVR INPUT dBm	RECEIVER NOISE OUTPUT dB	
		110 VAC - 47 Hz	120 VAC - 63 Hz
0	60		
10	70		
15	75		
20	80		

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#2 W.G. ATTEN. dB	RCVR INPUT dBm	RECEIVER NOISE OUTPUT dB	
		110 VAC - 47 Hz	120 VAC - 63 Hz
23	83		
24	84		
25	85		
26	86		
27	87		
28	88		
29	89		
30	90		
31	91		
32	92		
33	93		
34	94		
35	95		
36	96		
38	98		
40	100		
THRESHOLD (dBm)			
LIMIT: (dBm)		-88.5 MAX	-88.5 MAX

4.12.9. Meter Readings and Alarm Operation (See 3.13.8 and 3.13.9). Set up the equipment and connect to the receiver under test as shown in figure 7. Connect the Test Transmitter T-1054( )/GRC-144, tuned for an output frequency of 4700 MHz, in place of the H.P. 618 CR Signal Generator. Adjust Variable Attenuator No. 3 on test transmitter output, so that when Variable Attenuators No. 1 and No. 2 are set to "ZERO", the receiver input level measures -20 dBm ± 0.2 dB.

4.12.9.1. Tune the Receiver under test to a frequency of 4700 MHz.

4.12.9.2. Adjust the Test Transmitter for rated deviation (Traffic meter reading of 90 Divisions) at a modulating frequency of 50 kHz.

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4.12.9.3. Set the receiver Input Level to -60 dBm. Insure that the AGC SEL switch on I.F. Amplifier module (A5) is set to the "DIV OFF" position. Set receiver TRAFFIC LEVEL ADJ control on the AMPF, AF-RF module (A11) for a receiver Traffic meter reading of 100.

4.12.9.4. Reduce the receiver input level to -90 dBm. The CARR (IF) alarm light on the receiver meter panel shall turn RED. Observe CARR (IF) alarm light and record as indicated below.

4.12.9.5. Return the receiver input level to -60 dBm. The CARR (IF) alarm light shall turn GREEN. Observe and record as indicated below.

4.12.9.6. Connect the Receiver Test Set SM-B-684756-1 to the receiver cabinet connectors for Equipment Interconnection (J2) and Antenna Alignment (J3). On Test Set SM-B-684756-1, set the DC PWR switch (S3) to "ON" and the Test Selector Switch (S2) to "REMOTE ALARM".

4.12.9.7. Set the Synthesizer frequency controls to 5090 MHz. Note that the SYNTH LOCK (Alarm) light on the receiver meter panel turns RED and the SUMMARY ALARM indicator on Test Set SM-B-684756-1 is ON (lit). Observe and record as indicated below.

4.12.9.8. Reset the Synthesizer frequency controls to 4700 MHz. Note that the SYNTH LOCK (Alarm) light on the receiver meter panel turns GREEN and the SUMMARY ALARM indicator on Test Set SM-B-684756-1 is OFF. Observe and record as indicated below.

4.12.9.9. Set the Receiver Test Set DC PWR switch (S3) to its "OFF" position.

4.12.9.10. Remove the modulating signal source from the Test Transmitter Input and observe that the receiver TRAF Alarm Indicator on the meter panel turns RED. Record as indicated below.

4.12.9.11. Reconnect the modulating signal to the Test Transmitter and observe that the receive TRAF Indicator alarm turns GREEN. Record as indicated below.

RECORD: Receiver Alarm Operation

Verify Observation

Receiver CARR (IF) alarm RED _____

Receiver CARR (IF) alarm GREEN _____

SYNTH LOCK alarm RED _____

Test Set SUMMARY ALARM - ON _____

SYNTH LOCK alarm RED _____

Test Set SUMMARY ALARM - ON

SYNTH LOCK alarm GREEN _____

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Test Set SUMMARY ALARM - OFF _____

TRAF alarm - RED _____

TRAF alarm - GREEN _____

4.12.9.12. Functional Operation of Alarm Indicators shall be as shown for chart below.

ALARM INDICATOR	FUNCTION	INDICATOR COLOR	TEST CONDITION
CARRIER (IF)	STATUS OF CARRIER SIGNAL	GREEN	NORMAL CARRIER SIGNAL
		RED	CARRIER SIGNAL REMOVED
SYNTHESIZER LOCK	STATUS OF SYNTH LOCK	GREEN	NORMAL SYNTH LOCK CONDITION
		RED	SYNTH UNLOCKED
TRAFFIC	STATUS OF TRAFFIC OUTPUT	GREEN	NORMAL TRAFFIC OUTPUT SIGNAL
		RED	LOSS OF TRAFFIC OUTPUT SIGNAL

4.12.9.13. Observe and record the meter reading for each position of the meter function switch on the receiver meter panel.

RECORD: Meter Readings:

METER SWITCH POSITION	METER READING	(SEE NOTE) LIMIT	METER COLOR BAND
OFF (TRANSIT)		ZERO	
SYNTHESIZER		30 - FS	YELLOW
FREQ. MIXER CURRENT 1		30 - FS	YELLOW
FREQ. MIXER CURRENT 2		30 - FS	YELLOW
MULT OSC.		80 - 120	BLACK
AGC		NO LIMIT	
AMPL. MIXER CURRENT 1		30 - FS	YELLOW
AMPL. MIXER CURRENT 2		30 - FS	YELLOW

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METER SWITCH POSITION	METER READING	(SEE NOTE) LIMIT	METER COLOR BAND
CARRIER (IF)		NO LIMIT	
TRAFFIC		90 - 110	ORANGE
LOCAL OSC. POWER		50 - FS	YELLOW

NOTE: FS = FULL SCALE

4.12.9.14. Set the receiver Test Set meter selector switch to the ANT. ALIGN. A position. Vary the receiver input level from - 90 dBm to - 70 dBm and observe the readings on the receiver Test Set meter.

RECORD: Receiver (Antenna Alignment) output.

RCVR INPUT LEVEL dBm	RCVR TEST SET METER READING
-70	
-80	
-90	

LIMITS: Indication shall change with change in Receiver Input level from-70 to -90 dBm.

4.12.10. Receiver Output Capability and Stability (See 3.13.6). Set up the equipment and connect to the receiver under test as shown in figure 7 using the Test Transmitter (48 channel operation) as the signal source. Receiver set for normal operation.

4.12.10.1. Tune the test transmitter and the receiver under test to a 4700 MHz carrier frequency.

4.12.10.2. Set the test transmitter and test attenuators for a receiver input level of -88.5 dBm. Set the transmitter modulation frequency to 50 kHz. Set the TRAFFIC LEVEL ADJ control on the AMPL, AF-RF module (A4) in the transmitter to obtain a reading of 90 divisions on the meter with the function switch set in "TRAFFIC" position.

4.12.10.3. Set the TRAFFIC LEVEL ADJ control on the AMPL, AF-RF module (A11) in the receiver for a receiver traffic output level of 1 volt peak-to-peak (0.354 VRMS) into a 50-ohm load on the TFPM-43 Selective Voltmeter.

RECORD: Video Output set to 1.0 V P-P (0.354 VRMS) Verify:_____

Traffic Level meter reading _____ DIV

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LIMIT: 100 ± 10 Divisions

4.12.10.4. Slowly vary the receiver Input Level between -88.5 dBm and -38.5 dBm while observing the Selective Voltmeter (TFPM-43).

RECORD: Output Stability satisfactory (See Note) Verify:_____

NOTE: Output shall not vary more than ± 1 dB when Input Level is changed from -88.5 dBm to -38.5 dBm.

4.12.10.5. Set the receiver Input Level to -38.5 dBm. Disconnect the Selective Voltmeter from the receiver TRAFFIC OUTPUT connect (J4) and connect the Vertical Input of the Oscilloscope (P/O RCVR Test Station SM-B-684756) in its place. Maintain a 50-ohm termination on the receiver output connector.

4.12.10.6. Observe the traffic output display on the oscilloscope and if necessary make slight adjustment to the receiver traffic level with the control on the AMPL, AF-RF module (A11) for a 1 volt P-P display.

RECORD: Receiver traffic signal output - Verify: 1.0 volts (1000 mV) P-P _____

4.12.10.7. Remove the modulation from the Test Transmitter at the INPUT (J1) of the Modulator module (A8).

4.12.10.8. Check and measure the peak-to-peak amplitude of the Noise as displayed on the oscilloscope.

RECORD: Receiver Traffic Noise Output _____ mV P-P

LIMIT: 12 mV P-P Maximum

4.12.10.9. Disconnect all test equipment from the receiver under test and remove power from the equipment.

4.13. Transmitter/Receiver Test. The following electrical tests shall be performed as specified below on Receiver Radio R-1467( )/GRC-144 and Transmitter, Radio T-1054( )/GRC when interconnected in a one-way system.

4.13.1. Test equipment required.

4.13.1.1. Standard test equipment. Unless otherwise specified, the following standard test equipment shall be provided by the contractor in the quantities necessary to accomplish the testing herein.

Attenuator, Variable, Coax, MERIMAC AU-10A

Selective Voltmeter, WANDELL-GOLTERMANN TFPM-43

Isolator, SPERRY D44C2-2

Attenuator, Variable W/G, H.P. G382A

Adaptor, W/G to Coax, H.P. G281A

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Termination, Coax, 50-ohm, MICRO LAB TA5MN

Frequency Counter, H.P. 5246L

Frequency Converter, H.P. 5255A

Test Oscillator, H.P. 651B

Termination, Coax, 50-ohms, TEXSCAN TF-50

Phase Meter, DRANETZ Series 301

4.13.1.2. Special test equipment. Unless otherwise specified the following special test equipment shall be provided by the contractor in the quantities necessary to accomplish the testing herein.

Transmitter, T-1054( )/GRC-144

Shielded Test Room - 10' x 10' x 8', ACE R10H-10G

Variable Voltage & Frequency Generator - 110 VAC to 120 VAC at 47 Hz to 63 Hz, 1 Phase, 10 Amperes (Use variable source when required)

Adaptor, Coax, TNC(F) to BNC (F), GRFF #1367

4.13.1.3. Interconnecting cables. Interconnecting cables indicated below shall be provided by the contractor in the quantities necessary to accomplish the testing herein.

Power cables (Receiver and Transmitter AC Input) 6-feet long, 3-conductor cable, #16 wire.

Cable, Coax, RG-214( )/U, Terminated Type N(M) to N(M), 12-feet long (W1).

Cable, Coax, RG-214( )/U, Terminated Type N(M) to N(M), 3-feet long (W2).

Cable, Coax, RG-214( )/U, Terminated Type N(M) to N(F), 6-feet long (W3).

Cable, Coax, RG-58( )/U, Terminated Type BNC(M) to BNC(M), 6-feet long (W4).

Cable, Coax, RG-58( )/U, Terminated Type BNC(M) to BNC(M), 6-feet long W5, W6, and W7.

Cable, Coax, RG-58( )/U, Terminated Type BNC(M) to Dual Banana Plug, 6-feet long (W8 and W9).

4.13.2. Preliminary Set-up and Measurements.

4.13.2.1. Warm-up-time. Allow a warm-up period not to exceed thirty (30) minutes prior to conducting tests where data is recorded.

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4.13.2.2. Transmitter (T-1054( )/GRC-144) Check Out. The transmitter shall have been checked and calibrated prior to start of tests where the transmitter is used as the signal source. The transmitter Synthesizer frequency controls and the RF, B.P. Filter shall be tuned for a 4700 MHz carrier frequency at rated deviation (traffic meter reading of 90) at a modulating frequency of 50 kHz. When operating in a back-to-back mode, the transmitter shall normally be adjusted to a higher operating frequency than the receiver. (Set-up in accordance with procedures herein).

4.13.2.3. Module removal. When removing modules from either the transmitter or the receiver for replacement or change-over during test, the transmitter or receiver main power switch shall be set to "OFF" during removal and reinsertion of the module.

4.13.2.4. Test mode. The transmitter and receiver shall be in 48 channel configuration unless otherwise indicated.

4.13.2.4.1. The transmitter will be in 48 channel configuration by switching the ATTEN ASSY (A3) to the "48 CH" position and by actuating the TRAFFIC-SELECT switch on the Digital Data Modem chassis to the "48 CHAN" position.

4.13.2.4.2. The receiver will be in 48 channel configuration with the 1.0 MHz L.P. Filter (A3) installed and unless otherwise specified shall be set-up for Normal operation. Ensure that all module interconnect cables are in place and connected in accordance with SM-F-683619. Check each coax connector to insure tight connections.

4.13.2.4.3. Normal operation is further defined in 4.12.2.4.

4.13.2.4.4. Connect the transmitter, receiver, and test equipment as shown in figure 10.

4.13.2.4.5. After completion of Preliminary Set-up and Measurements and the 30 minute warm-up period, proceed with the actual test phase and record data as specified.

4.13.3. System PCM Phase Response (See 3.14.1).

4.13.3.1. Adjust the receiver for Normal operation at 4700 MHz, 48 channels. Set the transmitter for operation at 4800 MHz, 48 channels with the Radio Test Set operating and its Output Level select switch set to "HIGH".

4.13.3.2. Adjust the receiver Input Level to -61 dBm by setting Test Attenuator No. 1 to 10 dB and Test Attenuator No. 2 to 14 dB.

4.13.3.3. Set the transmitter modulation frequency thru adjustment of the test Oscillator H.P. 651B to 50 kHz at an output level of 0.5 VRMS.

4.13.3.4. Set the TRAFFIC LEVEL ADJ control on the transmitter AMPL, AF-RF module (A4) for a reading of 90 Divisions on the meter of the transmitter with the meter function switch set in the "TRAFFIC" position.

4.13.3.5. Set the TRAFFIC LEVEL ADJ control on the AMPL, AF-RF module (A11)

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in the receiver for a receiver traffic output level (Thru cable from connector J4 on RCVR) of 1 volt peak-to-peak (0.354 VRMS) into a 50-ohm load on the TFPM-43 Selective Voltmeter.

4.13.3.6. Connect the Channel "A" Input of the DRANETZ 301 Phase Meter to the Traffic Output cable from the J4 connector of the receiver (Thru a BNC TEE) and the Channel "B" Input of the Phase Meter to the H.P. 651B Output (50-ohm) thru a BNC TEE.

4.13.3.7. Set the H.P. 651B to 300 Hz at an output level of 0.5 VRMS. Check and measure the System Phase Shift and record as indicated below.

4.13.3.8. Check and measure the receiver Traffic Level Output Voltage of the 300 Hz signal on the Selective Voltmeter at the receiver output. (Use the voltmeter "dB" scale for convenience in making this and the following measurement). Record as indicated below.

4.13.3.9. Adjust the H.P. 651B Test Oscillator to 1000 Hz at an output level of 0.5 VRMS. Check and measure the Traffic Level Output Voltage as in 4.13.3.8 above, and record as indicated below.

System PCM Phase Response and Phase Shift -48 Channel

MEASUREMENT	LIMIT
Phase Shift at 300 Hz _____ DEGREES	37 ± DEGREES
Output at 300 Hz _____ VRMS _____ dB	1.0 dB Max. below 1000 Hz level
Output at 1000 Hz _____ VRMS _____ dB	

4.13.4. Video Frequency Response (See 3.14.2). Connect the transmitter, receiver, and test equipment as shown in figure 10.

4.13.4.1. Adjust the receiver for Normal operation at 4700 MHz, 48 channels. Set the transmitter for operation at 4800 MHz, 48 channels with the Radio Test Set operating and its Output Level select switch set to "HIGH".

4.13.4.2. Adjust the receiver Input Level to -78 dBm by setting Test Attenuator No. 1 to 30 db and Test Attenuator No. 2 to 11 dB.

4.13.4.3. Adjust the H.P. 651B Test Oscillator supplying the modulation to the transmitter for a frequency of 50 kHz at a level of 0.5 VRMS using the TFPM-43 Selective Voltmeter switched to HI Impedance Input. MAINTAIN this level throughout this test.

4.13.4.4. Set the TRAFFIC LEVEL ADJ control on the transmitter AMPL, AF-RF module (A4) for a reading of 90 Divisions on the meter of the transmitter with the meter function switch set in the "TRAFFIC" position. DO NOT touch this control for the balance of this test.

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4.13.4.5. Set the TRAFFIC LEVEL ADJ control on the receiver AMPL, AF-RF module (A11) for a reading of 100 Divisions on the meter of the receiver with the meter function switch set in the "TRAFFIC" position. DO NOT touch this control for the balance of this test.

4.13.4.6. Check and measure the receiver Traffic Output Level as indicated on the TFFM-43 Selective Voltmeter connected thru a BNC-TEE and cable to the receiver Traffic Output connector (J4). The Selective Voltmeter shall be tuned to 50 kHz and its input impedance switch set to "HI IMPED". Record the Traffic Output Level (Eo) in dB as indicated below.

4.13.4.7. Repeat procedures of 4.13.4.3. and 4.13.4.6 for the following frequencies: 460, 650, 920, 1300, 1840, 2600, 3300 kHz. Check and measure the Traffic Output Level (Eo) and record as indicated below.

System Frequency Response - 48 Channel Operation

FREQUENCY (kHz)	EO (dB)	RESPONSE ATTENUATION (dB)	RESPONSE ATTENUATION LIMIT (dB)
50			0 REFERENCE
460			1.0 dB MAX
650			1.5 ± 0.5
920			3.0 ± 0.6
1300			6.0 ± 0.9
1840			12.0 ± 1.2
2600			23.0 ± 2.4
3300			42.0 dB MAX

4.13.4.8. Remove all power and disconnect all test equipment and cables from the transmitter and receiver under test.

4.14. Indicator, Antenna Alignment ID-1708( )/GRC (See 3.15). The following electrical tests shall be performed on each Indicator, Antenna Alignment ID-1708( )/GRC as specified below.

4.14.1. Test equipment required.

4.14.1.1. Standard Test Equipment. Unless otherwise specified, the following test equipment or equivalents shall be provided by the contractor in the quantities necessary to accomplish the testing herein.

Lambda Power Supply - LH-125 FM, 28 VDC

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Receiver Test Set - SM-B-684756-1

4.14.1.2. Interconnecting cables. Interconnecting cables indicated below shall be provided by the contractor in the quantities necessary to accomplish the testing herein.

Set, cables between Power Supply and Test Set SM-B-684756-1

Set, Test Leads from Test Set SM-B-684756-1 to Indicator under test.

4.14.2. Operational Test (See 3.15.1). On Test Set SM-B-684756-1 switch S2 to TFC position. Switch S1 to position 5 TFC adj. Connect the Receiver Test Set SM-B-684756-1 terminals TFC and GND to Pins A and B respectively of Cable W-1 on the alignment indicator under test. Connect the Power Supply to the Test Set Power Input terminals.

4.14.2.1. Turn the DC power On/Off switch on the Test Set to ON and vary the TFC control on the receiver test set until the meter on the receiver test reads 40μ amps, the antenna alignment indicator shall be 80 ± 16 div. Verify and record on data sheet. Turn the ON/OFF (transit) switch on the antenna alignment indicator to OFF. The meter shall read zero, and the input current on the receiver test set shall remain ON. Verify and record on data sheet.

4.14.2.2. Turn off the Power Supply and disconnect the Antenna Alignment Indicator from the Test Set.

4.14.3. Continuity Check (See 3.15.2). Connect the SIMPSON 270 to Pins F and G of Cable W1 on the Alignment Indicator, then connect a jumper between the Field Telephone binding posts E1 and E2 on the Indicator. Then remove jumper.

RECORD: Continuity at E1 - E2, Verify: _____
Open at E1 - E2 Verify: _____

4.15. Operational test (Radio Set) (See 3.17).

4.15.1. Warm-up period. A warm up period of thirty (30) minutes is required before taking measurements and data to allow for frequency stabilization of the R.F. Standard OSC. within the transmitter and receiver Frequency Synthesizers.

4.15.2. Test equipment required.

4.15.2.1. Standard test equipment. Unless otherwise specified, the following standard test equipment shall be provided by the contractor in the quantities necessary to accomplish the testing herein

Variable Attenuator, PRD G101
Oscilloscope, Tektronix 535 with Type "H"
Plug-In Coax to Waveguide Adaptor, H.P. G261A or UG-1683( )/GRC per SM-B-683716
Test Oscillator, H.P. 651B*
AC VTVM Ballantine 310
Termination,, MICROLAB, TA-5MN
Thermistor Mount, H.P. 478A**

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Microwave Power Meter, H.P. 431C**
Attenuator, 20 dB; 1 Watt Min, Type H, calibrated at 4.4 and 5.0 GHz**
Termination, Coax, 50 ohm, TEXSCAN TF-50 or DA541( )/GRC per SM-B-683715
Frequency Counter, H.P. 5246L
Signal Generator, H.P. 618 CR
Adaptor, Double-Ended Female, UG-29( )/U
BNC-TEE, UG-274( )/U
Adaptor, N(M) to BNC(F)
Frequency Converter, H.P. 5255A

References:

* Test Oscillator H.P 651B is required only if noise problems exist with the multiplex equipment (see Alternate Test Method under 4.15.5.1.5 herein).

** These test equipment items may be replaced with Calorimetric Power Meter, H.P 434A and a calibrated RG-214( )/U Cable for RF PWR measurement. See figure 1 for this alternate test set up.

4.15.2.2. Special test equipment. Unless otherwise specified, the following special test equipment shall be provided by the contractor in the quantities necessary to accomplish the testing herein.

TD-204( )/U Cable Combiner (GLE)
TD-353( )/U Multiplexer (GLE)
Headset-Microphone H-91( )/U (GLE)
Earphone with Dual Banana Plug
Coupler, Directional CU-1890( )/GRC per SM-B-651395
System Test Set and Interconnect Cable per SM-A-684766-2

This test set is used to allow necessary control of alarm and other signal during the transmitter-receiver operational test.

4.15.2.3. Interconnecting cables. Unless otherwise specified, the interconnecting cable indicated below shall be provided by the contractor in the quantities necessary to accomplish the testing herein.

Cable, RG-214( )/U with 2-Type "N" Male Connectors
Cable, RG-53( )/U with 2-BNC Male Connectors
Power Cable (Transmitter and Receiver AC Input) 6-feet long, 3-conductor cable, #16 wire
CG-2437( )/TCC Cable Assembly per SC-DL-531016 (10 ft 0 in length)
CG-2438( )/TCC Cable Assembly per SC-DL-531017 (10 ft 0 in length)
1/4 Mile CX-11230( )/G Cable Assembly, Coaxial (GLE)
Cable, RG-62( )/U with 2-BNC Male Connectors
Cable, Coax, w/Dual Banana Plug to CGU-161/U Connector (Pins A & B)

4.15.3. Preliminary set-up and adjustment.

4.15.3.1. Connect the equipment as shown in figures 3 and 11 before connecting to the AC power line. Insure that all power switches on the equipment are in their respective "OFF" positions.

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4.15.3.2. Adjust the receiver synthesizer and the three bandpass filters to 4700 MHz.

4.15.3.3. Adjust the synthesizer and RF bandpass filter on the transmitter to 4800 MHz; throw the switch on the Oscillator, RF 100 MHz, in the transmitter to "ON" and insure that Plug P2 is connected to J13 (Inside Radio Patch Panel on top of the transmitter). NOTE: On the System Test Set, throw the Reset switch to the "OFF" position (Red light On). This will prevent the audible alarm from sounding while the system is set-up. The Waveguide switch shall be set to the "ANTENNA" position.

4.15.3.4. Turn on the power for the transmitter and receiver by throwing their respective AC power switches to "ON".

4.15.3.5. Set the switch on the attenuator assembly to the 48 Chan position.

4.15.3.6. Pull the TD-204( )/U Cable Combiner partly out of its case. Set the TRAFFIC SEL switch on the internal right-hand side facility panel of the TD-204( )/U to the 48 Chan position.

4.15.3.7. Turn on the AC power to the multiplex equipment and the scope. If the alarm sounds on the multiplex equipment, depress the Buzzer Off switch on the front panel of the TD-204( )/U and or TD-353( )/U.

4.15.3.8. On the transmitter, turn the meter switch on the front panel to the "TRAFFIC" position and if the meter does not indicate 100, adjust the TRAFFIC LEVEL control on the Amplifier, AF-RF module within the transmitter for an indication of 100 on the meter.

4.15.3.9. Set the TONE-OFF switch located on the right-inside of the TD-204( )/U to the "TONE" position. Insure that the TALK-OFF-SIG switch on the front panel of the TD-204( )/U is in the "OFF" position. Adjust the THRU OW RECEIVE LEVEL control (green) on the digital data modem chassis of the transmitter for a panel meter reading of 100 in the "OW TEST" position, then return the TONE-OFF switch in the TD-204( )/U to its "OFF" position. Replace the TD-204( )/U in it's case.

4.15.3.10. With the meter switch on the Radio Test Set in the transmitter actuated to the OSC LEVEL position, set the LEVEL ADJ control on the Oscillator, RF 100 MHz module to the "SET OSC LEVEL ADJ" mark on the Radio Test Set meter.

4.15.3.11. Actuate the OUTPUT LEVEL switch on the Radio Test Set to the "LOW" output position.

4.15.3.12. On the receiver, set the meter switch on the front panel to "CARRIER (IF)".

4.15.3.13. Set the switch on the I.F. Amplifier to the "TEST" position. NOTE: The variable attenuator in the receiver-transmitter RF test line, if connected, shall be set and remain in the 6.0 dB position.

4.15.3.14. Adjust the directional coupler "ATTEN ADJ" control on the input of the receiver until the meter on the receiver front panel indicates 150.

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4.15.3.15. Set the switch on the IF Amplifier to "DIV OFF". NOTE:

1. If the tests on this system are for Operation Test only, continue with paragraph 4.15.3..17 herein.

2. If there has been receiver module change after the initial receiver alignment (cabinet level), then the IF Gain Set and Carrier Alarm verification of 4.15.3.16 should be performed.

4.15.3.16. Receiver IF Gain Set and Carrier Alarm verification.

4.15.3.16.1. Set the switch on the I.F. Amplifier to "TEST" and the receiver meter selector switch to "CARRIER (IF)".

4.15.3.16.2. Turn off the Radio Test Set by actuating switch on Osc. RF, 100 MHz module (in Transmitter) to "OFF" position.

4.15.3.16.3. Using the Frequency Counter H.P. 5246L, the Microwave Power Meter H.P. 431C and Thermistor Mount H.P. 478A, adjust and calibrate the H.P. 618CR Signal Generator for 4700 MHz ± 100 kHz and an output level of 0 dBm at the receiver RF Input, (include the cable to RF input), with the signal generator output attenuator set at 0 dBm.

4.15.3.16.4. Adjust the signal generator Output Attenuator to -91 dBm, then if necessary adjust the IF Amplifier Gain control for 150 + 10 on the receiver meter. Set the IF Amplifier AGC SEL switch to "DIV OFF" position.

RECORD: IF Gain Set is verified.

4.15.3.16.5. Adjust the signal generator level for a receiver input of -60 dBm (Output Attenuator), the "CARR (IF)" alarm is Green.

4.15.3.16.6. Decrease the signal generator level for a receiver input of -90 dBm, the "CARR (IF)" alarm shall be Red. If the "CARR (IF)" alarm is still Green, proceed with the following paragraphs, otherwise the "CARR (IF)" alarm is considered as verified and proceed with 4.15.3.16.10.

4.15.3.16.7. Set the signal generator to a receiver input level of -87 dBm.

4.15.3.16.8. Adjust the ALM SET control (Green) on the IF Amplifier for a "CARR (IF)" alarm indication (Red).

4.15.3.16.9. Repeat procedures of 4.15.3.16.5 and 4.15.3.16.6.

RECORD: Carrier (IF) Alarm is verified.

4.15.3.16.10. Turn on the Radio Test Set by actuating switch on Osc. RF, 100 MHz module (In Transmitter) to "ON" position, then repeat procedures 4.15.3.12 through 4.15.3.15.

4.15.3.17. On the Radio Test Set in the transmitter, set the Output Level switch to the "HIGH" position. 4.15.3.18. Adjust the "TRAFFIC LEVEL" control on the Amplifier, AF-RF module within the receiver for an indication of 100 on the receiver meter with the meter switch in the "TRAFFIC" position.

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RECORD: Receiver adjustments are completed.

4.15.4. Transmitter tests.

4.15.4.1. Meter readings.

4.15.4.1.1. Set the meter function switch on transmitter front panel to the "OW TEST" position.

4.15.4.1.2. Pull the TD-204( )/U Cable Combiner partly from its case and actuate the TONE-OFF switch, located on the right side, to the "TONE" position. Insure that the TALK-OFF-SIG switch is in the "OFF" position.

RECORD: Transmitter front panel "OW TEST" meter reading.

4.15.4.1.3. Return the TONE-OFF switch in the TD-204( )/U to its "OFF" position. Replace the TD-204( )/U in its case.

4.15.4.1.4. Rotate the meter function switch on the transmitter front panel through all switch positions as indicated below.

RECORD: Indication of the meter for each switch position.

METER SWITCH POSITION	LIMIT & METER BAND
OFF (TRANSIT)	ZERO
OW TEST 1/	90-110 (ORANGE)
TEST LEAD (+)	ZERO, UNLESS USED
TEST LEAD (-)	ZERO, UNLESS USED
MODULATOR CONTROL	80-120 (BLACK)
MODULATOR OUTPUT	80-120 (BLACK)
FREQ. CONTROL CTR FREQ.	20-0-20 (BLUE)
FREQ. CONTROL OUTPUT	80-120 (BLACK)
TRAFFIC	90-110 (ORANGE)
SYNTHESIZER	30-F.S. 2/ (YELLOW)
AMPL-MULTIPLIER	30-F.S. (YELLOW)
2ND MULTIPLIER	30-F.S. (YELLOW)
3RD MULTIPLIER (LOCAL OSC)	30-F.S. (YELLOW)
RF POWER	30-F.S. (YELLOW)
RF POWER	0-30 (NONE)

NOTES:
1/ See instructions in procedures under 4.15.3.9
2/ F.S. = Full Scale meter indication

4.15.4.2. Alarm conditions.

4.15.4.2.1. Check and verify the status of all alarm indicator lights on the transmitter as indicated below.

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RECORD: Check for normal color indication.

ALARM INDICATOR	STATUS
RADIO TEST SET	RED
WAVEGUIDE SW	GREEN
RING	GREEN
CNTRL ALARM	GREEN
TRAFFIC	GREEN
RF POWER	GREEN
SYNTH LOCK	GREEN
REFL RF POWER	GREEN
CABLE TO RADIO MODEM	GREEN
RADIO TO CABLE MODEM	GREEN

4.15.4.3. Indicator and order wire tests.

4.15.4.3.1. Check and verify the status of the Power Supply, Synthesizer, Cable Terminal, Oscillator RF 100 MHz, Digital Data Modem indicator and Remote Status indicator lights as shown below. Depress the "TEST" switch on the synthesizer to check the status of the module lamps. The SYNTH OVEN indicator lamp can be checked by Press-To-Test function.

RECORD: Check for normal color indication.

INDICATOR	STATUS
POWER SUPPLY DS-1 THRU DS-4 AND DS-6 THRU DS-12	ON (GREEN)
SYNTHESIZER 28V	ON (GREEN)
SYNTHESIZER 5V	ON (GREEN)
SYNTHESIZER MODULES (6)	OFF (NO LIGHT), ON (RED) WITH TEST SWITCH DEPRESSED
RADIO REPEATER OPERATION	LIGHTS (WHITE) WHEN RADIO IS CONNECTED IN RADIO REPEATER CONFIGURATION
CABLE TERMINAL OPERATION	LIGHTS (WHITE) WHEN RADIO IS CONNECTED IN CABLE
INDICATOR	STATUS
CABLE TERMINAL OPERATION - CONTINUED	TERMINAL CONFIGURATION

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INDICATOR	STATUS
OSC, RF, 100 MHz "ON" LAMP	ON (RED)
SYNTH OVEN	ON (WHITE) - DEPRESS INDICATOR TO TEST LAMP
DIGITAL DATA MODEM CARDS A2, A3, A4 (2), A8, A9 AND A10	ON (GREEN)
DIGITAL DATA MODEM CARD A5 - NORMAL/TEST INDICATOR LAMP	OFF - INDICATES SWITCH IN NORMAL POSITION
	ON - (RED) - INDICATES SWITCH IN TEST POSITION
DIGITAL DATA MODEM CABLE LG (Miles) 1/4 Mile Indicator 1/2 Mile Indicator 3/4 Mile Indicator 1 Mile Indicator	 ON (WHITE) ON (WHITE) ON (WHITE) ON (WHITE)
DIGITAL DATA MODEM REMOTE "TRAFFIC IN" ON SYSTEM TEST SET	ON (GREEN) OFF (RED)
DIGITAL DATA MODEM REMOTE "TRAFFIC OUT" ON SYSTEM TEST SET	ON (GREEN) OFF (RED)

4.15.4.3.2. On the System Test Set SM-A-684766-2, actuate the RESET switch to the "ON" (Green Light) position to remove the audible alarm disable condition. Rotate the "MODULE TEST" switch on Daughter Board No. 1 of the Orderwire unit through positions 1 to 12 and observe the indications listed below. After completion of this test, reset the System Test Set to the audible alarm disable condition, and return the Module Test switch to position No. 1.

RECORD: Indication for each switch position.

BOARD NO 1 MODULE TEST SWITCH POSITION	BOARD NO. 1 MODULE TEST INDICATOR	CENTRAL AIR INDICATOR	SPEAKER OUTPUT
1	OFF	GREEN	NONE
1 DEPRESS MODULE TEST LAMP	ON (RED)	GREEN	NONE
2	OFF	GREEN	4 kHz
3	OFF	GREEN	4 kHz
4	OFF	GREEN	4 kHz
5	OFF	GREEN	4 kHz
6	OFF	GREEN	4 kHz

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BOARD NO 1 MODULE TEST SWITCH POSITION	BOARD NO. 1 MODULE TEST INDICATOR	CENTRAL AIR INDICATOR	SPEAKER OUTPUT
7	OFF	GREEN	4 kHz
8	OFF	GREEN	4 kHz
9	OFF	GREEN	4 kHz
10	OFF	GREEN	1.6 kHz
11	OFF	GREEN	1.1 kHz
12	OFF	RED	ALTERNATE
12 Momentarily Depress ON (RED) Reset Switch on Transmitter Meter Panel		RED	1.1 & 1.6 kHz NONE

NOTE: Turn speaker volume fully CW for the above tests.

4.15.4.3.3. Check and verify the indication of the meter on the Radio Test Set for the switch positions below. (Adjust the "LEVEL ADJ" control on the 100 MHz RF Oscillator if required to obtain the proper meter indication).

RECORD: CR-1 Level: Limit: Yellow range to full scale
CR-2 Level: Limit: Yellow range to full scale

4.15.4.3.4. Disconnect the cable from J6 (on top of the transmitter), then depress the TALK-OFF-SIG switch on the TD-204( )/U. The "RING" light on the transmitter shall turn amber and the speaker shall sound a 1600 Hz tone. Verify and record these indications and release the switch.

4.15.4.3.5. Depress the "RING" button on the transmitter. The "RING" light shall turn amber and the speaker shall sound. On the TD-204( )/U, the "CALL" indicator shall light and the buzzer shall sound. Verify and record these indications, release the Ring button, and reconnect the cable to J6.

4.15.4.3.6. Throw the TD-204( )/U TALK-OFF SIG switch to "TALK" and talk into the Head Set Microphone at the TD-204( )/U. The message shall be clearly understandable on the earphonephone of the handset at the transmitter. Record.

4.15.4.3.7. Depress the push-to-talk switch on the handset of the transmitter and talk into the handset microphone. The message shall be clearly understandable at the TD-204( )/U. Record.

4.15.4.4. Transmitter power output.

4.15.4.4.1. Adjust the transmitter controls for 4400 MHz output frequency and disconnect cable W9 from the ATTEN ASSY (A3) PCM INPUT (J1); ignore all alarms due to this change.

4.15.4.4.2. Calibrate the microwave power meter, and get the range to the 10 mW position, then connect the thermistor mount through the 20 dB attenuator to the transmitter RF output (coax adaptor at CP5) in place of the 50 ohm termination. NOTE: If the H.P. 434A Calorimetric Power Meter is used, set

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the "Zero" and check the calibration using the calibration output before measuring the RF power.

4.15.4.4.3. Measure and record the RF output power (Microwave Power Meter Indication Plus Calibrated Attenuator Value). Also verify and record the transmitter front panel" RF Power" meter reading.

Power meter Limit: 250 mW Minimum

Panel Meter Limit: 50 Min to full scale

4.15.4.4.4. Adjust the transmitter frequency to 5000 MHz, then measure and record the RF output power and the RF Power meter reading as above.

Power Meter Limit: 250 mW Minimum

Panel Meter Limit: 50 Min to full scale.

4.15.4.4.5. Replace the 50 ohm termination to the transmitter RF output, reconnect cable W9 to the attenuator input and reset the transmitter frequency to 4800 MHz. All alarms shall be normal.

4.15.5. Receiver and antenna alignment indicator tests.

4.15.5.1. Meter readings and Traffic Level test.

4.15.5.1.1. Rotate the meter function switch on the receiver front panel through all switch positions as indicated below.

RECORD: Indication of the meter for each switch position.

METER SWITCH POSITION	METER BAND & LIMIT
OFF (TRANSIT)	ZERO
SYNTHESIZER	30-F.S.1/(YELLOW)
FREQ MIXER CURRENT -1	30-F.S. (YELLOW)
FREQ MIXER CURRENT -2	30-F.S. (YELLOW)
MULT-OSC.	80-120 (BLACK)
AGC	NO LIMIT
AMPL-MIXER CURRENT -1	30-F.S. (YELLOW)
AMPL-MIXER CURRENT -2	30-F.S. (YELLOW)
CARRIER (IF)	NO LIMIT
TRAFFIC	90-110 (ORANGE)
LOCAL OSC POWER	50-F.S. (YELLOW)
TEST LEAD (-)	ZERO UNLESS USED
TEST LEAD (+)	ZERO UNLESS USED

1/ F.S. = Full Scale meter indication.

4.15.5.1.2. Disconnect the cable from J4 on top of the receiver and connect it to the "TEE" on the input of the oscilloscope. Next, connect another RG-58/U cable from the other side of the "TEE" to J4 on top of the receiver.

4.15.5.1.3. Adjust the traffic levels in the transmitter and then in the

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receiver to exactly 100 on the transmitter and receiver meters.

4.15.5.1.4. Verify and record the traffic level as indicated by the oscilloscope against the following acceptance limits.

Initial Limit: 1.0 + 0.12
- 0.11 VOLTS P-P
Oper. and Final Limit during or after Environment Test:
1.0 + 0.26
- 0.21 VOLTS P-P

4.15.5.1.5. (Alternate Method - see Notes following 4.15.2.1). If the signal shown on the oscilloscope is too noisy to obtain a definite traffic level measurement (due to noisy multiplex equipment), disconnect the cable (W9) to the input of the attenuator assembly on the transmitter and connect the cable from the H.P. 651B Test Oscillator in its place. Ignore all alarm indications due to this connection and proceed as follows.

4.15.5.1.5.1. Adjust the test oscillator to 50 kHz and 0.5 VRMS Output Level.

4.15.5.1.5.2. Adjust the traffic level in the transmitter to 90 and then in the receiver to 100 as indicated by the transmitter and receiver meters.

4.15.5.1.5.3. Verify and record the traffic level as indicated by the oscilloscope against the following acceptance limits.

Initial Limit: 1.0 + 0.12
- 0.11 VOLTS P-P
Oper. and Final Limit during or after Environmental Test:
1.0 + 0.26
- 0.21 VOLTS P-P

4.15.5.1.5.4. Disconnect the cable from the input of the attenuator assembly and replace the original cable (W9). All alarms shall return to Normal.

4.15.5.2. Alarm conditions.

4.15.5.2.1. Check and verify the status of all alarm indicator lights on the receiver as indicated below.

RECORD: Check for normal color indication.

ALARM INDICATOR	STATUS
CARRIER (IF)	GREEN
SYNTH LOCK	GREEN
TRAFFIC	GREEN
SPARE	NORMALLY GREEN-NOT CURRENTLY ACTIVE

4.15.5.3. Indicator tests.

4.15.5.3.1. Check and verify the status and operation of the Power Supply. Synthesizer and Oven indicator lights as shown below. Depress the "TEST"

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switch on the synthesizer to check the status on the module lamps. The TDA and SYNTH OVEN indicator lamps can be checked by Press-To-Test function.

RECORD: Check for normal color indication.

INDICATOR	STATUS
TDA OVEN	OFF OR CYCLING ON/OFF (WHITE) DEPENDING ON AMBIENT TEMPERATURE IF OFF-DEPRESS TO TEST
SYNTH OVEN	ON (WHITE). IF OFF-DEPRESS TO TEST
POWER SUPPLY DS-1 THRU DS-4	ON (GREEN)
SYNTHESIZER 26V	ON (GREEN)
SYNTHESIZER 5V	ON (GREEN)
SYNTHESIZER MODULES (6)	OFF (NO LIGHT) ON (RED WITH TEST SWITCH DEPRESSED)

4.15.5.4. Antenna alignment indicator test.

4.15.5.4.1. Actuate the ON-OFF (TRANSIT) switch on the Antenna Alignment Indicator ID-1708( )/GRC to the "ON" position.

4.15.5.4.2. Observe the indication on the meter, then set the OUTPUT LEVEL switch on the transmitter RADIO TEST SET TO "LOW" output. The meter indication of the Antenna Alignment Indicator shall decrease.

RECORD: Meter varies with change of signal level.

4.15.5.4.3. Actuate the switch on the Antenna Alignment Indicator to the "OFF (TRANSIT)" position.

4.16. Operating test frequency selection. The Government reserves the right at any time to select any harmonically compatible operating test frequencies, similar to those shown in figure A3, of Appendix A, to assure total compliance with specified requirements for the following tests:

a. Paragraphs 4.11.5 and 4.11.9 Transmitter Power Output.

b. Paragraphs 4.11.10 Frequency Accuracy.

c. Paragraphs 4.12.3 and 4.12.4 Local Oscillator Power and Frequency Accuracy.

d. Paragraph 4.12.5 Receiver Noise Figure Measurements.

5. PACKAGING

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5.1. Packaging requirements. The packaging requirements for the desired level(s) of protection shall be as specified by the acquisition activity.

6. NOTES.

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1. Intended use. Radio Set AN/GRC-144( ) and other items covered by this specification are intended for configuration into Radio Repeater Set AN/TRC-138( ) to provide simultaneous 48/96 channel PCM radio repeater and 48/96 channel PCM radio-to-cable repeater operation. When reconfigured with an asynchronous data modem (ADM), the AN/GRC-144( ) radio can be employed in the AN/TRC-138( ) shelter as an ADM radio repeater capable of operation with asynchronous digital combiners (ADC), all part of, and which satisfy US Army requirements for multi-channel line-of-sight communications within the field Army's high-channel capacity Army Area Communication System (AACOMS).

6.2. Acquisition requirements. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (See 2.1.1 and 2.2).

c. First Article Inspection.

(1) Eight (8) sample units of each applicable item cited in Section I are generally required so that lengthy Environmental tests, Electrical characteristic performance measurements, and Electromagnetic compatibility testing may be initiated and completed within a nominal period of time. Four (4) of these sample units shall be further subjected to Reliability Testing. (See 3.2 and 4.3).

d. Marking shipping of samples.

e. Place of final inspection.

f. Technical literature, tools and running spare parts (See 3.18).

g. MIL-STD-461A dated 1 Aug 68 and MIL-STD-462 dated 31 Jul 67 are the specific issues to be used where these standards are referenced in this specification.

6.3. Color. The color chips furnished by the procuring agency (See 3.4) will be those cited on the applicable equipment drawings. The color chips may be obtained upon request to Commander, US Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, Pennsylvania, 19120.

6.4. Group C Inspection. Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the

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adequacy of corrective action (See 4.5.3.2).

6.5. Location of operational test. It is desirable that the Operational Test (See 4.15) be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspections is completed. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging and packing.

6.6. Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter symbol identifying the particular design; for example: "AN/GRC-144". The contractor should apply for nomenclature in accordance with the applicable clause in the contract (See 1.1).

6.7. Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product (See 4.5).

6.8. Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.9. Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.10. Subject term (keyword) listing.

Army Area Communication System (AACOMS)
Asynchronous Data Modem (ADM) Attenuator Assembly, SM-D-683635
Control-Comparator, Cable, SM-D-683689
Daughter Board
Orderwire Signal-to-noise (S/N) ratio
Receiver, Radio R-1467/GRC-144
Transmitter, Radio T-1054/GRC/144

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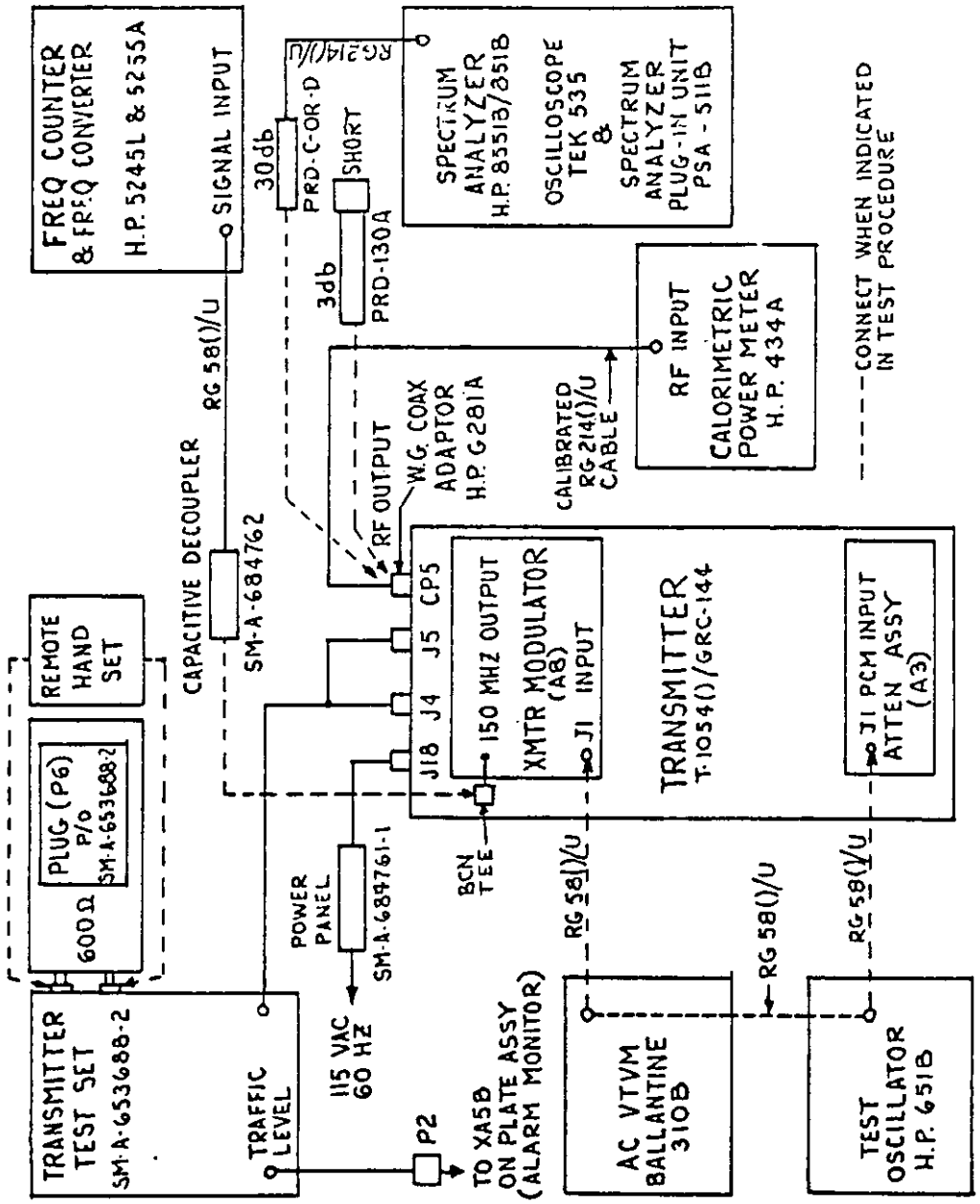


FIGURE 1

TEST SET-UP FOR
PRELIMINARY, TRANSMITTER DEVIATION, AND
POWER OUTPUT TESTS

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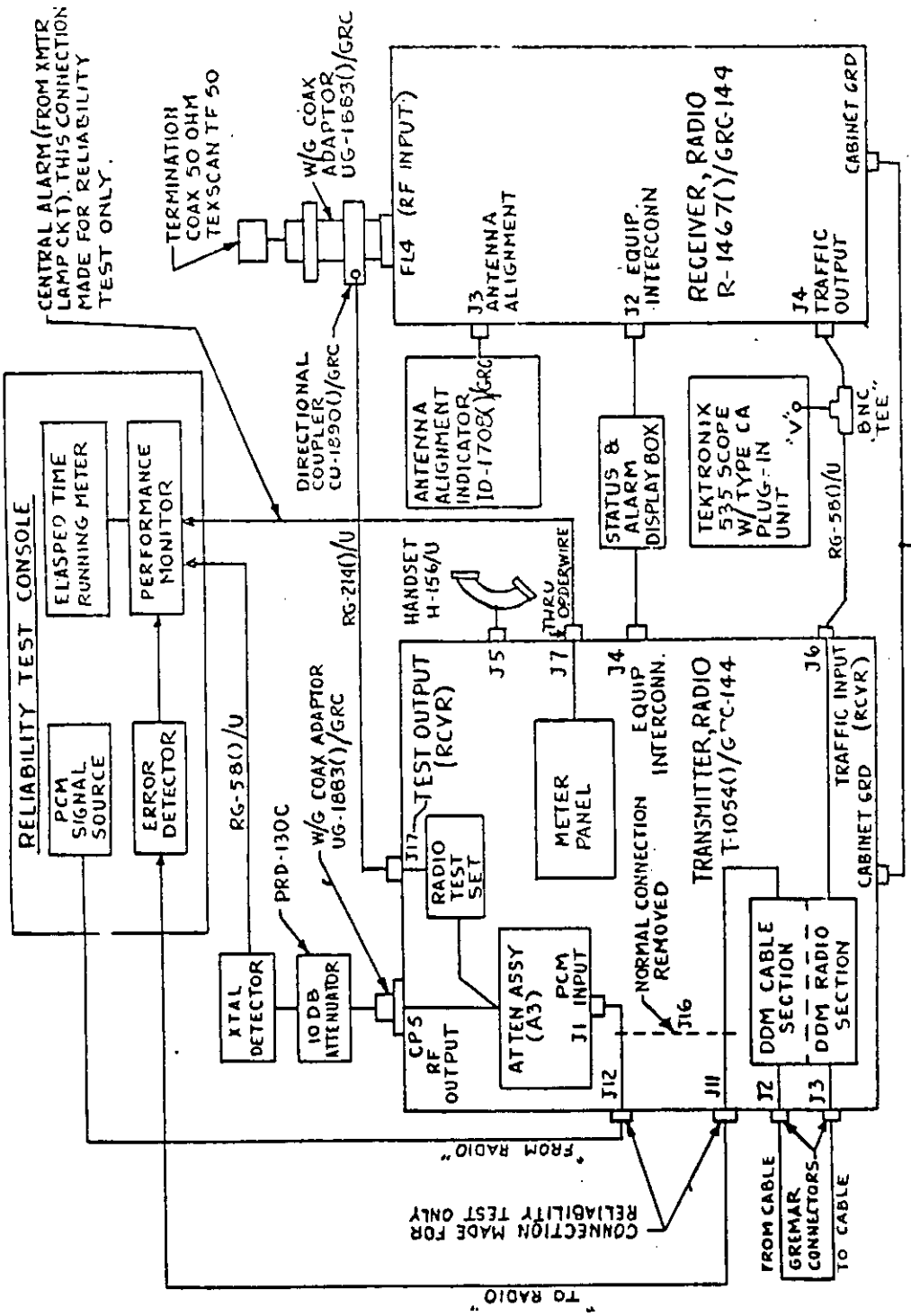


FIG. A1
RELIABILITY TEST SET-UP

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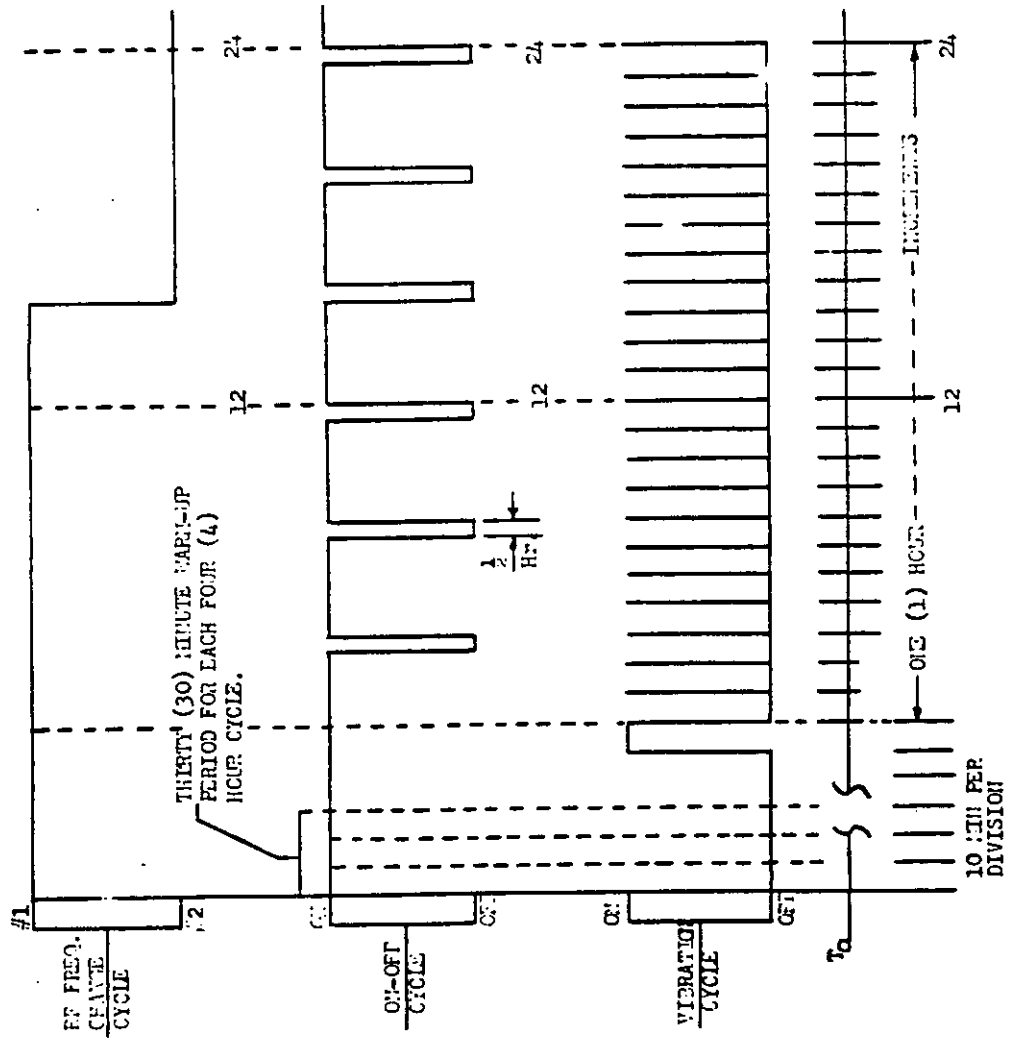


FIGURE A2
EQUIPMENT TEST CYCLE

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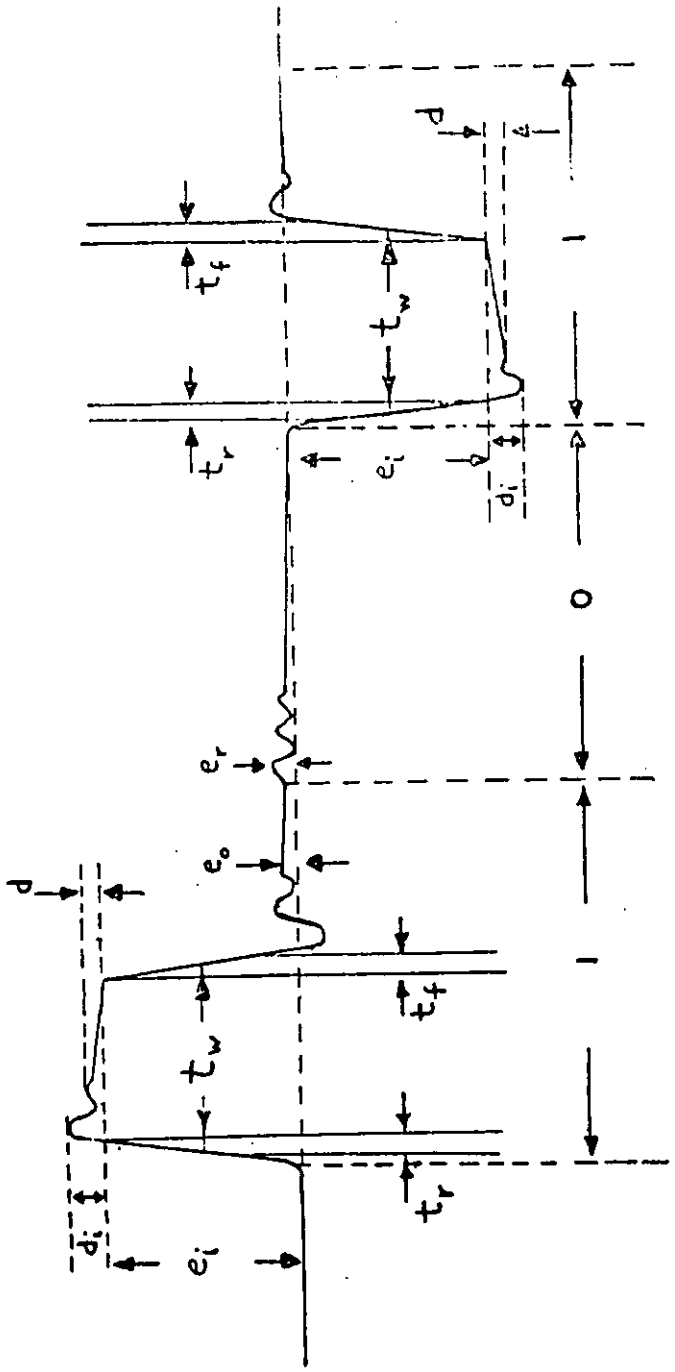


FIG. B1

1/2 BAUD BIPOLAR OUTPUT WAVE SHAPE TO CABLE FROM RADIO GRC-144, MULTIPLEXER TD-976 AND PULSE RESTORER TD-982. NO BUILD OUT NETWORK. 56 OHM LOAD

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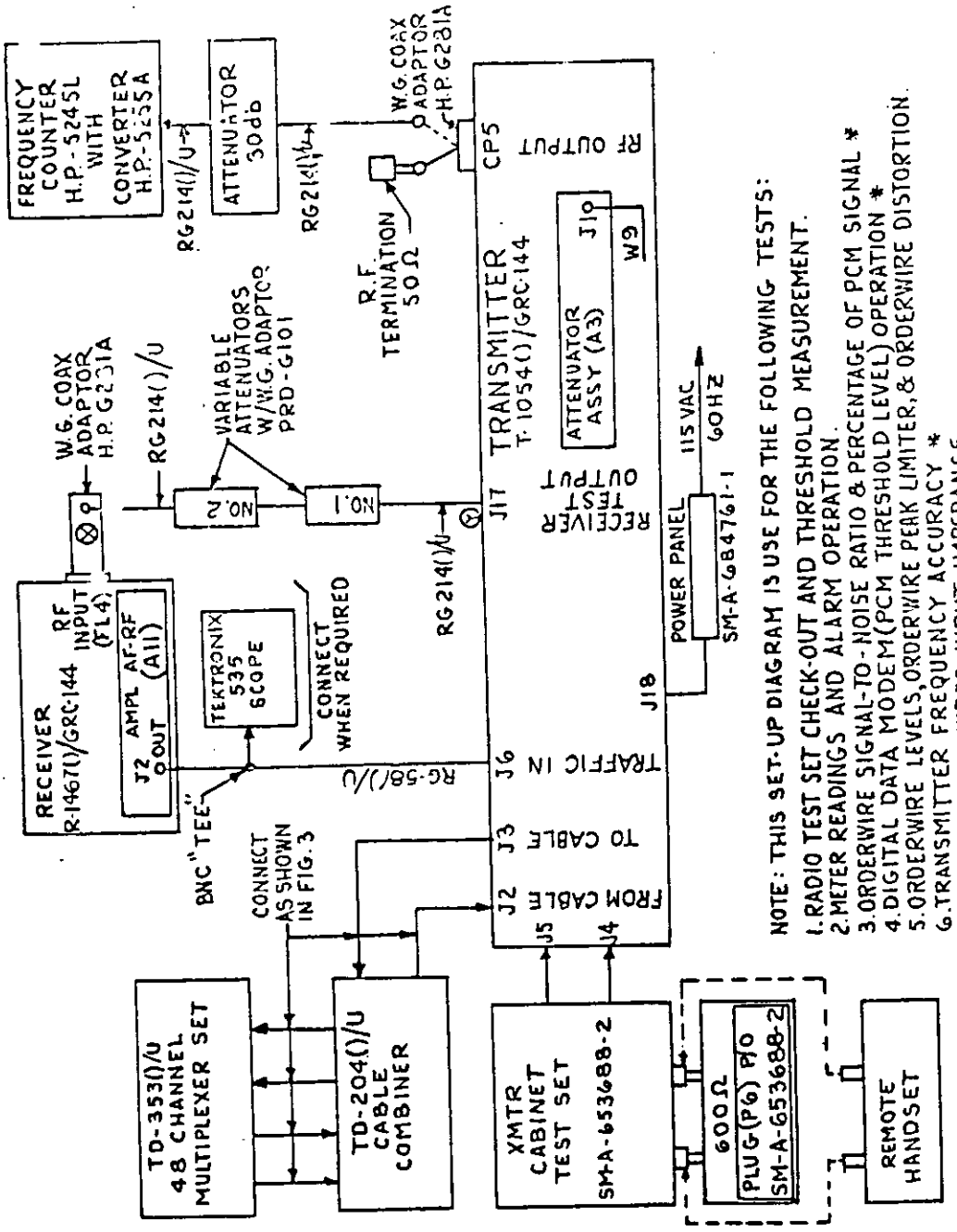


FIGURE 2
GENERAL TRANSMITTER TEST SET-UP

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NOTE: THIS SET-UP DIAGRAM IS USED FOR THE FOLLOWING TESTS:

1. RADIO TEST SET CHECK-OUT AND THRESHOLD MEASUREMENT
2. METER READINGS AND ALARM OPERATION
3. ORDERWIRE SIGNAL-TO-NOISE RATIO & PERCENTAGE OF PCM SIGNAL *
4. DIGITAL DATA MODEM (PCM THRESHOLD LEVEL) OPERATION *
5. ORDERWIRE LEVELS, ORDERWIRE PEAK LIMITER & ORDERWIRE DISTORTION

* INCLUDES SUPPLY LINE VARIATIONS

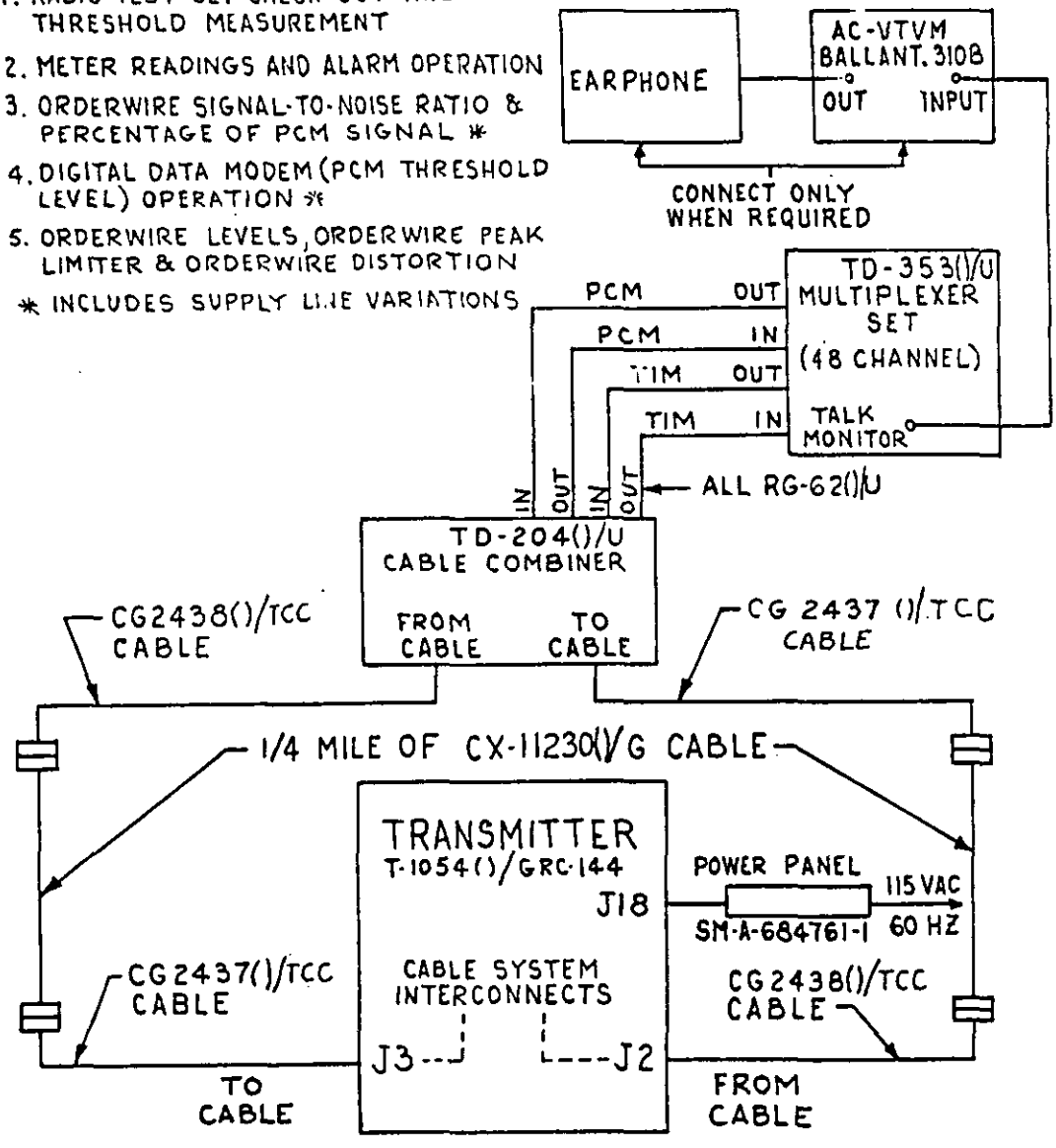


FIGURE 3

GENERAL TEST SET-UP FOR
DIGITAL DATA MODEM & ORDERWIRE

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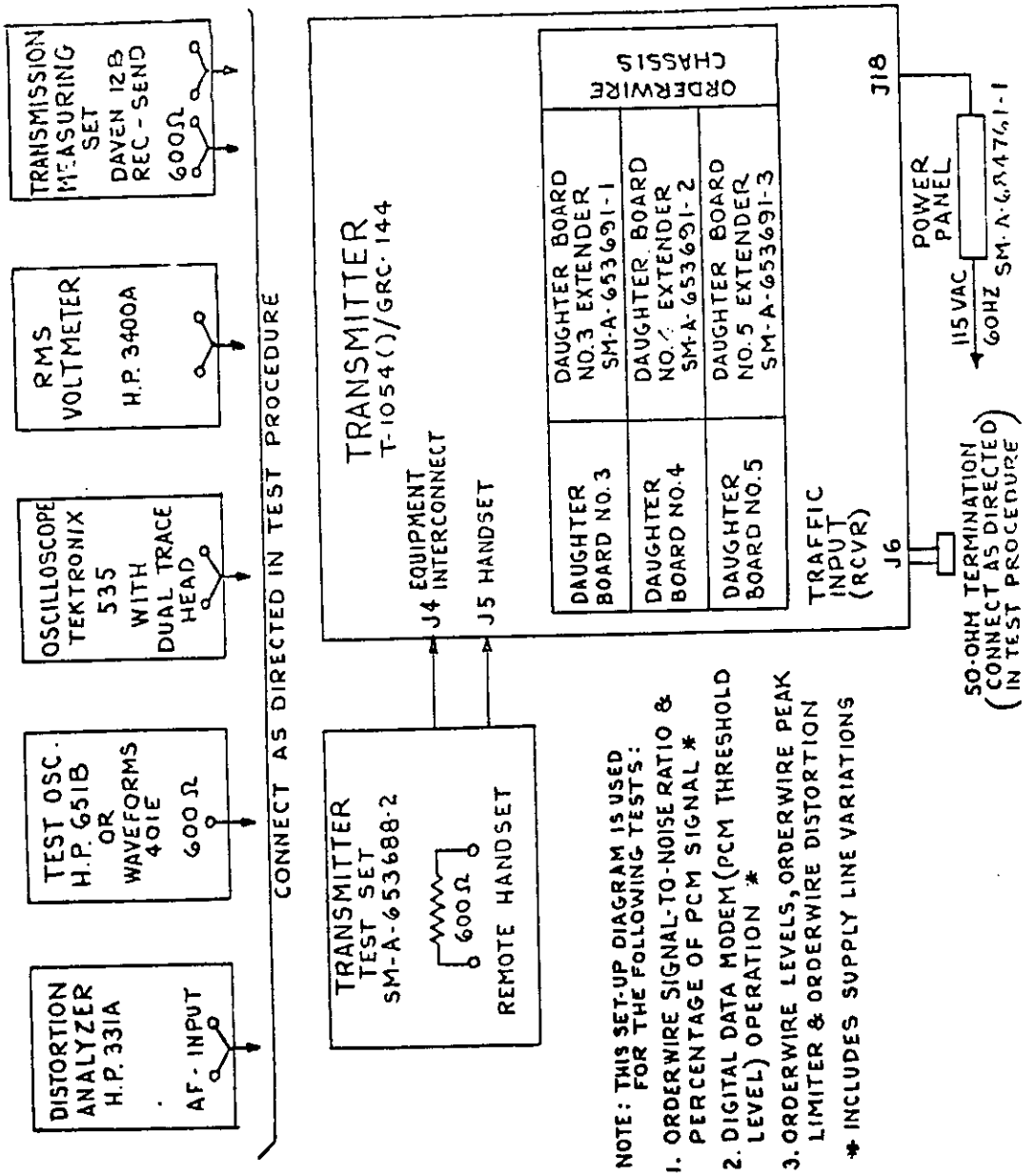


FIGURE 4
EXTENDER BOARD & TEST EQUIPMENT
SET-UP FOR
DIGITAL DATA MODEM & ORDERWIRE

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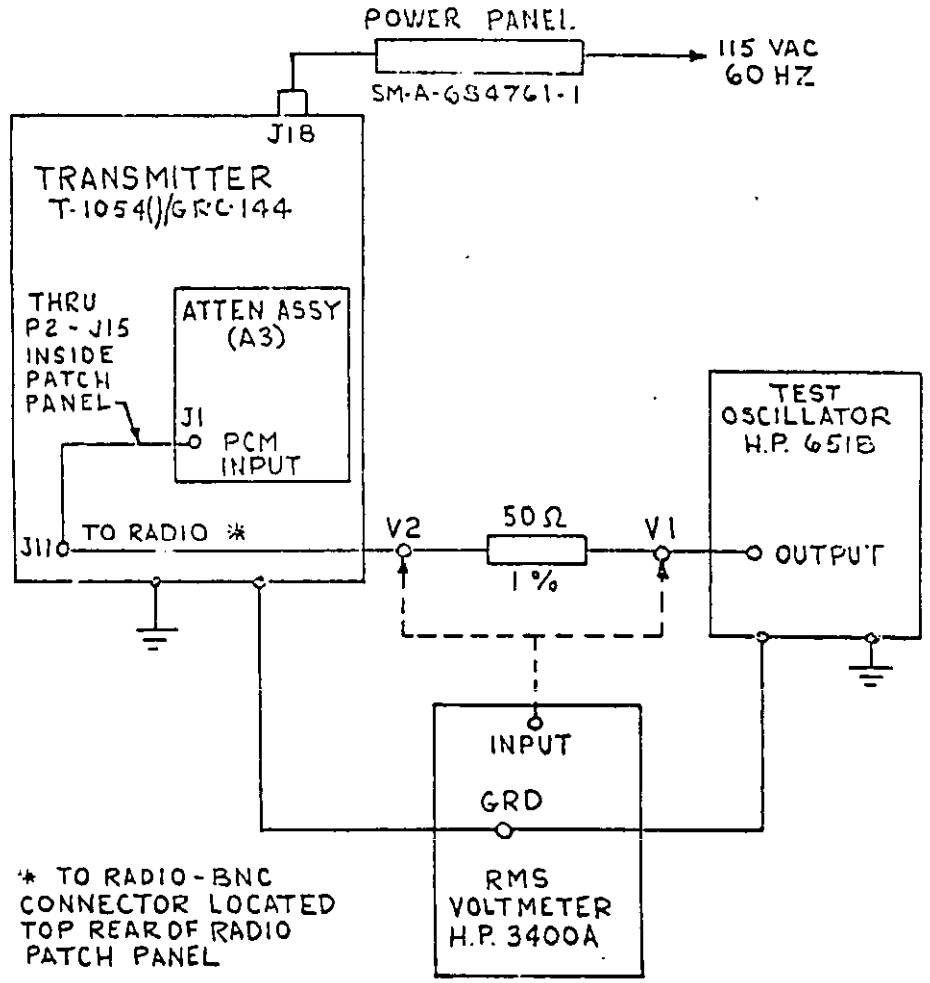


FIGURE 5
TRANSMITTER VIDEO INPUT IMPEDANCE
TEST SET-UP

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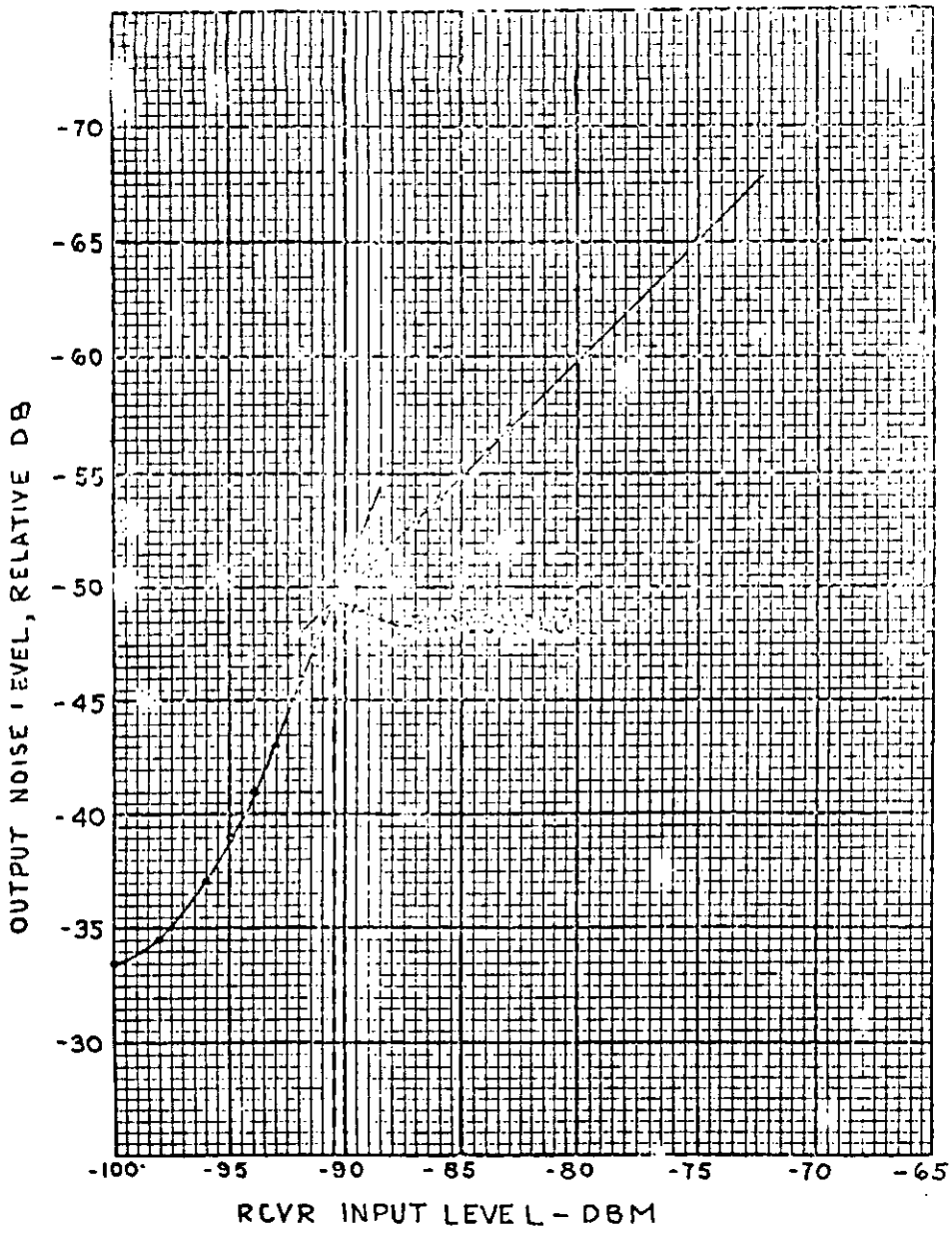


FIGURE 6
TYPICAL THRESHOLD CURVE

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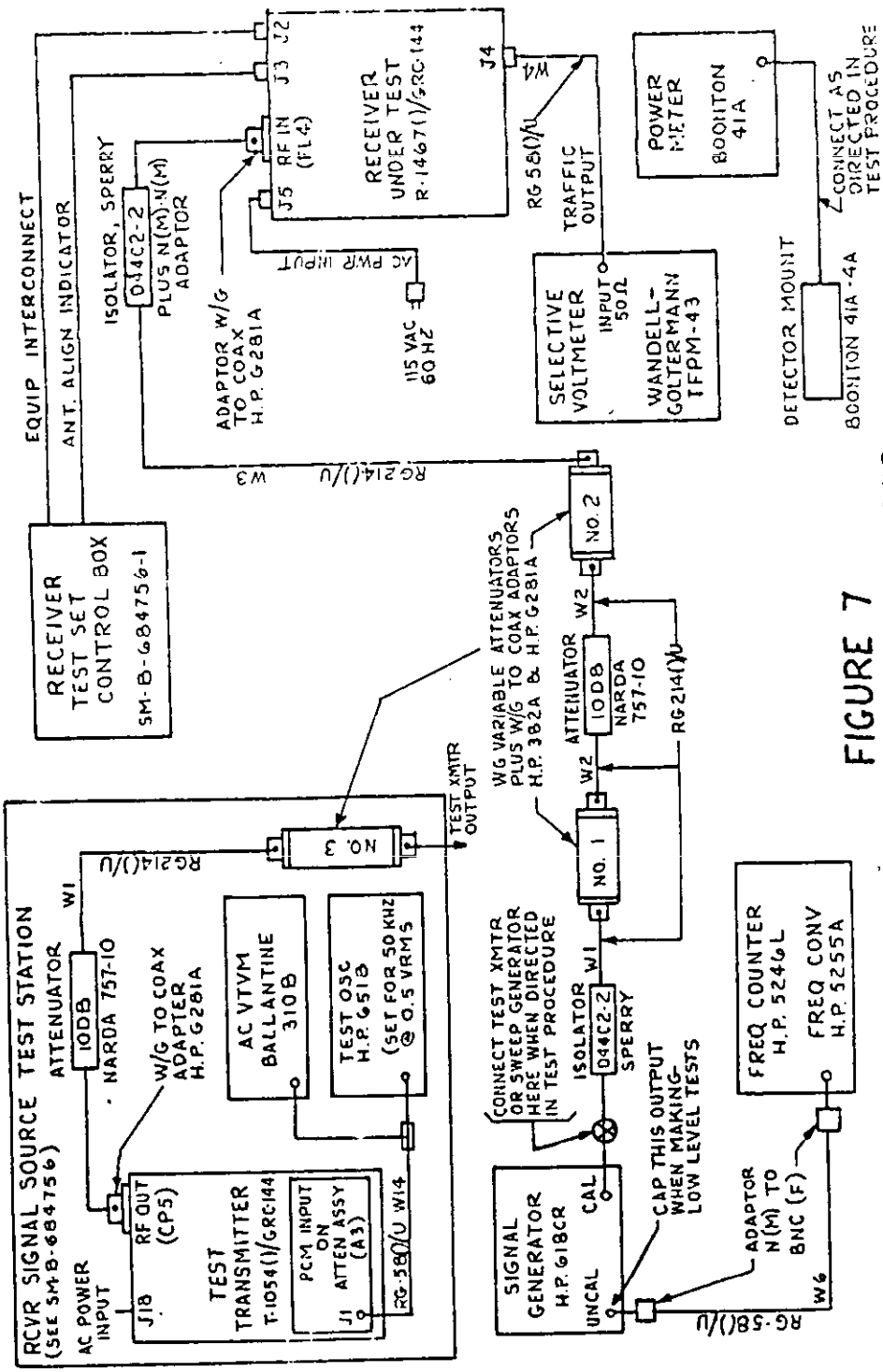


FIGURE 7
RECEIVER BASIC TEST SET-UP

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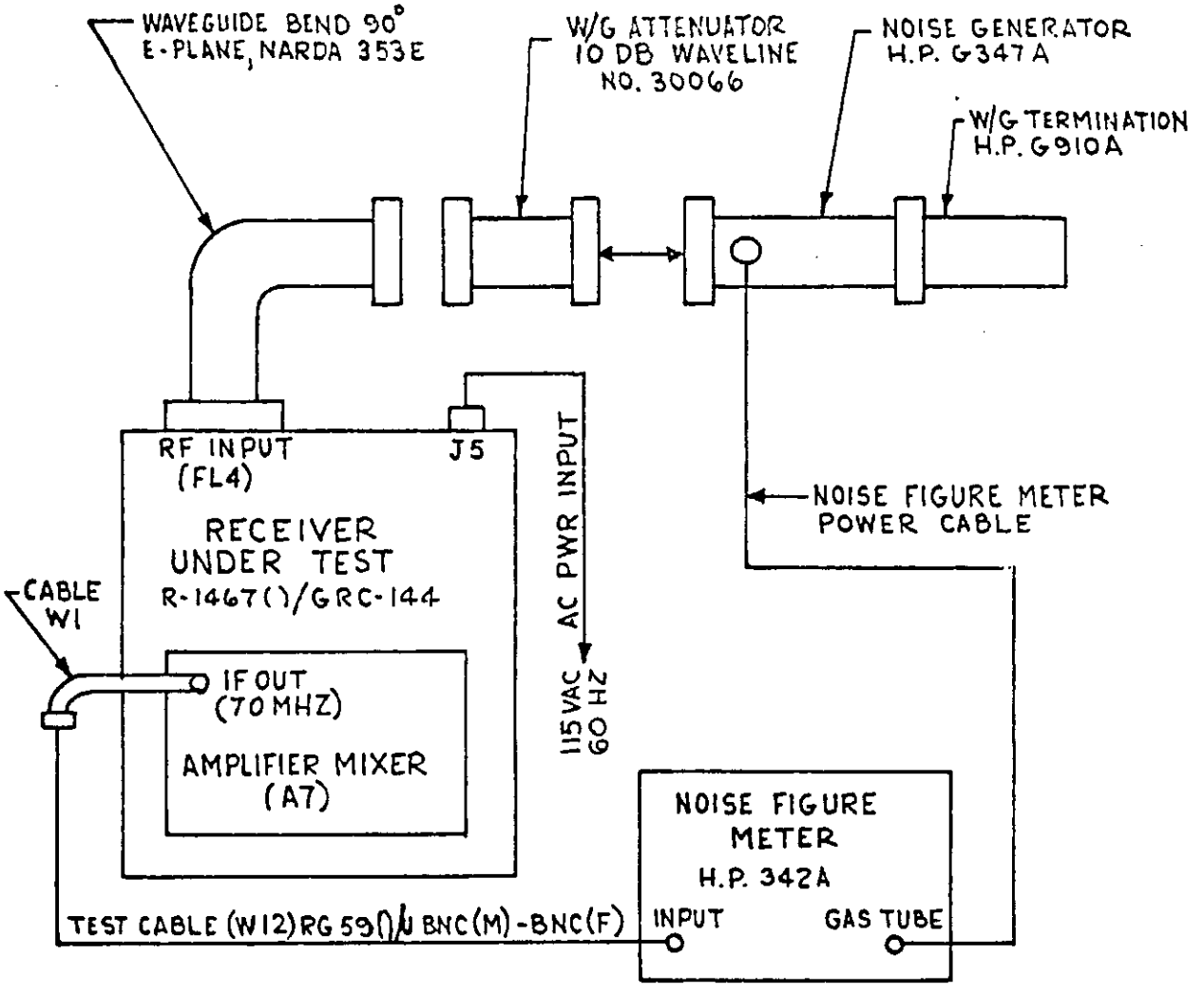


FIGURE 8
NOISE FIGURE MEASUREMENT TEST SET-UP

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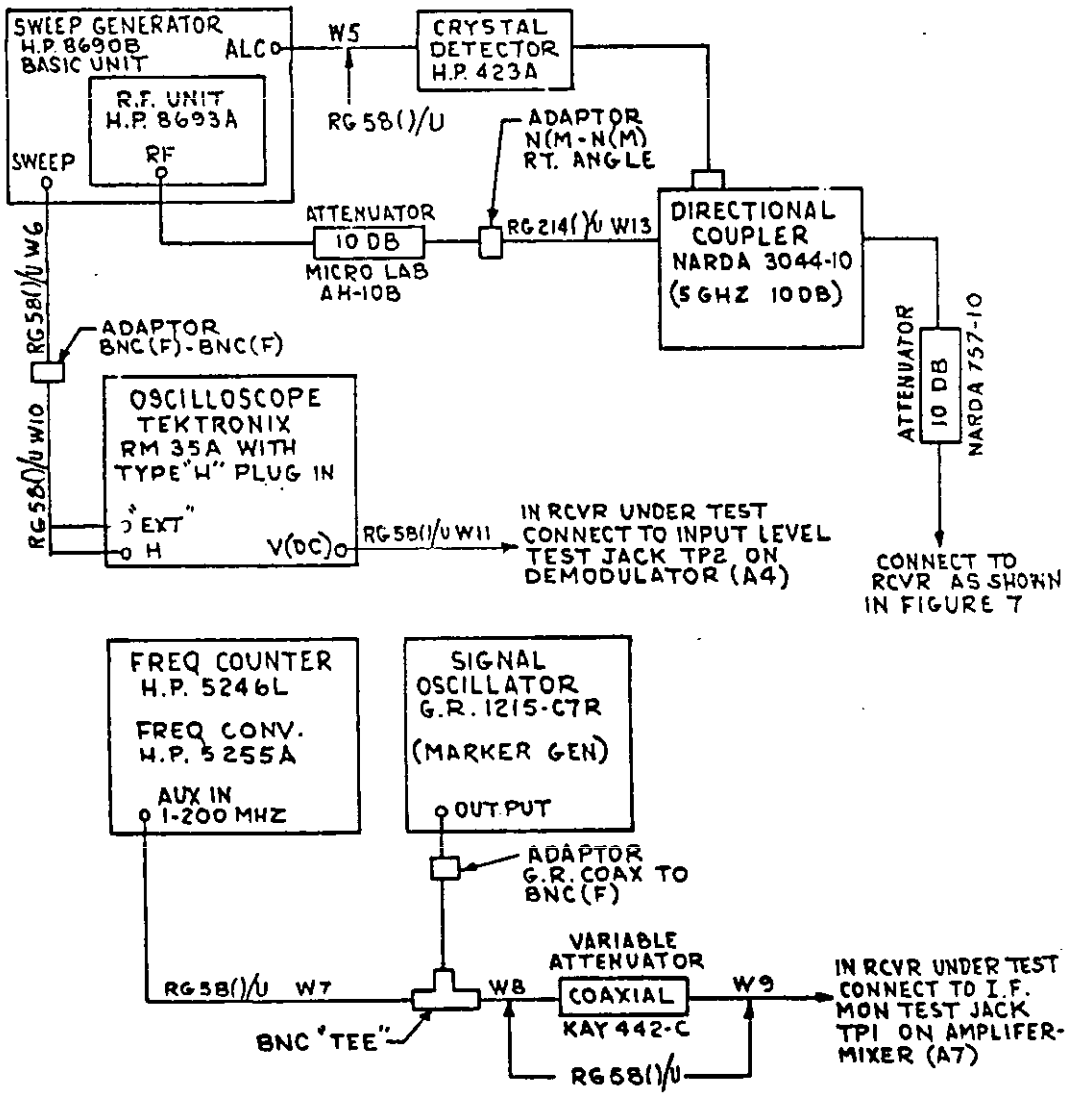


FIGURE 9
BANDWIDTH MEASUREMENT
TEST SET-UP

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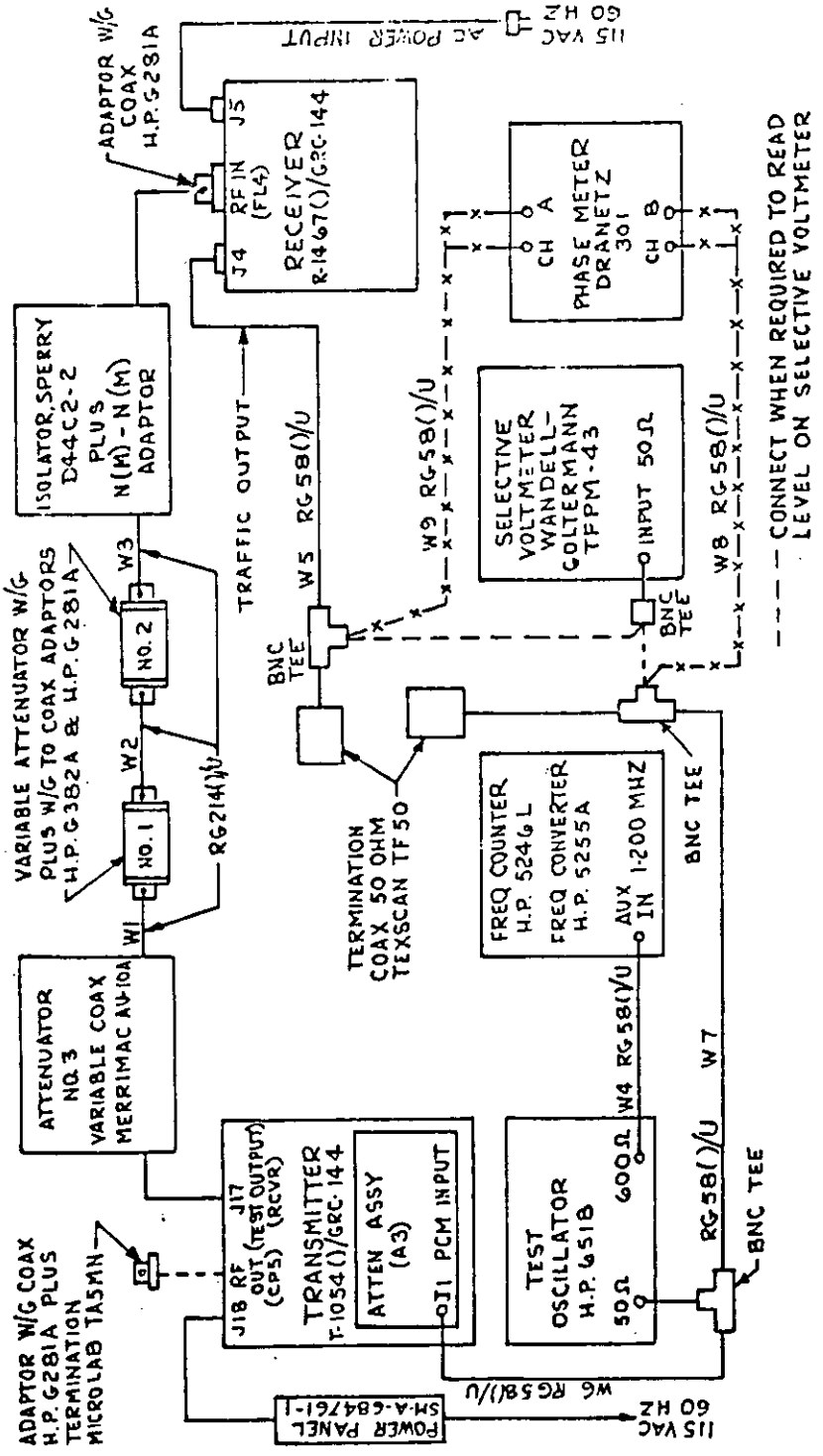


FIGURE 10
TEST SET-UP FOR
SYSTEM PCM PHASE RESPONSE,
PHASE SHIFT AND FREQUENCY
RESPONSE - 48 CHANNELS

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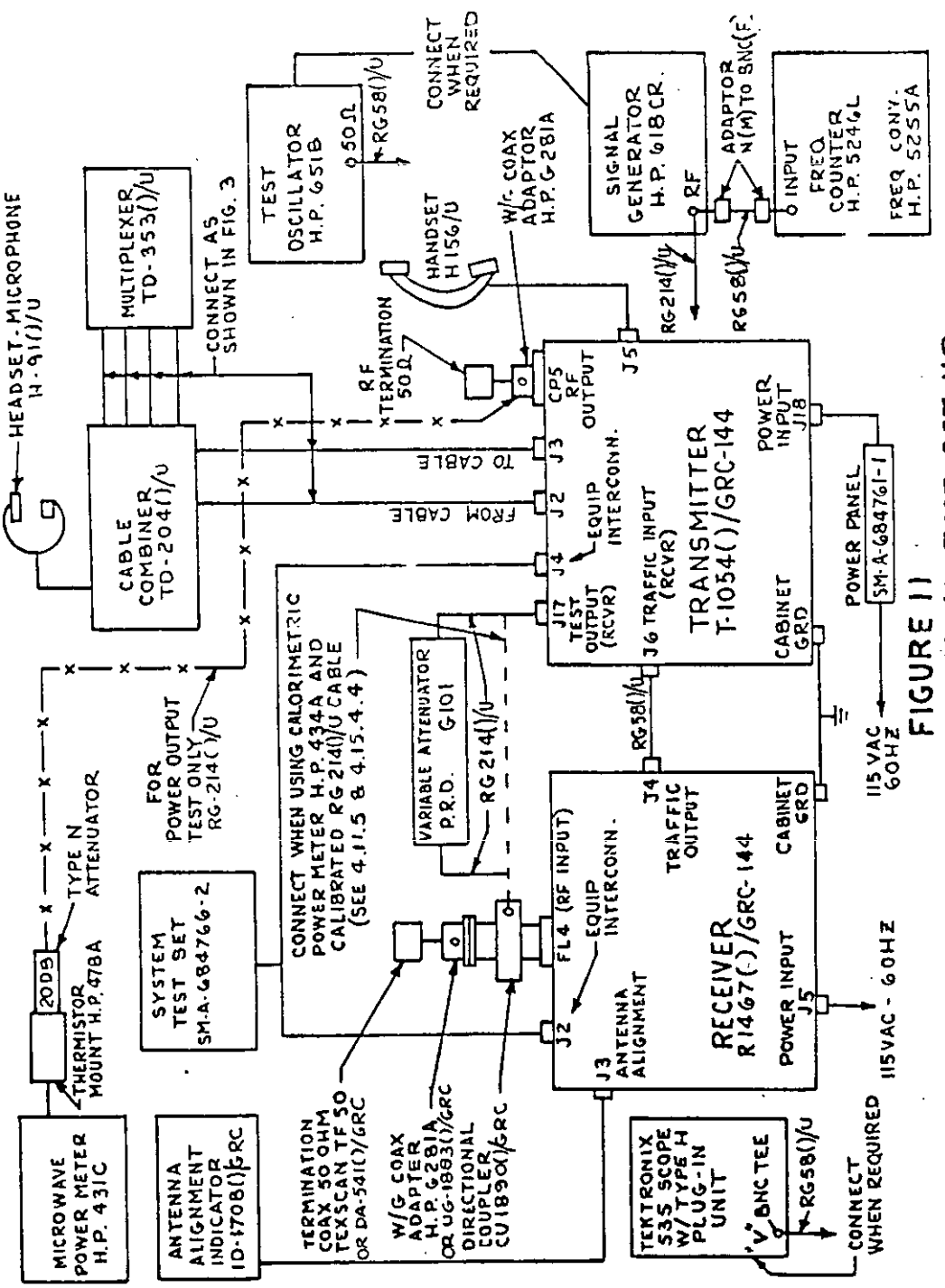


FIGURE 11
SYSTEMS OPERATIONAL TEST SET-UP

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APPENDIX A

RELIABILITY TEST REQUIREMENT FOR RADIO SET AN/GRC-144( )

10. SCOPE

10.1. Scope. This test requirement defines the general requirements, responsibilities, details, and operations necessary to conduct the reliability test program for Radio Set AN/GRC-144( ). For purposes of reliability testing, a radio set shall consist of one (1) transmitter and one (1) receiver.

10.2. Purpose. The purpose of this test requirement is to provide the information necessary to conduct the Reliability Test Program for Radio Set AN/GRC-144( ) by subjecting the equipment to an extended period of monitored operation under the conditions specified herein. The test program included accept/reject criteria for production lots.

20. APPLICABLE DOCUMENTS.

20.1 Government documents. The following publication forms a part of this document to the extent referenced herein:

MIL-STD-781 - Reliability Tests, Exponential Distribution

30. RELIABILITY REQUIREMENTS.

30.1. Test prerequisites. The test shall be conducted in accordance with Test Plan IV of MIL-STD-781. Equipment quantities selected for reliability testing shall be in accordance with paragraph 4.2.4 of MIL-STD-781.

30.1.1. Acceptance of equipment for testing. All equipment selected for reliability testing shall be from lots that have been subjected to and met Groups A and B inspections, and the following electrical inspections listed under Subgroup I of Group C Inspection (See Table V):

For Receiver R-1467( )/GRC-144:
Frequency Accuracy
Receiver Output Capability and Stability

For Transmitter/Receiver Characteristics:
System PCM Phase Response
Video Frequency Response

30.1.2. Test time indicators. Each radio set station shall be provided with an elapsed time indicator which shall accurately show accumulated test time and time of equipment failures or malfunctions.

30.1.3. Reliability electrical preconditioning. Each radio set shall have been subjected to the specified electrical preconditioning, paragraph 4.7.2. Failures that occurred during this period will be considered non-relevant.

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30.1.4. Test facility. The test facility shall be in accordance with paragraph 5.3 of MIL-STD-781 to the extent required by this test requirement. The test equipment and support facility shall be in proper operating condition and calibration prior to the commencement of testing. All test equipment must be maintained in proper calibration throughout the test. (See figure A1 for set-up of test equipment and equipment under test).

30.1.5. Test equipment required. A list of test equipment shall be supplied indicating manufacturer's type, serial number, calibration date, and calibration due date.

30.2. Test procedures. The following tests shall be performed on all units at least once during each twenty-four (24) hour period:

Electrical Tests	Insp. Para.
Receiver Meter Readings	30.7.5.1
Transmitter Meter Readings	30.7.5.2
Orderwire Performance Monitor Check	30.7.5.3
Transmitter Alarm Status	30.7.5.4
Receiver Alarm Status	30.7.5.5
Receiver Indicator Lamp Status	30.7.5.6
Transmitter Indicator Lamp Status	30.7.5.7
Transmitter Power Output Meter	30.7.5.8
Receiver Video Output Level	30.7.5.9
Orderwire Operation	30.7.5.10
Antenna Alignment Indicator Meter Reading	30.7.5.11
Operator Controls Functional	30.7.5.12

Transmitter Power Output (3.12.2), Digital Data Modem Operation (3.12.4), Transmitter alarm system (3.12.14), and Receiver Alarm System (3.13.8), shall be monitored during all periods of on-time (excluding warm-up), as further delineated in 30.4.4.

30.3. Environmental and operational stresses. The vibration and on-off cycling requirements shall be employed as stated in figure A2.

30.3.1. Temperature test condition. 40° C ± 5° C.

30.3.2. Vibration test condition. The peak acceleration value shall be 2.2G ± 10% at any non-resonant frequency between 20 and 40 Hz measured at the mounting points on the equipment. The duration of vibration shall be at least ten(10) minutes during each hour of equipment operating time.

30.3.3. ON-OFF cycle test condition. The "ON-OFF" cycle shall be in accordance with figure A2. The equipment ON-OFF cycle, vibration and RF frequency sequencing and their relationship are shown in figure A2. This sequencing shall be followed throughout the test.

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30.4. Test condition.

30.4.1. Cycle. The testing shall be conducted on a twenty-four (24) hour, seven (7) days-per-week basis. Paragraph 5.4.7 of MIL-STD-781 shall apply.

30.4.2. Test frequencies. The radio sets shall be cycled sequentially (see figure A3) in accordance with equipment test cycle (figure A2). The operating frequencies of each AN/GRC-144 transmitter and receiver shall be harmonically compatible with each other and shall be adjusted so operation of one radio set will not interfere with operation and testing of another radio set in the same test area. The reliability test plan shall define the spread of carrier frequencies at which the equipment shall be tested.

30.4.3. Test measurements. Measurements during reliability testing and criteria of satisfactory performance shall be as described in 30.7.2. The full sequence of tests specified in 30.7.5 shall be performed at least once during each twenty four (24) hour period on each radio set as soon as practical after thermal equilibrium has been reached during the "ON" period of the on-off cycling. These test measurements shall be taken at different test frequencies each day in accordance with the schedule of figure A3.

30.4.4. Monitoring. The radio sets shall be monitored to detect failures. The operating parameters of transmitter power output, error rate and central alarm shall be continuously monitored during all periods of ON time (except warm-up) so that any operational malfunction of the equipment will be detected at the time of occurrence.

30.4.5. Preventive maintenance. Preventive maintenance shall not exceed the procedures delineated under chapter 4, paragraphs 4-1 through 4-11 of TM 11-5820-695-12.

30.5. Failure recording and analysis. A failure recording and analysis system shall be implemented in accordance with paragraph 5.5 of MIL-STD-781.

30.6. Accept-reject criteria. The accept-reject criteria shall be per Test Plan IV of standard MIL-STD-781.

30.7. Performance.

30.7.1. Test prerequisite procedures.

30.7.1.1. Selection of equipment. Equipments shall be selected at random by the Government inspector from the monthly production lot. Paragraph 5.4.1 of MIL-STD-781 shall apply.

30.7.1.2. Installation of equipment. The equipment to be tested and necessary instrumentation to meet the requirements for the test shall be installed to provide for the safety of the equipment, the test facility and the personnel.

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30.7.1.3. Test-time indicator. The elapsed time indicator shall be located in the system test console and shall be energized only when the radio set is in one of the operational modes and is de-energized during the OFF and warm-up period. Test time shall be accumulated only during the operational period.

30.7.1.4. Pre-test measurements. The pre-test measurements shall be performed in accordance with the electrical test procedure of 30.7.5. Measurements shall be at any set of frequencies specified in figure A3 while the radio set is being operated in accordance with the temperature and ON-OFF cycle conditioning of 30.3. Each transmitter-receiver pair in the reliability test group shall be pre-tested on a different frequency.

30.7.2. Minimum performance limits. The performance limits for proper equipment operation shall be in accordance with the applicable paragraphs of figure A4. Relevant failures include the inability to meet the limits on power output (125 mW Min), error rate (1 in 106 maximum), and alarm circuit operation. (Operation of any alarm circuit indicating reduction in traffic carrying capability is a relevant failure. A reduction in transmitter traffic carrying capability is indicated by operation of all transmitter alarms except "RADIO TEST SET", "WAVEGUIDE SW", "RING", and "REFL RF POWER". A reduction in receiver traffic carrying capability is indicated by operation of all receiver alarms except "SPARE").

30.7.3. Test frequencies. The test frequencies used during the reliability test shall be those listed in figure A3.

30.7.4. Test equipment. The test and monitoring equipment for the implementation of this reliability program is as follows: (see note 1).

H.P. 5355B Universal Counter
H.P. 211A Square Wave Generator
H.P. 3430A Digital Voltmeter
Tektronix - Type 545A Oscilloscope
Tektronix - Type CA Plug-In Unit
Tektronix - Type 013-017 Probe
H.P. 618 Signal Generator
H.P. 5246L Frequency Counter
H.P.5253B Frequency Converter
BNC-TEE
H.P. 5255A Frequency Converter
H.P. 431C Power Meter
H.P.478A Thermistor Mount
H.P. 400D VTVM
Tektronix - Type 535 Oscilloscope
Sperry - D44C2 Isolator
FRD - 130C Attenuator
Vibration Fixture (SPCL)
BRUEL KJAEAR Power Supply Type 2801
BRUEL KJAEAR Pre-Amplifier Model 1606

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BRUEL KJAEAR Accelerometer Kit 4313
BRUEL KJAEAR Accelerometer Type 433
Control Box (Unit Power and Vibration) (SPCL)
Display Box (SPCL)
NARDA Microline Attenuator 757C
Microlab/FXR XTAL Detector XPA37
Microlab Termination TA-5 MN
Connector, Plug, Electrical (Type BNC) (J6-Transmitter) per SM-B-652910
Connector, Plug, Electrical (Type MHV) (J2-Transmitter) per SM-B-652971
Connector, Plug, Electrical (Type MHV) (J3-Transmitter) per SM-B-652972
Connector, Plug, Electrical (J7-Transmitter) per SM-B-683838-2
Connector, Receptacle, Electrical (J16-Transmitter) per SM-B-683846-1
Cable, RG-58( )/U (BNC to BNC) 20" length
Cable, RG-214( )/U (N to N) 26" length
Cable, RG-58( )/U (BNC to BNC) - for "TO RADIO" and "FROM RADIO" connections - length as required.

NOTE 1: Tolerances shall be as specified by the instrument manufacturer and substitution of equivalent instrumentation is permitted.

30.7.5. Test procedures. The following performance parameters will be measured on each equipment at least once each twenty-four (24) hour period. The performance limits, below which a failure shall be deemed to have occurred, shall be as referenced in 30.7.2. The test time for each radio set shall be measured by the elapsed time meters which record the equipment's "ON" time. The equipment shall be operated on a seven (7) day per week, twenty-four (24) hour per day basis until an accept/reject decision has been reached.

30.7.5.1. Receiver meter readings. Record all meter readings for the switch positions shown in figure A4. The switch shall be left in the OFF (TRANSIT) position when complete.

30.7.5.2. Transmitter meter readings. Record all meter readings for the switch positions shown in figure A4. Depress the OW TEST switch on the orderwire board No. 1 to obtain a reading in the OW TEST position. The metering switch shall be left in the OFF (TRANSIT) position when complete.

30.7.5.3. Orderwire performance monitor check. Adjust the display box so that orderwire alarms are functional. Rotate the MODULE TEST Switch on Daughter Board No. 1 through positions 1 to 12 and verify that the indications are as listed in figure A4. Readjust the display box to the condition which inhibits alarms due to orderwire failure, and return the module test switch to position No. 1.

30.7.5.4. Transmitter alarm status. Operation of all alarms except for RADIO TEST SET, WAVEGUIDE SW, RING and REFL RF POWER indicates a reduction in traffic carrying capability. Verify that the alarm status is as listed in figure A4.

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30.7.5.5. Receiver alarm status. Operation of all alarms except for SPARE indicates a reduction in traffic carrying capability. Verify that the alarm status is as listed in figure A4.

30.7.5.6. Receiver indicator lamp status. Verify that the indicators are as listed in figure A4.

30.7.5.7. Transmitter indicator lamp status. Verify that the indicators are as listed in figure A4.

30.7.5.8. Transmitter power out. Verify that the RF power monitored on the test console is in accordance with the output power requirement stated in figure A4.

30.7.5.9. Receiver video output level. Connect the oscilloscope to the BNC "TEE" connector at the receiver TRAFFIC OUTPUT (J4), Verify the peak to peak video output is in the range indicated in figure A4.

30.7.5.10. Orderwire operation.

a. Depress the "RING" button on the transmitter meter panel. The 1600 Hz ring tone shall appear on the scope with the PCM data superimposed: a 1600 Hz tone shall be audible at the local handset and at the transmitter speaker (adjust volume if required), and the "RING" lamp on the transmitter meter panel shall change from green to amber (it may be necessary to temporarily remove the traffic cable between the transmitter and receiver to observe the amber light).

b. Depress the push-to-talk switch on the local handset and talk or whistle into the mouthpiece. With the meter switch in the OW TEST position, the monitor meter shall indicate the orderwire modulation, and the orderwire voice signal shall appear on the scope with the PCM traffic superimposed. Return switch to OFF (TRANSIT) position and remove the scope.

30.7.5.11. Antenna alignment indicator operation. Switch the output level switch on the transmitter Radio Test Set from HIGH LEVEL to LOW LEVEL and back to HIGH LEVEL. Verify that the antenna alignment indicator meter reading decreases at LOW LEVEL and increases at HIGH LEVEL.

30.7.5.12. Frequency change.

a. Readjust the transmitter and receiver synthesizer and band pass filter controls to the new assigned frequency.

b. Vary the remaining operator controls and adjust for the proper frequency listed in figure A3.

c. Repeat and record the data for all tests, 30.7.5.1 through 30.7.5.11.

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d. At the completion of the test, insure that the monitoring circuits have been returned to the normal positions (example: reset switch on the radio set control box to off (red lamp on). Master, unit power, and test switches on the master control panel to off, and reset switch on alarm display panel to off, down.

30.7.6. Environmental and operational stresses. The environmental and operational stress facilities will be provided as follows:

30.7.6.1. Temperature test condition. A thermostatically controlled ambient temperature of 40° C ± 5° C shall be provided for the test area.

30.7.6.2. Vibration test condition. Vibration tables, located at the test area, will provide the vibration of 2.26 ± 10% at any non-resonant frequency between 20 and 40 Hz measured at the mounting points on the equipment. The vibration tables will be energized for ten (10) minutes out of each hour of "ON" time.

30.7.6.3. ON-OFF cycle test condition. The units shall be automatically programmed for the ON-OFF cycle described in figure A2.

30.7.7. Test conditions.

30.7.7.1. Cycle. The testing will be conducted on a twenty-four (24) hour, seven (7) days-per week basis to determine accept-reject criteria in conformance with 30.7.

30.7.7.2. Monitoring. The transmitter power output (3.12.2), digital data modem (3.12.4), and traffic status alarm (3.12.14 and 3.13.8) for the transmitter and receiver, respectively, shall be automatically monitored. When a failure of one of the above parameters occurs an alarm system will be activated which will turn off the vibration equipment and the elapsed operating time meter.

30.7.7.3. Maintenance measures to be performed. No adjustment or repair will be made on the radio set except that necessary to enable a failed radio set to perform as specified.

30.7.8.1. Operating time log. An operating time log shall be maintained for each unit starting with the initial receipt of the radio set and continuing through the completion of testing. The following data shall be recorded:

Elapsed time meter reading
Operate time
Accumulated time
Name of test or operation
Failure report number (if applicable)
Signature of operator

These entries will provide the necessary elapsed time information, the chronology of the test sequences and a record of all test interruptions. In

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addition, the operator shall enter the reason for any test interruption and any significant events or notes applicable to the period of operation.

30.7.8.2. Failure tag. Paragraph 5.10.4 of MIL-STD-781 shall apply.

30.7.8.3. Failure identification report. Paragraph 5.10.5 of MIL-STD-781 shall apply.

30.7.9. Restoration of failed equipments. Paragraph 5.9 of MIL-STD-781 shall apply.

30.8. Test reports and failure reporting and analysis. Test reports, failure reporting and failure analysis shall be as specified in the contract.

30.9. Frequency Plan. The transmitter and receiver frequencies shall be changed twice each day. When changing frequencies, Station No. 1 frequencies are changed to that of Station No. 2, and Station No. 2 frequencies become the new frequencies for Station No. 1. The same exchange of frequencies takes place for Stations No. 3 and No. 4.

	STATION NO. 1		STATION NO. 2	
	XMTR	RCVR	XMTR	RCVR
1	4400.0	4500.0	4700.0	4800.0
2	4815.5	4919.5	4800.5	4900.5
3	4410.9	4510.9	4710.9	4810.9
4	4825.5	4925.5	4810.5	4910.5
5	4420.0	4520.0	4720.0	4820.0
6	4850.2	4950.2	4835.2	4935.2
7	4500.0	4600.0	4800.0	4900.0
8	4800.0	4900.0	4900.5	5000.0
9	4700.9	4800.9	4710.9	4810.9
10	4810.0	4910.0	4410.0	4510.0
11	5000.0	4900.0	4500.0	4400.0
12	4933.3	4833.3	4633.3	4533.3
13	4511.7	4411.7	4988.7	4888.7
14	4522.2	4422.2	4999.2	4899.2
15	4744.4	4844.4	4444.4	4544.4
16	4666.0	4566.0	4555.0	4455.0
17	4488.8	4588.8	4988.8	4888.8
18	4625.0	4525.0	4475.0	4575.0
19	4900.4	5000.4	4600.4	4500.4
20	4475.0	4575.0	4950.0	4850.0

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APPENDIX A

FIGURE A3. Frequency plan.

	STATION NO. 3		STATION NO. 4	
	XMTR	RCVR	XMTR	RCVR
1	4750.0	4850.0	4900.0	5000.0
2	4460.5	4560.5	4410.5	4510.5
3	4760.9	4860.9	4845.9	4945.9
4	4570.5	4670.5	4420.5	4520.5
5	4760.0	4860.0	4865.0	4965.0
6	4595.2	4695.2	4445.2	4545.2
7	4850.0	4950.0	4825.0	4925.0
8	4750.5	4850.5	4700.5	4800.0
9	4750.9	4850.9	4790.9	4890.9
10	4430.0	4530.0	4450.0	4550.0
11	4550.0	4450.0	4510.0	4410.0
12	4566.3	4466.3	4522.3	4422.3
13	5000.7	4900.7	4966.7	4866.7
14	4988.2	4888.2	4933.2	4833.2
15	4422.4	4522.4	4912.4	4812.4
16	4417.0	4517.0	4999.0	4899.0
17	4966.8	4866.8	4877.8	4777.8
18	4600.0	4500.0	5000.0	4900.0
19	4525.4	4625.4	4944.4	4844.4
20	4825.0	4925.0	4560.0	4660.0

NOTE: This figure (A3) constitutes a scheme for operation of four (4) Radio Sets under test at the same time. The test plan to be submitted by the contractor for government approval shall delineate the actual test frequencies proposed for use.

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FIGURE A4. Reliability Test Limits.

Receiver Meter Readings	Normal
OFF (TRANSIT)	0 SYNTH
30 - Full Scale	
FREQ. MIXER CURRENT 1	30 - Full Scale
FREQ. MIXER CURRENT 2	30 - Full Scale
MULT. OSC	80 - 120
AGC	No Limit
AMPL - MIXER CURRENT 1	30 - Full Scale
AMPL - MIXER CURRENT 2	30 - Full Scale
CARRIER (IF)	No Limit
TRAFFIC	90 - 110
LOCAL OSC. POWER	30 - Full Scale

Transmitter Meter Readings	Normal
OFF (TRANSIT)	0
OW TEST	90 - 110
MODULATOR CONTROL	80 - 120
MODULATOR OUTPUT	80 - 120
FREQ. CONTROL-CTR FREQ	20 - 0 - 20
FREQ. CONTROL-OUTPUT	80 - 120
TRAFFIC	90 - 110
SYNTHESIZER	30 - Full Scale
AMPL-MULT	30 - Full Scale
2ND MULT	30 - Full Scale
3RD MULT (LOCAL OSC)	30 - Full Scale
RF POWER	30 - Full Scale
REFL RF POWER	0 - 30

Radio Test Set	
OFF-TRANSIT	0
OSC. LEVEL	SET OSC LEVEL ADJ
CR-1	YELLOW RANGE
CR-2	YELLOW RANGE

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FIGURE A4. Reliability Test Limits - Continued.

Receiver Meter Readings		Normal	
ORDERWIRE BOARD NO. 1	BOARD NO. 1 MODULE	CENTRAL	SPEAKER
MODULE TEST SWITCH POSITION	TEST INDICATOR	ALARM	OUTPUT
1	Off	Green	None
1 DEPRESS MODULE TEST LAMP	On (Red)	Green	None
2	Off	Green	4 kHz
3	Off	Green	4 kHz
4	Off	Green	4 kHz
5	Off	Green	4 kHz
6	Off	Green	4 kHz
7	Off	Green	4 kHz
8	Off	Green	4 kHz
9	Off	Green	4 kHz
10	Off	Green	1.6 kHz
11	Off	Green	1.1 kHz
12	Off	Red	1.1/1.6 kHz
12 DEPRESS XMTR RESET	On (Red)	Red	None

Transmitter Alarm Status	Normal
RADIO TEST SET	RED
WAVE GUIDE SW	GREEN
RING	GREEN
CNTRL ALARM	GREEN
TRAFFIC	GREEN
RF POWER	GREEN
SYNTH LOCK	GREEN
REFL RF POWER	GREEN
CABLE TO RADIO MODEM	GREEN
RADIO TO CABLE MODEM	GREEN
REMOTE TRAFFIC IN	ON - GREEN
REMOTE TRAFFIC OUT	ON - GREEN

Receiver Alarm Status	Normal
CARR (IF)	GREEN
SPARE	GREEN
SYNTH LOCK	GREEN
TRAF	GREEN

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FIGURE A4. Reliability Test Limits - Continued.

Receiver Indicator Lamp Status	Normal
Oven TDA	Cycling-On-Off
Oven Synthesizer	On-White
Power Supply DS1 Thru DS4	On-Green
Synthesizer 28V	On-Green
Synthesizer 5V	On-Green
Synth Modules (6)	Off
Synth Modules (6) Depress Test Switch	On-Red
Transmitter Indicator Lamp Status	Normal
Power Supply DS1 Thru 4, 6 Thru 12	On-Green
Synthesizer 28V	On-Green
Synthesizer 5V	On-Green
Synth Modules (6)	Off
Synth Modules (6) Depress Test Switch	On-Red
Transmitter Power Output	125 mW Minimum
Receiver Video Output Level (Volts P-P)	.71 to 1.41
Depress Ring Button	1600 Hz on Scope and Speaker, Ring Amber
Talk At Handset	Voice on Scope, OW Test Meter Moves
Antenna Alignment Indicator Varies with Level	
All Operator Controls Are Functional	

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APPENDIX B

RECONFIGURATION KIT REQUIREMENTS FOR ASYNCHRONOUS DIGITAL COMBINING (ADC) MODE OF OPERATION FOR RADIO SET AN/GRC-144( )

10. SCOPE

10.1. Scope. This appendix defines the ADM Reconfiguration Kit used to configure Radio Set AN/GRC-144( ) to provide 48/96 channel asynchronous digital data modem (ADDM) operation with an asynchronous digital combining (ADC) system. This Appendix is a mandatory part of this specification. The information contained herein is intended for compliance.

10.2. Purpose. The purpose of this appendix is to provide general information, specify requirements, define examinations, and tests for the ADM Reconfiguration Kits.

10.3. Cross-Reference. This appendix is cross-referenced to existing paragraphs of this specification and Appendix A thereto. Additions and revisions to requirements and test paragraphs are stated where applicable in order to adequately define the asynchronous data modem (ADM) modules when installed in Radio Set AN/GRC-144( ).

20. APPLICABLE DOCUMENTS.

20.1 Government drawings. The following publications form a part of this document to the extent referenced herein:

GL-SM-B-404159	
GL-SM-B-404179	
GL-SM-B-404187	
GL-SM-B-404195	
GL-SM-B-405816	
SM-B-650580	Oscillator, Voltage Controlled (4.608 MHz)
DL-SM-B-653326	Regenerator, Digital, Cable
DL-SM-B-653334	Alarm-Monitor
SM-B-683698	Oscillator, Voltage Controlled (9.830400 MHz)
SM-B-684088	Dummy Panel
SM-A-684333	Crystal Unit, Quartz (4.60800 MHz)
SM-A-684334	Crystal Unit, Quartz (4.91520 MHz)
DL-SM-B-684639	Control-Comparator Cable
DL-SM-B-684642	Processor, Digital, Radio
DL-SM-B-684644	Regenerator, Digital, Radio
DL-SM-B-684645	Regenerator, Digital, Radio
DL-SM-B-684646	Control-Comparator, Cable
DL-SM-B-684647	Alarm-Monitor
DL-SM-B-684648	Processor, Digital, Radio
DL-SM-B-684658	Filter, Low Pass (1.0 MHz)
DL-SM-B-684659	Filter, Low Pass (3.8 MHz)

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APPENDIX B

30. REQUIREMENTS.

30.1. Applicability of requirements and test procedures. All requirements stated in Section 3 and all tests and examinations in Section 4 of this specification apply to the ADDM mode of operation (48 and 96 channels) except as noted herein.

30.1.1. Special test equipment. All test procedures (including figures 2, 3, and 11) in this document (MIL-R-55333(CR)) describe use of TD-204( )/U Cable Combiner and TD-353( )/U Multiplexer Set for DDM mode of operation. For ADDM mode testing, these items shall be replaced by TD-660( )/G and TD-976( )/G Multiplexer Sets. In addition, TD-754( )/G Multiplexer and TD-982( )/G Pulse Restorer are required. If these items are not available as Government Loaned Equipment (GLE) for testing purposes, the supplier shall provide alternate ADDM test equipment (multiplexer simulators) capable of reproducing those signals required by the individual test procedures (see 50. of this Appendix).

30.2. DDM Mode to ADDM Mode changeover procedure for 48 channel data rate. The transmitter and the receiver of the AN/GRC-144( ) are pressured to be initially configured in the 48 channel synchronous mode of operation, usually referred to as the DDM Mode, therefore the following steps shall be accomplished prior to operation and testing of the AN/GRC-144( ) in the ADDM Mode with the ADDM modules:

30.2.1. Prepare the Cable Digital Regenerator module (1A12A2) and Radio Digital Processor module (1A12A10) for use by placing Cable Length Jumper Plug (P2) in the 1/4 mile or 1/2 mile position on the Cable Length connector as required.

30.2.2. Remove the following modules from the Digital Data Modem (1A12) in Transmitter T-1054( )/GRC-144:

Oscillator, Voltage Controlled (4.608), (1A12A1 and 1A12A11) per SM-B-650580
Regenerator, Digital, Cable (1A12A2) per DL-SM-B-653326
Control-Comparator, Cable (1A12A3) per DL-SM-B-684639
Alarm-Monitor (1A12A5) per DL-SM-B-653334
Processor, Digital, Radio (1A12A10) per DL-SM-B-684642

30.2.3. Install the following modules in the open slots in the Digital Data Modem (1A12):

Dummy Panel (1A12A11) per SM-B-684088
Regenerator, Digital Cable (1A12A2) per DL-SM-B-684645
Control-Comparator, Cable (1A12A3) per DL-SM-B-684646
Alarm-Monitor (1A12A5) per DL-SM-B-684647
Processor, Digital, Radio (1A12A10) per DL-SM-B-684648
Oscillator, Voltage Controlled (9.830400 MHz) (1A12A11) per SM-B-683698

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30.2.4. Using the card extractor tool, remove the DL-SM-B-684644 Regenerator, Digital, Radio module (1A12A8) from the Modem chassis. Remove crystal unit (4.60800 MHz) (Y1), per SM-A-684333 and replace with crystal unit (4.91520 MHz) (Y1), per SM-A-684334. Next, reinsert the 1A12A8 module into its proper slot.

30.2.5. On ATTN ASSY (1A3) place the 48/96 CH selector switch in the "48 CH" position.

30.2.6. On the modem chassis, place the ADDM TRAFFIC SELECT 48 CHAN/96 CHAN switch in the "48 CHAN" position.

30.3. Changeover procedure to ADDM 96 channel data rate. If changeover is from DDM Mode to ADDM Mode at the 96 Channel Data Rate, perform all steps from 30.3.1 through 30.3.4 of the following procedure. To change from 48 Channel Data Rate to 96 Channel Data Rate when the radio set is already in the ADDM Mode, follow only steps 30.3.2 through 30.3.4 below.

30.3.1. Perform steps 30.2.1 through 30.2.4 above.

30.3.2. Remove Filter, Low Pass, 1.0 MHz (2A3) per DL-SM-B-684658 from Receiver R-1467( )/GRC-144( ) and replace with Filter, Low Pass, 3.8 MHz (2A3) per DL-SM-B-684659.

30.3.3. On ATTEN ASSY (1A3) in Transmitter T-1054( )/GRC-144, place the 48/96 CH selector switch in the "96 CHAN" position.

30.3.4. On the modem chassis (1A12), place the ADDM TRAFFIC SELECT 48 CHAN/96 CHAN switch in the "96 CHAN" position.

40. ADDM requirements and test procedures. The following data are revisions or changes which cross-reference directly to existing paragraphs of this specification and are further expressed by parenthetical notations.

40.1.1. Digital data modem (PCM threshold level) operation (see 3.12.4). The transmitters and receivers operating in both 48 and 96 channel ADDM modes in a closed loop through the cable and radio combiner (data modem in transmitter) shall have a bit error rate of less than 1 in 10^6 (see 4.11.7).

40.1.2. Carrier level threshold (48 CHNL), (see 3.13.4). Add the following: This requirement also applies to the 48 and 96 channel ADDM modes of operation.

40.1.3. Carrier level threshold with supply line variations (see 3.13.5). Revise end of last sentence to read as follows: shall be met for 48 channel DDM and 48 and 96 channel ADDM modes: (See 4.12.8).

40.1.4. Video frequency response (see 3.14.2 and 4.13.4). Add the following column of eight (8) frequencies to coincide with the 8 responses:

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APPENDIX B

96 CHANNEL
50
950
1330
1900
2660
3800
5320
6800

40.1.5. Operational test (radio set) (see 3.17 and 4.15). Add the following sentence: The radio set shall be operated in a 48 channel DDM or 48 channel ADDM mode depending on the configuration being tested at the time.

40.1.6. Technical literature, tools and running spare parts (see 3.18). Add the following: Literature for the ADDM mode of operation shall be furnished as specified in the contract.

40.1.7. First article inspection (see 4.3). Add new sub-paragraphs "d" and "e" under NOTES as follows:

d. When the ADDM modules are procured at the same time as the DDM modules with the end item, the Temperature and Humidity tests do not necessarily require installation of ADDM modules in separate, complete cabinets. As a minimum, these modules shall be operationally tested in the end item radio set before and after temperature and humidity cycling. The ADDM modules may be mounted in a separate, non-cabinet mounted, modem sub-assembly which is placed in the test chamber near the test cabinets. Likewise, the fungus test need not be on separate, complete cabinets. It shall be sufficient to place a DDM configured radio set in the fungus test chamber with a second transmitter cabinet (or simulated cabinet) which houses an ADDM sub-assembly only.

e. Resonance-search, bounce-vehicular, altitude, and EMI testing shall be performed on cabinets outfitted in the DDM mode only. If only ADDM configured radios are being procured, then all the testing will be done with ADDM modules in place in the cabinets.

40.1.8. Group A inspection (see 4.5.1). The following shall be added to Table III - (Group A Inspection, visual and mechanical) at the end of the items listed under Modem, Digital Data Chassis Assembly for Transmitter, Radio T-1054( )/GRC-144:

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Dummy Panel	3.19 - 4.10 - III-1
Regenerator, Digital, Cable	3.19 - 4.10 - III-1
Control-Comparator, Cable	3.19 - 4.10 - IV-1
Alarm-Monitor	3.19 - 4.10 - III-1
Processor, Digital, Radio	3.19 - 4.10 - III-1
Oscillator, Voltage Controlled (9.830400-MHz)	3.19 - 4.10 - III-1
Crystal Unit (4.91520 MHz)	3.19 - 4.10 - III-1

The following shall be added to Table III - (Group A Inspection, Visual and Mechanical) at the end of the items listed under Receiver, Radio R-1467()/GRC-144:

Filter, Low Pass, 3.8 MHz	3.19 - 4.10 - III-1

40.1.9. Subgroup II (see 4.5.3.1.2). Add the following sentence: Tests shall be performed alternately on DDM and ADDM configured radio sets. (First Article Tests (see 4.3) shall be considered as first configuration).

40.1.10. Subgroup III (see 4.5.3.1.3). Add the following sentence: Tests shall be performed on radio sets configured in the ADDM mode (which were note tested during First Article).

40.1.11. Sampling for reliability testing of equipment (see 4.5.5.1). Add the following sentences: When the ADDM kit hardware is being supplied concurrently with a basic DDM configured radio, the reliability selection will be made using non-ADDM configured equipment. When only ADDM mode radios are being procured, reliability testing shall be completed for the 48 ch mode only.

40.1.12. Electrical preconditioning (burn-in) (see 4.7.2). Add the following sentence: When the ADDM modules are being supplied concurrently with the basic DDM configured radio, the ADDM modules listed in 40.1.8, Appendix B, with the exception of the Dummy Panel and Filter, Low Pass, 3.8 MHz shall be preconditioned in the same manner as the spare modules (see 4.7.2.7).

40.1.13. Spare Modules (see 4.7.2.7). Add the following sentence: The ADDM kit hardware, except the Dummy Panel and the 3.8 MHz Low Pass Filter, when supplied concurrently with DDM configured radios, shall be tested in a manner similar to that used for spare modules.

40.1.14. Fungus test (see 4.8.5). Add the following sentences: When the ADDM kit hardware is being supplied concurrently with basic DDM configured radios, the ADDM modules shall be mounted in an extra modem subassembly, as a minimum, which is placed in an otherwise empty transmitter cabinet (or simulated cabinet) inside the test chamber. Note that the 3.8 MHz Low Pass Filter is mounted in the storage bracket inside the receiver cabinet.

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40.1.15. Inspection for mechanical (dimension) interchangeability (see 4.9.1). Add the following sentence: The gage points identified on gage lists (being procured concurrently or as separately contractually identified) for the ADDM modules shall be measured for compliance with mechanical interchangeability requirements.

40.1.16. Inspection for electrical interchangeability (see 4.9.2). Under transmitter modules, add the following grouping:

ADDM Modules

Regenerator, Digital Cable DL-SM-B-684645, GL-SM-B-404159
Control Comparator, Cable DL-SM-B-684646, GL-SM-B-404179
Alarm Monitor, DL-SM-B-684647, GL-SM-B-404187
Processor, Digital Radio DL-SM-B-684648, GL-SM-B-404195
Oscillator, Voltage Controlled (9.830400 MHz) SM-B-683698, GL-SM-B-405816

40.1.16.1. Under Receiver Modules, add the following:

Filter, Low Pass, 3.8 MHz**, DL-SM-B-684659, **96 Channel ADDM use only.

40.1.17. Transmitter, Radio T-1054( )/GRC-144 Tests (see 4.11). Add the following sentence: For the 96 channel ADDM mode, only the following transmitter tests are required: 4.11.3, 4.11.4, 4.11.6 (Orderwire Signal-to-Noise Ratio only), 4.11.7, 4.11.11, 4.11.12, 4.11.14, and 4.11.15.

40.1.18. Special test equipment (see 4.11.1.2). Add the following items to the list of test equipment:

TD-976( )/G Multiplexer (GLE)
TD-982( )/G Restorer, Pulse Form (GLE)
TD-660( )/G Multiplexer (GLE)
TD-754( )/G Multiplexer (GLE)
ADDM Test Equipment - (see 30.1.1)

40.1.19. Multiplex alarms, adjustments, and normal switch positions (see 4.11.2.3 and 4.11.2.3.1). Refer to 30. of this appendix.

40.1.20. Test mode (see 4.11.2.5, 4.11.2.5.1 and 4.11.2.5.2). Refer to 30.2 and 30.3 of this appendix.

40.1.21. Under table of transmitter meter readings (see 4.11.3.2.1, 4.11.3.3, 4.11.2.4.1, 4.11.3.4.2 and 4.11.3.4.7). Refer to 30. of this appendix.

30.1.22. In section line of 4.11.3.4.8 and 4.11.3.4.9, change DDM to read ADDM.

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40.1.23. To use 4.11.3.4.10 in the ADDM mode, refer to 30. of this appendix.

40.1.24. Orderwire Signal-to-Noise Ratio and Percentage of PCM Signal (see 4.11.6). Refer to 30. of this appendix.

40.1.25. To use 4.11.6.8 and 4.11.6.11 in the ADDM mode, refer to 30. of this appendix.

40.1.26. To use 4.11.7.3, 4.11.7.4, 4.11.7.9, and 4.11.7.10 in the ADDM mode, refer to 30. of this appendix.

40.1.27. To use 4.11.11.3 and 4.11.11.4 in the ADDM mode, refer to 30. of this appendix.

40.1.28. Orderwire signal-to-noise ratio with supply line variation (see 4.11.12). Refer to 30. of this appendix.

40.1.29. Orderwire peak limiter (see 4.11.13). Refer to 30. of this appendix.

40.1.30. Orderwire distortion (see 4.11.14). Refer to 30. of this appendix.

40.1.31. To use 4.11.15.6 in the ADDM mode, refer to 30. of this appendix.

40.1.32. Transmitter video input impedance (see 4.11.16). For the transmitter, receiver, multiplexer sets and coax termination interconnected as in figure 2, refer to 30. of this appendix.

40.1.33. Receiver, Radio R-1467( )/GRC-144 tests (see 4.12). Add the following sentence: For the 96 channel ADDM mode, only meter readings and alarm operation of 4.12.9 shall be performed.

40.1.34. Test mode - (see 4.12.2.4, 4.12.2.4.1, and 4.12.2.4.2). Refer to 30.2 and 30.3 of this appendix.

40.1.35. Transmitter/Receiver test (see 4.13). Add the following sentence: For the 96 channel ADDM mode, only video frequency response of 4.13.4 shall be performed.

40.1.36. Test mode - (see 4.13.2.4, 4.13.2.4.1, and 4.13.2.4.2). Refer to 30.2 and 30.3 of this appendix.

40.1.37. Paragraph 4.13.4.7 shall be changed to read as follows: Repeat procedures of 4.13.4.3 and 4.13.4.6 at frequencies of 460, 650, 920, 1300, 1840, 2600, and 3300 kHz for 48 channels: and 950, 1330, 1900, 2660, 3800,5320, and 6800 kHz for 96 channels.

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40.1.37.1. Change title of table to read as follows:

System Frequency Response - 48 and 96 channel operation

40.1.37.2. In the frequency response table, add a second FREQ (kHz) column adjacent to the first column to coincide with the respective response limits as follows:

FREQ (kHz)
50
950
1330
1900
2660
3800
5320
6800

40.1.38. Special test equipment (see 4.15.2.2). Add the following items to the list of test equipment:

TD-976( )/G Multiplexer (GLE)
TD-982( )/G Restorer, Pulse Form (GLE)
TD-660( )/G Multiplexer (GLE)
TD-754( )/G Multiplexer (GLE)
ADDM Test Equipment - (see 20.0 herein)

40.1.39. Preliminary set-up and adjustment (see 4.15.3.1). For the equipment interconnected to multiplexers as shown in figure 3 and figure 11, refer to 30. of this appendix.

40.1.40. To use 4.15.3.6, 4.15.3.7, and 4.15.3.9 in the ADDM mode, refer to 30. of this appendix.

40.1.41. To use 4.15.4.1.2 and 4.15.4.1.3 in the ADDM mode, refer to 30. of this appendix.

40.1.42. To use 4.15.4.3.4, 4.15.4.3.5, 4.15.4.3.6, and 4.15.4.3.7 in the ADDM mode, refer to 30. of this appendix.

40.1.43. Antenna alignment indicator test (see 4.15.5.4). Add the following sentences to this paragraph: This test is not required if previously performed in the DDM mode. If necessary to perform in the 48 channel ADDM mode, it is not required to be performed in the 96 channel ADDM mode.

40.1.44. Selection of equipment (see 30.7.1.1 of Appendix A). Add the following sentences: When the ADDM kit hardware is being supplied concurrently with a basic DDM configured radio, the reliability selection will be made using non-ADDM configured equipment. When only ADDM mode radios are being procured, reliability testing shall be completed for the 48 channel mode only.

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APPENDIX B

50. Interfaces for ADDM configured AN/GRC-144 with multiplex simulators.

50.1. The following data delineates the necessary characteristics of Multiplexer TD-976( )/G which must be simulated to properly check interfaces with the ADDM configured Radio Set. It is not considered necessary to test the radio set in a mode which includes cable repeaters to adequately affirm the ADDM capability.

50.1.1. Super group interface. The equipment shall provide 48/96 channel operation when connected up to 1/2 mile of CX-11230( )/G cable at a 4.9152 MB/s rate without restorers.

50.1.2. Super group output - input signals. Circuits shall be provided to drive up to 1/2 mile of CX-11230( )/G cable at a 4.9152 MB/s rate without restorers.

50.1.3. Super group output signals. The signals appearing at the output connector to the cable system shall be a composite of half baud bipolar data, analog voice orderwire, and constant current restorer power. When there are more than four (4) inputs to the Multiplexer TD-976( )/G (simulator), the information output will always be at the 96 channel information rate. When there are four or less inputs, the information output will be at the 48 channel rate. When the multiplexer (simulator) is transmitting at the 48 channel information rate, each bit shall be transmitted twice.

50.1.4. Data. The data signal at the termination is specified with the output connector to the cable system terminated into 56 ohms ± 2% and the building out network set for 1/2 mile of actual cable. Digital "1" signals shall be alternate positive and negative half baud pulses. Pulse amplitude shall be +0.9 ± 0.1 volts for positive pulses and -0.9 ± 0.1 volts for negative pulses. Pulse width shall be 100 ± 10 nanoseconds measured at the 50% amplitude points. Rise and fall times shall be 20 nanoseconds maximum. Pulse droop shall be 0.2 volts maximum. Backswing shall be 0.2 volts maximum. Ripple plus noise shall be less than 5% of pulse amplitude. The above parameters are defined in figure B1 and 50.1.9.

50.1.5. Orderwire. The cable orderwire signal appearing at the output connector to the cable system shall be as follows when the connector is terminated in 830 ohms ± 2%. With 0 dBm at 1000 Hz applied to the microphone terminals of the handset connector and the switches set for analog voice orderwire, signal shall be 2.75 volts rms ± 10%. With no signals applied, the residual noise level shall be less than 20 millivolts rms.

50.1.6. Super group input signals. The signals appearing at the input connector of Multiplexer TD-976( )/G (Simulator) from the cable system shall be a composite of the bipolar data, analog voice orderwire, and constant current restorer power after these signals have passed through the zero to the five mile restorer cable system.

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50.1.7. Analog voice. This orderwire shall be superimposed on the super group cable signal to the radio or a remote Multiplexer TD-976( )/G (Simulator).

50.1.8. Signalling circuits. The multiplexer (Simulator) shall contain an oscillator which operates at 1600 ± 30 Hz for signalling purposes for the analog orderwire. The detector shall operate with tone signals originating from a 1600 Hz oscillator at the opposite end of the link and it shall be adequately protected from speech "talk down". When the detector is activated, an audible signal shall sound. The activated conditions shall also light an indicator lamp which has been labeled as "CABLE O.W." Answer by the operator at the TD-976( )/G (Simulator) front panel or at the remote alarm display and orderwire access equipment shall remove the signal.

50.1.9. Signal and pulse mnemonics. Symbol abbreviations and descriptions associated with figure B1 are as follows:

SYMBOL	DESCRIPTION	VALUES AND LIMITS
tw	Pulse Width	100 ns ± 10 ns
ei	Pulse Amplitude	0.9 ± 0.1 volts
d	Droop	0.2 volts Max.
tr	Rise Time	20 nsec Max.
tf	Fall Time	20 nsec Max.
eo	Back Swing	0.2 volts Max.
er	Ripple	5% ei
di	Peak to Level	0.3 volts

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3. DOCUMENT TITLE: Radio Set AN/GRC-144 ( ) (Units of) and Indicator, Antenna Alignment ID-1708()/GRC

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

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